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REPORT

Report No. 61291

AS-506 "G" MANNED LUNAR LANDING MISSION
POST-LAUNCH OPERATIONAL TRAJECTORY
FOR JULY 16, 1969

Northrop Corporation
Huntsville, Alabama

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16. ABSTRACT The AS-506 post launch trajectory contains the Real-Time Operational Trajectory using the actual Greenwich Mean Time of Guidance Reference Release as supplied by the Manned Spacecraft Center, Houston, Texas. Presented in this report are the following: (a) Critical Sequence of Events (b) AS-506 Trajectory Data (c) Translunar Midcourse Correction (TLMC) (d) Trajectory Summary (e) S-IC and S-II Spent Stage Trajectory Data (f) S-IVB Slingshot Trajectory Data					
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FOREWORD

This report was prepared under contract NAS8-20082 and presents the AS-506 "G" Post-Launch operational trajectory for 16 July 1969.

The work was conducted in coordination with G. Wittenstein, Section Chief, S&E-AERO-MFT, Aero-Astroynamics Laboratory, George C. Marshall Space Flight Center. The data presented in this report were prepared by Messrs. J. Bradford, R. Maples, J. Reaves, R. Miller, J. Oliver, R. Stewart, and R. Wells.

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NOMENCLATURE LIST

<u>Printout Symbol</u>	<u>Trajectory Key Event</u>
TIME BASE 1	Liftoff
MACH 1	Mach One
S-IC CECO	S-IC Center-Engine Cutoff
S-IC OECO	S-IC Outboard-Engine Cutoff
S-IC/S-II SEP	S-IC/S-II Separation
S-II ENGINE START	S-II Engines Ignition
S-II 90 THRUST	S-II Engines 90-Percent Thrust
INITIATE IGM	Initiate Iterative Guidance Mode
S-II CUTOFF	S-II Engines Cutoff
S-IVB ULL IGN	S-IVB Ullage Rocket Ignition
S-II/S-IVB SEP	S-II/S-IVB Separation
S-IVB ENGINE START	S-IVB Engine Ignition
S-IVB 90 THRUST	S-IVB Engine 90-Percent Thrust
S-IVB ULLAGE JETT	S-IVB Ullage Case Jettison
S-IVB GCS1	S-IVB Guidance Cutoff Signal 1
EPOI	Earth Parking Orbit Insertion
TB6	Time Base 6
S-IVB REIGNITION	S-IVB Engine Reignition
S-IVB 90 THRUST	S-IVB 90-Percent Thrust
S-IVB GCS2	S-IVB Guidance Cutoff Signal 2
TLI	Translunar Injection

1.0 PURPOSE

This report is prepared to provide post-launch trajectory data of the AS-506 "G" Manned Lunar Landing Mission to those working groups that require immediate information about the real-time trajectory.

2.0 PRESENTATION OF DATA

The data presented herein is for the launch date of July 16, 1969, and is designated "G" mission profile. The Greenwich Mean Time of Guidance Reference Release (GRR) is 13:31:43.02 and is based on data received in real-time from the Houston Guidance Flight Controller.

Table 1 contains the critical sequence of events from Guidance Reference Release to Translunar Injection (TLI) + approximately 3 hours.

Table 2.2 contains trajectory listings which begin at time of first motion and include launch vehicle boost to orbit, parking orbit, TLI (first opportunity) and post-TLI (TLI + 25 minutes). The trajectory data are presented in both metric and English units.

Table 2.3 presents the Translunar Midcourse Correction (TLMC) results for mid-course delta velocities applied at Time Base 7 (TB7) + 2 hours.

Summarized in Table 2.4 are key trajectory parameters including data for the S-IC, S-II, and S-IVB first and second burns (first opportunity).

Presented in Table 2.5 are the spent stage trajectories for the S-IC and S-II from the time of S-IC/S-II separation and S-II/S-IVB separation until each stage impacts.

Table 2.6 contains the trajectory listing for the slingshot maneuvers of the S-IVB stage to place the stage in a heliocentric orbit.

TABLE 2-1
CRITICAL EVENT SEQUENCE
AS-506, 72.06° DEGREE AZIMUTH, JULY 16, 1969 LAUNCH DATE

NOMINAL TIME FROM FIRST MOTION (HR-MIN-SEC)	NOMINAL TIME FROM FIRST MOTION (SEC)	PROGRAM TIME REFERENCE (SEC)	PREDICTED SEQUENCE OF EVENTS	GUIDANCE REFERENCE RELEASE (GRR)
-0 00 17.3	-17.3			FIRST MOTION (HOLDOWN ARM RELEASE)
0 00 00.0	0.0			LIFT-OFF, TIMEBASE 1 SET
0 00 00.40	0.40	0.0(TB1)		BEGIN TOWER CLEARANCE YAW MANEUVER
0 00 01.4	1.4	1.0(TB1)		END YAW MANEUVER
0 00 10.4	10.4	10.0(TB1)		PITCH AND ROLL INITIATION
0 00 13.5	13.5 *			S-IC RADIAL ENGINE CANT
0 00 20.4	20.4	20.0(TB1)		ROLL MANEUVER COMPLETE
0 00 31.5	31.5			BEGIN SECOND SEGMENT OF PITCH POLYNOMIAL
0 00 34.0	34.0 *			MACH 1
0 01 05.8	65.8			BEGIN THIRD SEGMENT OF PITCH POLYNOMIAL
0 01 09.0	69.0 *			MAXIMUM DYNAMIC PRESSURE
0 01 21.0	81.0 **			BEGIN FOURTH SEGMENT OF PITCH POLYNOMIAL
0 01 36.0	96.0 *			COMPUTER SWITCH POINT NO. 1
0 01 45.4	105.4	105.0(TB1)		COMPUTER SWITCH POINT NO. 2
0 02 10.4	130.4	130.0(TB1)		S-IC CENTER ENGINE CUTOFF
0 02 15.00	135.00	134.6(TB1)		TIMEBASE 2' SET
0 02 15.05	135.05	0.0(TB2)		BEGIN TILT ARREST
0 02 40.5	160.5 *			S-IC OUTBOARD ENGINE CUTOFF, TIMEBASE 3 SET
0 02 40.82	160.82	0.0(TB3)		S-II ULLAGE IGNITION
0 02 41.3	161.3	0.5(TB3)		SIGNAL TO FIRE SEPARATION DEVICES AND RETROCKET MOTORS
0 02 41.5	161.5	0.7(TB3)		SEPARATION STRUCTURE COMPLETELY SEVERED
0 02 41.6	161.6			S-IC RETROCKET THRUST BUILD-UP BEGINS
0 02 41.6	161.6			

* DENOTES SIMULATION MAJOR CYCLE OCCURRENCE TIME. THESE SIMULATED DATA POINTS
MAY NOT BE EXACTLY CONSISTENT WITH THE SWITCH SELECTOR TCD SEQUENCE AND
ACTUAL LVDC PRESETTINGS, BUT THE EFFECTS OF DIFFERENCES ARE CONSIDERED
NEGLECTIBLE.

** NEAREST TIME POINT AVAILABLE

TABLE 2.1
CRITICAL EVENT SEQUENCE
AS-506, 72.06° DEGREE AZIMUTH, JULY 16, 1969 LAUNCH DATE

NOMINAL TIME FROM FIRST MOTION (HR-MIN-SEC)	NOMINAL TIME FROM FIRST MOTION (SEC)	PROGRAM TIME REFERENCE (SEC)	PREDICTED SEQUENCE OF EVENTS
0 02 42.2	162.2	1.4(TB3)	S-II ENGINE START SEQUENCE INITIATED
0 02 43.2	163.2		S-II IGNITION
0 02 45.2	165.2		S-II ENGINE AT 90 PERCENT THRUST
0 02 45.8	165.8		S-II ULLAGE THRUST TERMINATION
0 02 47.7	167.7	6.9(TB3)	S-II P. U. VALVE COMMANDED TO 5.5 MIXTURE RATIO
0 03 11.5	191.5	30.7(TB3)	S-II SECOND PLANE SEPARATION (JETTISON S-II AFT INTERSTAGE)
0 03 17.2	197.2		LET JETTISON
0 03 21.5	201.5 *		IGM ENABLE
0 03 42.2	222.2	61.4(TB3)	COMPUTER SWITCH POINT NO. 3
0 03 42.5	222.5 *		SMC TURN-ON
0 04 21.1	261.1	100.3(TB3)	LX STEP PRESSURIZATION
0 05 52.2	352.2	191.4(TB3)	COMPUTER SWITCH POINT NO. 4
0 07 39.8	459.8	299.3(TB2)	S-II CENTER ENGINE CUTOFF
0 08 08.5	488.5 **		TII LE C, SIGNAL PU T2 INITIATE MRS, START ARTIFICIAL TAU MODE
0 08 19.0	499.0 *		END ARTIFICIAL TAU MODE, BEGIN IGM PHASE 2
0 09 11.38	551.38	0.0(TB4)	S-II ENGINE CUTOFF, TIMEPHASE 4 SET, ENABLE CHI FREEZE
0 09 12.1	552.1	0.7(TB4)	S-IVB ULLAGE IGNITION
0 09 12.2	552.2	0.8(TB4)	SIGNAL T0 FIRE SEPARATION DEVICES AND RETROCKET MOTORS
0 09 12.3	552.3		SEPARATION STRUCTURE COMPLETELY SEVERED
0 09 12.4	552.4	1.0(TB4)	S-IVB ENGINE START SEQUENCE INITIATED
0 09 15.4	555.4		S-IVB IGNITION
0 09 17.9	557.9		S-IVB ENGINE AT 90 PERCENT THRUST

*DENOTES SIMULATION MAJOR CYCLE OCCURRENCE TIME. THESE SIMULATED DATA POINTS MAY NOT BE EXACTLY CONSISTENT WITH THE SWITCH SELECTOR ICD SEQUENCE AND ACTUAL LVDC PRESETTINGS, BUT THE EFFECTS OF DIFFERENCES ARE CONSIDERED NEGLIGIBLE.

**THERE IS APPROXIMATELY 6 SECONDS DELAY AFTER T11 GOES TO ZERO BEFORE THRUST DROP IS SENSED.

TABLE 2.1
CRITICAL EVENT SEQUENCE
AS-506, 72.063 DEGREE AZIMUTH, JULY 16, 1969 LAUNCH DATE

NOMINAL TIME FROM FIRST MOTION (HR-MIN-SEC)	NOMINAL TIME FROM FIRST MOTION (SEC)	PROGRAM TIME REFERENCE (SEC)	PREDICTED SEQUENCE OF EVENTS
0 09 19.9	559.9		S-IVB ULLAGE THRUST TERMINATION
0 09 21.0	561.0 *		END CHI FREEZE (BEGIN IGM PHASE 3), START ARTIFICIAL TAL MODE
0 09 24.2	564.2	12.8(TB4)	S-IVB ULLAGE CASE JETTISON
0 09 26.4	566.4		SMC TURN-ON
0 09 31.0	571.0 *		END ARTIFICIAL TAL MODE
0 11 05.1	665.1 *		BEGIN CHI STEERING
0 11 32.1	692.1		BEGIN CHI FREEZE, END IGM
0 11 39.2	699.2		S-IVB FIRST GUIDANCE CUTOFF (GCS1)
0 11 39.43	699.43	0.0(TB5)	SET TIMEBASE 5
0 11 39.7	699.7	0.3(TB5)	S-IVB APS ULLAGE ENGINE NO. 1 IGNITION
0 11 39.8	699.8	0.4(TB5)	S-IVB APS ULLAGE ENGINE NO. 2 IGNITION
0 11 49.2	709.2	9.8(TB5)	PARKING ORBIT INSERTION
0 11 59.4	719.4		BEGIN ORBITAL GUIDANCE
0 12 38.4	758.4	59.0(TB5)	H2 CONTINUOUS VENT ON
0 13 06.4	786.4	87.0(TB5)	S-IVB APS ULLAGE ENGINE NO. 1 CUTOFF
0 13 06.5	786.5	87.1(TB5)	S-IVB APS ULLAGE ENGINE NO. 2 CUTOFF
0 13 19.4	799.4		FIRST ORBITAL NAVIGATION CALCULATED
2 34 37.00	9277.00	0.0(TB6)	BEGIN S-IVB RESTART PREPARATIONS, TIMEBASE 6 SET
2 35 19.0	9319.0	42.0(TB6)	HELIUM HEATER ON
2 35 19.2	9319.2	42.2(TB6)	H2 CONTINUOUS VENT OFF
2 42 53.3	9773.3	496.3(TB6)	S-IVB APS ULLAGE ENGINE NO. 1 IGNITION
2 42 53.4	9773.4	496.4(TB6)	S-IVB APS ULLAGE ENGINE NO. 2 IGNITION
2 42 53.8	9773.8	496.8(TB6)	HELIUM HEATER OFF

*DENOTES SIMULATION MAJOR CYCLE OCCURRENCE TIME. THESE SIMULATED DATA POINTS
MAY NOT BE EXACTLY CONSISTENT WITH THE SWITCH SELECTOR ICD SEQUENCE AND
ACTUAL LVDC PRESETTINGS, BUT THE EFFECTS OF DIFFERENCES ARE CONSIDERED
NEGLECTIBLE.

TABLE 2.1

CRITICAL EVENT SEQUENCE

AS-506, 72.063 DEGREE AZIMUTH, JULY 16, 1969 LAUNCH DATE

NOMINAL TIME FROM FIRST MOTION (HR-MIN-SEC)	NOMINAL TIME FROM FIRST MOTION (SEC)	PROGRAM TIME REFERENCE (SEC)	PREDICTED SEQUENCE OF EVENTS
2 44 07.0	9847.0	570.0(TB6)	INITIATE J-2 FUEL LEAD
2 44 10.0	9850.0	573.0(TB6)	S-IVB APS ULLAGE ENGINE NO. 1 CUTOFF
2 44 10.1	9850.1	573.1(TB6)	S-IVB APS ULLAGE ENGINE NO. 2 CUTOFF
2 44 15.0	9855.0	578.0(TB6)	S-IVB REIGNITION
2 44 17.5	9857.5		S-IVB 90 PERCENT THRUST
2 44 21.0	9861.0		INITIATE OUT-OF-ORBIT IGM, PHASE 4
2 44 32.8	9872.8		SMC TURN-ON
2 46 12.2	9972.2		IGM MR SHIFT, START ARTIFICIAL TAU MODE
2 46 20.5	9980.5		END ARTIFICIAL TAU MODE, BEGIN IGM PHASE 5
2 48 35.0	10115.0 *	838.0(TB6)	COMPUTER SWITCH POINT NO. 5
2 49 33.5	10173.5 *		BEGIN CHI STEERING
2 50 00.5	10200.5		BEGIN CHI FREEZE, END IGM
2 50 03.8	10203.8		S-IVB SECOND GUIDANCE CUTOFF SIGNAL (GCS2)
2 50 04.00	10204.00	0.0(TB7)	SET TIMEBASE 7
2 50 04.5	10204.5	0.5(TB7)	H2 CONTINUOUS VENT ON
2 50 13.8	10213.8		TRANSLUNAR INJECTION
2 50 24.0	10224.0		BEGIN ORBITAL GUIDANCE, FIRST CALCULATION IN ORBITAL NAVIGATION
3 05 03.8	11103.8	899.8(TB7)	H2 CONTINUOUS VENT OFF
3 05 04.0	11104.0		BEGIN MANEUVER TO TRANSDUCTION AND DOCKING ATTITUDE
3 10 04.0	11404.0		SIMULATED CSM SEPARATION
4 50 04.00	17404.00	7200.0(TB7)§	BEGIN MANEUVER TO AND MAINTAIN THE SLINGSHOT ATTITUDE,
4 50 04.00	17404.00	0.0(TB8)	TIMEBASE 8 SET

*DEVOTES SIMULATION MAJOR CYCLE OCCURRENCE TIME. THESE SIMULATED DATA POINTS MAY NOT BE EXACTLY CONSISTENT WITH THE SWITCH SELECTOR ICD SEQUENCE AND ACTUAL LVDC PRESETTINGS, BUT THE EFFECTS OF DIFFERENCES ARE CONSIDERED NEGLIGIBLE.

§THE IU DOES NOT RECOGNIZE GROUND COMMAND TO RELEASE THE INHIBIT PRIOR TO TB7 + 2 HOURS.

TABLE 2.1
CRITICAL EVENT SEQUENCE
AS-506, 72.063 DEGREE AZIMUTH, JULY 16, 1969 LAUNCH DATA

NOMINAL TIME FROM FIRST MOTION (HR-MIN-SEC)	NOMINAL TIME FROM FIRST MOTION (SEC)	PROGRAM TIME REFERENCE (SEC)	PREDICTED SEQUENCE OF EVENTS
4 50 04.4	17404.4	0.4(TB8)	H2 CONTINUOUS VENT ON
5 02 04.0	18124.0	720.0(TB8)	BEGIN LEX DUMP
5 03 52.2	18232.2	828.2(TB8)	END LEX DUMP
5 36 44.0	20204.0	2800.0(TB8)	APS ULLAGE IGNITION
5 41 24.0	20484.0	3080.0(TB8)	APS ULLAGE CUTOFF
5 41 44.0	20504.0	3100.0(TB8)	INITIATE MANEUVER T2 AND MAINTAIN COMMUNICATIONS ATTITUDE

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
0.00	34046046.00	2902326.66	55.7	408.64	90.00	10.607
0.40	34205456.00	2897073.63	55.6	408.64	89.92	11.812
5.00	34339774.50	2836333.50	79.2	408.76	88.49	12.129
10.00	34336414.00	2776424.69	161.1	409.63	86.76	12.413
15.00	34468396.00	2704511.19	308.6	410.55	84.82	12.759
20.00	34616934.50	2635543.69	530.3	412.65	82.65	13.125
25.00	34795146.00	2572534.13	835.4	416.63	80.27	13.518
30.00	35033816.00	2506476.16	1233.6	423.20	77.71	13.952
35.00	35317584.50	2440363.44	1735.3	433.20	75.03	14.422
40.00	35634274.00	2374208.06	2351.2	447.48	72.33	14.926
45.00	35990157.00	2308008.53	3092.4	466.87	69.71	15.469
50.00	36375300.50	2241767.31	3968.9	492.00	67.29	16.050
55.00	36775154.50	2175488.38	4990.2	523.33	65.16	16.673
60.00	37180718.50	2105184.03	6165.0	561.10	63.40	17.303
65.00	37573365.50	2042826.41	7499.7	605.20	62.05	17.872
65.75	37630312.00	2032850.09	7713.8	612.31	61.89	17.935
70.00	37955338.00	1976316.88	8996.3	655.07	61.16	18.481
75.00	38344152.50	1905765.09	10656.3	711.47	60.61	19.358
80.00	38718815.50	1843126.09	12484.3	774.96	60.36	20.374
81.00	38787107.50	1829821.41	12870.5	788.53	60.34	20.575
85.00	39044678.00	1776568.38	14485.3	845.68	60.36	21.418
90.00	39322030.50	1705910.59	16662.4	923.72	60.58	22.515
95.00	39544724.50	1643138.13	19017.8	1009.15	60.98	23.682
100.00	39716511.00	1576263.63	21554.0	1102.01	61.49	24.903
105.00	39851298.50	1509287.16	24272.5	1202.58	62.11	26.186
110.00	39963828.00	1442215.53	27173.8	1311.06	62.80	27.550
115.00	40058686.00	1375038.03	30261.9	1427.56	63.49	29.016
120.00	40145785.50	1307759.50	33537.6	1553.10	64.27	30.612
125.00	40238781.00	1240353.50	36998.7	1688.11	65.07	32.377
130.00	40334532.50	1172814.13	40649.6	1833.33	65.84	34.351
135.00	40427432.00	1105109.03	44497.3	1989.82	66.59	36.554
140.00	41778193.00	1056610.53	48509.5	2120.69	67.40	30.233
145.00	31819812.75	997349.63	52639.4	2257.02	68.18	31.902
150.00	31863600.25	944019.09	56892.3	2402.86	68.92	33.760
155.00	31930868.25	890589.53	61276.3	2559.25	69.62	35.868
160.00	31915897.25	837403.78	65802.2	2727.28	70.26	38.133
160.82	31160189.25	826691.91	66557.7	2755.61	70.37	37.624
161.59	1718261.80	824990.88	67270.1	2765.96	70.48	2.112

TABLE 2-2 (BT0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST POSITION (SEC)	*** RADIAL DISTANCE OF THE C G (M)	MAGNITUDE OF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE 0N SPHERICAL EARTH (DEG)	LONGITUDE 0N EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (KM)
0.00	637353.56	0.000	28.6084	-80.60	0.00
0.40	637353.69	0.590	28.6084	-80.60	0.00
5.00	6373379.38	10.768	28.6084	-80.60	0.00
10.00	6373463.44	23.183	28.6084	-80.60	0.00
15.00	6373613.31	37.127	28.6083	-80.60	0.01
20.00	6373837.25	52.845	28.6082	-80.60	0.02
25.00	6374144.69	70.476	28.6082	-80.60	0.03
30.00	6374545.19	90.216	28.6081	-80.60	0.03
35.00	6375049.19	112.314	28.6082	-80.60	0.06
40.00	6375667.44	137.038	28.6083	-80.60	0.12
45.00	6376410.75	164.710	28.6087	-80.60	0.24
50.00	6377289.50	195.699	28.6092	-80.60	0.43
55.00	6378313.31	230.398	28.6099	-80.60	0.72
60.00	6379490.44	269.207	28.6111	-80.59	1.13
65.00	6380827.44	312.225	28.6127	-80.59	1.70
65.75	6381042.00	319.009	28.6129	-80.59	1.80
70.00	6382326.44	359.195	28.6147	-80.58	2.45
75.00	6383988.94	411.502	28.6174	-80.57	3.41
80.00	6385819.38	470.218	28.6208	-80.56	4.63
81.00	6386206.06	482.771	28.6216	-80.56	4.91
85.00	6387822.81	535.734	28.6250	-80.54	6.14
90.00	6390002.56	608.323	28.6302	-80.53	7.99
95.00	6392360.63	688.299	28.6363	-80.50	10.20
100.00	6394899.44	775.916	28.6436	-80.48	12.82
105.00	6397620.88	871.421	28.6522	-80.45	15.90
110.00	6400525.00	975.100	28.6621	-80.41	19.47
115.00	6403616.06	1087.310	28.6736	-80.37	23.57
120.00	6406894.75	1208.554	28.6866	-80.33	28.25
125.00	6410359.19	1339.545	28.7013	-80.28	33.56
130.00	6414013.44	1481.138	28.7178	-80.22	39.55
135.00	6417864.63	1634.332	28.7363	-80.15	46.27
140.00	6421880.50	1762.021	28.7567	-80.08	53.70
145.00	6426014.19	1895.443	28.7788	-80.00	61.79
150.00	6430271.06	2038.628	28.8027	-79.92	70.57
155.00	6434659.25	2192.610	28.8286	-79.82	80.10
160.00	6439189.38	2358.469	28.8565	-79.72	90.43
160.82	6439945.69	2386.463	28.8613	-79.71	92.20
161.59	6440658.81	2396.459	28.8658	-79.69	93.89

*** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN. SYS, (M)	Y CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN. SYS, (M)	Z CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN. SYS, (M)	X CØMPONENT ØF VEL. IN THE XXE SYSTEM (M/S)	Y CØMPONENT ØF VEL. IN THE XXE SYSTEM (M/S)	Z CØMPONENT ØF VEL. IN THE XXE SYSTEM (M/S)
0.00	59.125	-0.056	0.006	0.000	-0.000	-0.000
0.40	59.250	-0.051	0.004	0.590	0.000	0.001
5.00	84.938	-0.146	-0.083	10.767	0.128	-0.067
10.00	169.063	4.498	-0.750	23.122	1.672	-0.198
15.00	318.813	14.402	-2.125	37.078	1.856	-0.373
20.00	542.813	22.985	-3.379	52.819	1.654	0.081
25.00	850.125	30.723	0.267	70.444	1.434	1.578
30.00	1250.625	37.408	15.025	90.089	1.229	4.638
35.00	1754.625	42.991	50.341	111.871	1.005	9.923
40.00	2372.875	47.397	119.270	135.820	0.767	18.210
45.00	3116.375	50.676	238.770	161.900	0.549	30.287
50.00	3995.313	52.923	429.731	189.991	0.350	46.917
55.00	5019.375	54.234	716.669	219.890	0.173	68.785
60.00	6196.750	54.669	1127.328	251.323	0.016	96.483
65.00	7534.063	54.429	1691.885	283.693	-0.115	130.396
65.75	7748.688	54.340	1791.780	288.558	-0.133	136.019
70.00	9033.563	53.575	2441.602	316.107	-0.232	170.580
75.00	10596.375	52.146	3409.320	349.186	-0.336	217.722
80.00	12527.188	50.167	4631.369	383.296	-0.452	272.377
81.00	12913.938	49.731	4909.638	390.230	-0.479	284.233
85.00	14530.688	47.583	6146.059	418.214	-0.583	334.825
90.00	16709.938	44.396	7993.178	453.552	-0.678	405.397
95.00	19066.813	40.718	10213.630	489.255	-0.820	484.132
100.00	21603.188	35.924	12848.022	525.352	-1.118	571.007
105.00	24320.313	29.651	15937.949	561.474	-1.367	666.424
110.00	27217.688	22.407	19526.906	597.617	-1.618	770.501
115.00	30298.438	12.661	23656.450	634.651	-2.219	882.868
120.00	33562.125	1.884	28372.772	670.684	-1.959	1005.377
125.00	37005.313	-6.449	33726.955	706.667	-1.381	1137.980
130.00	40630.188	-11.901	39769.933	743.470	-0.792	1281.024
135.00	44441.688	-14.283	46556.206	781.292	-0.183	1435.487
140.00	48405.313	-13.826	54068.173	803.024	0.403	1568.398
145.00	52472.000	-9.947	62253.106	823.777	1.169	1707.072
150.00	56644.375	-1.965	71154.498	845.302	2.036	1855.119
155.00	60926.875	10.601	80821.519	867.940	3.018	2013.507
160.00	65326.125	28.360	91308.590	891.997	4.121	2183.277
160.82	66058.250	31.823	93108.427	895.948	4.313	2211.891
161.59	66747.938	35.191	94818.007	893.543	4.446	2223.641

TIME BASE 1

MACH 1

MAX Q

S-IC CECØ

S-IC ØECØ
S-IC/S-II SEP

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	COMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN PITCH (DEG)	COMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN YAW (DEG)	COMMANDED ROLL ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN ROLL (DEG)
0.00	0.000	-0.0602	0.000	0.019472	-17.941	-17.90646720
0.40	-0.000	-0.0605	0.000	0.018234	-17.941	-17.90665102
5.00	-0.000	0.0311	1.250	1.083350	-17.941	-17.94329882
10.00	-0.000	0.0106	0.398	1.211642	-17.941	-17.94140601
15.00	-0.455	-0.0567	0.000	-0.300452	-17.339	-17.91676998
20.00	-1.223	-1.0160	0.000	-0.032072	-12.339	-12.96141338
25.00	-2.371	-2.0379	0.000	-0.056187	-7.339	-7.95028305
30.00	-3.976	-3.5489	0.000	-0.053513	-2.339	-2.95075205
35.00	-6.067	-5.5475	0.000	-0.057094	0.000	-0.04340151
40.00	-8.612	-8.0190	0.000	-0.048697	0.000	-0.00134260
45.00	-11.556	-10.9161	0.000	-0.039106	0.000	0.00000304
50.00	-14.813	-14.1472	0.000	-0.030898	0.000	-0.00007119
55.00	-18.296	-17.6231	0.000	-0.026270	0.000	-0.00013201
60.00	-21.921	-21.2570	0.000	-0.022511	0.000	-0.00018454
65.00	-25.607	-24.9714	0.000	-0.018885	0.000	-0.00029461
65.75	-26.161	-25.5282	0.000	-0.020716	0.000	-0.00032593
70.00	-29.277	-28.6634	0.000	-0.024904	0.000	-0.00046113
75.00	-32.871	-32.2826	0.000	-0.029853	0.000	-0.00066216
80.00	-36.342	-35.7437	0.000	-0.047231	0.000	-0.00115376
81.00	-37.018	-36.4228	0.000	-0.049226	0.000	-0.00119968
85.00	-39.656	-39.0863	0.000	-0.056556	0.000	-0.00130693
90.00	-42.792	-42.2980	0.000	-0.053196	0.000	-0.00102629
95.00	-45.734	-45.2038	0.000	-0.097992	0.000	-0.00190414
100.00	-48.468	-47.9198	0.000	-0.139355	0.000	-0.00212825
105.00	-51.008	-50.5639	0.000	-0.111802	0.000	-0.00155601
110.00	-53.364	-52.6032	0.000	-0.286405	0.000	-0.00182584
115.00	-55.544	-54.8616	0.000	-0.228863	0.000	-0.00034027
120.00	-57.556	-57.3569	0.000	-0.004493	0.000	0.00025871
125.00	-59.408	-59.2269	0.000	-0.017150	0.000	0.00002773
130.00	-61.109	-60.9037	0.000	-0.039859	0.000	-0.00003884
135.00	-62.669	-62.5270	0.000	-0.095340	0.000	-0.00009962
140.00	-64.099	-63.9310	0.000	-0.059533	0.000	-0.00006093
145.00	-65.411	-65.2527	0.000	-0.055902	0.000	-0.00004295
150.00	-66.619	-66.4665	0.000	-0.056383	0.000	-0.00004540
155.00	-67.734	-67.5891	0.000	-0.058090	0.000	-0.00004794
160.00	-68.773	-68.6338	0.000	-0.058739	0.000	-0.00004798
160.82	-68.873	-68.7982	0.000	-0.058797	0.000	-0.00003508
161.59	-68.873	-68.9424	0.000	-0.058897	0.000	0.00024871

TIME BASE 1

MACH 1

MAX Q

S-IC CECØ

S-IC ØECØ
S-IC/S-II SEP

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MØTTON (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTØR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTØR (KG)
0.00	0.00	0.23	89.9999	0.2479	2.0784	2.6061
0.40	0.00	0.43	46.4985	0.4728	2.0473	4.9692
5.00	0.03	68.30	2.8933	573.0772	20.1500	128.5485
10.00	0.07	313.69	3.5136	1342.1770	112.3922	718.9962
15.00	0.11	792.93	3.4406	2774.5317	278.1964	1786.8495
20.00	0.15	1573.59	2.0342	5014.2706	326.4039	2127.0305
25.00	0.21	2719.39	1.3926	8220.4706	386.1637	2454.7604
30.00	0.26	4288.57	0.9761	12559.0575	426.8661	2713.4970
35.00	0.33	6327.19	0.6836	18203.4446	441.0261	2803.5089
40.00	0.41	8855.95	0.4568	25317.0327	412.5026	2622.1909
45.00	0.49	11865.21	0.2959	34044.1367	358.0639	2276.1359
50.00	0.59	15306.78	0.1789	44497.9580	279.2355	1775.0405
55.00	0.70	15096.35	0.1084	55117.1084	211.1810	1342.4328
60.00	0.83	23085.39	0.0950	73051.4863	223.7084	1599.8252
65.00	0.98	27017.95	0.1301	111261.4287	358.5265	2848.8461
70.00	1.01	27574.59	0.1364	122109.3994	383.5135	3047.3925
75.00	1.16	30459.63	0.1630	148122.2441	506.2599	3821.5964
80.00	1.60	34654.57	0.3024	120214.5029	1068.4655	5084.3427
85.00	1.66	34725.24	0.3106	117273.8135	1099.8737	7216.5085
90.00	1.87	34093.45	0.3251	102149.1348	1130.1521	7184.1365
95.00	2.14	31161.01	0.2930	84525.8828	931.0210	5918.3020
100.00	2.41	26479.97	0.5300	64718.3350	1431.0999	9097.1974
105.00	2.68	21853.85	0.6696	47416.6582	1492.1756	9485.4426
110.00	2.97	17584.08	0.5622	33594.1196	1008.1338	6408.4916
115.00	3.29	13835.72	0.8322	23592.4438	1174.0683	7463.3020
120.00	3.62	10582.29	0.5853	16455.2366	631.6106	3011.2607
125.00	3.95	7890.84	0.0897	11532.8325	72.1861	344.1538
130.00	4.31	5801.97	0.1320	8111.6811	78.1127	372.4096
135.00	4.68	4199.42	0.1798	4850.9570	77.0058	367.1321
140.00	5.08	2015.55	0.2542	3238.1631	78.1674	310.5584
145.00	5.41	2070.21	0.1859	1568.8853	39.2494	155.9375
150.00	5.88	1459.97	0.1710	252.0057	25.4528	101.1240
155.00	6.45	1013.15	0.1550	-671.9060	16.0179	63.6391
160.00	7.11	681.70	0.1457	-1301.3289	10.1288	36.2176
165.00	7.87	438.89	0.1449	-1685.4988	6.4870	23.1957
170.00	8.00	406.13	0.1456	-1734.6624	6.0295	21.5596
175.00	8.07	371.97	0.1517	-1776.1748	5.7529	20.5707

TIME BASE I

MACH I

MAX Q

S-IC CECØ

S-IC ØECØ
S-IC/S-II SEP

TABLE 2.2 (BT0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Y PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Z PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	X VEL. COMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Y VEL. COMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Z VEL. COMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)
0.00	6373324.5	19255.8	1174.6	-0.451	125.641	388.843
0.40	6373324.4	19305.8	1329.4	0.128	125.636	388.846
5.00	6373347.7	19883.8	3118.8	10.185	125.707	388.814
10.00	6373428.6	20516.1	5062.7	22.408	127.192	388.735
15.00	6373574.4	21153.5	7006.1	36.234	127.322	388.626
20.00	6373793.9	21789.3	8950.1	51.843	127.072	389.163
25.00	6374096.0	22424.0	10899.4	69.333	126.808	390.762
30.00	6374490.6	23057.5	12860.3	88.836	126.561	393.945
35.00	6374987.9	23689.6	14842.6	110.466	126.298	399.380
40.00	6375598.8	24320.4	16859.2	134.249	126.022	407.843
45.00	6376333.9	24949.9	18927.4	160.139	125.762	420.127
50.00	6377203.5	25578.0	21068.1	188.009	125.514	436.996
55.00	6378217.0	26205.0	23306.1	217.649	125.277	459.136
60.00	6379382.5	26830.8	25669.3	248.774	125.042	487.141
65.00	6380706.2	27455.4	28188.0	280.778	124.812	521.394
65.75	6380918.6	27549.0	28581.1	285.583	124.776	527.070
70.00	6382190.0	28078.9	30893.6	312.761	124.567	561.947
75.00	6383834.9	28701.1	33819.1	345.333	124.302	609.492
80.00	6385644.9	29321.8	37001.1	378.845	123.986	664.585
81.00	6386027.2	29445.8	37671.6	385.648	123.914	676.533
85.00	6387624.5	29940.8	40478.0	413.067	123.609	727.507
90.00	6389776.1	30557.9	44289.8	447.596	123.215	798.587
95.00	6392100.9	31172.8	48477.5	482.370	122.715	877.863
100.00	6394600.3	31784.7	53082.0	517.405	121.995	965.313
105.00	6397274.7	32392.7	58145.0	552.321	121.251	1061.337
110.00	6400122.9	32997.2	63710.1	587.103	120.427	1166.052
115.00	6403147.4	33596.1	69819.0	622.613	119.171	1279.095
120.00	6406346.7	34190.4	76518.2	656.932	118.676	1402.307
125.00	6409716.4	34783.1	83858.9	690.998	118.392	1535.645
130.00	6413257.7	35374.2	91892.1	725.666	118.008	1679.463
135.00	6416974.3	35963.1	100672.6	761.115	117.521	1834.740
140.00	6420831.3	36549.3	110182.6	780.543	117.009	1968.342
145.00	6424779.6	37133.3	120369.0	798.812	116.578	2107.717
150.00	6428820.4	37715.1	131275.4	817.629	116.127	2256.488
155.00	6432957.3	38294.6	142951.1	837.313	115.660	2415.631
160.00	6437195.4	38871.6	15450.8	858.145	115.170	2586.190
160.82	6437899.6	38965.9	157580.6	861.545	115.089	2614.935
161.59	6438562.7	39054.5	159600.6	858.782	115.035	2626.743

S-IC CECØ

S-IC ØECØ
S-IC/S-II SEP

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED PITCH CHANL (DEG)	GIMBAL DEFLECTION ANG REQUIRED YAW CHANNEL (DEG)	GIMBAL DEFLECTION ANG REQUIRED ROLL CHANNEL (DEG)	EULERIAN ATTITUDE RATE IN PITCH (DEG/S)	EULERIAN ATTITUDE RATE IN YAW (DEG/S)	EULERIAN ATTITUDE RATE IN ROLL (DEG/S)
TIME BASE 1	0.00	0.0000	0.00000	-0.00366798	0.00000000	0.00200055
0.40	-0.0261	-0.000798	0.002564	0.00567030	-0.00032230	-0.00937538
5.00	-0.0426	0.191529	-0.002564	-0.13706344	0.42336835	-0.00284825
10.00	-0.0562	0.189391	-0.000106	0.05065309	-0.15279737	-0.00129102
15.00	0.1887	-0.123611	-0.026242	-0.13683445	0.01410904	0.17937678
20.00	0.0108	-0.025652	-0.000048	-0.14093527	-0.06522282	1.01946378
25.00	0.0226	-0.020498	0.000011	-0.26076571	-0.02666029	1.00017834
30.00	0.0266	-0.029037	-0.000030	-0.34838488	-0.01866175	1.00031622
35.00	0.0231	-0.029114	-0.013553	-0.44895082	-0.00405367	-0.15339026
40.00	0.0217	-0.026180	0.000132	-0.53945705	0.00344738	0.00526924
45.00	0.0175	-0.026117	-0.000005	-0.61585466	0.00131322	0.00035672
50.00	0.0146	-0.023604	-0.000008	-0.67370060	0.00155769	0.00036256
55.00	0.0122	-0.022802	-0.000021	-0.71343417	0.00021666	0.00031437
60.00	0.0099	-0.020653	-0.000035	-0.73789839	0.00149455	0.00027338
65.00	0.0060	-0.020231	-0.000061	-0.74290618	-0.00101982	0.00022610
65.75	0.0068	-0.020773	-0.000067	-0.74166379	-0.00116567	0.00022145
70.00	0.0121	-0.022531	-0.000098	-0.73211418	-0.00133737	0.00026483
75.00	0.0194	-0.024844	-0.000144	-0.71091814	-0.00157937	0.00027549
80.00	0.0507	-0.035202	-0.000256	-0.67978097	-0.00267320	0.00048856
81.00	0.0520	-0.035654	-0.000266	-0.67797603	-0.00143375	0.00056900
85.00	0.0584	-0.040666	-0.000284	-0.65461238	-0.00174552	0.00049592
90.00	0.0443	-0.039951	-0.000226	-0.61406037	-0.00324606	0.00045952
95.00	0.0969	-0.068831	-0.000424	-0.55666297	-0.01179380	0.00080526
100.00	0.1232	-0.088367	-0.000488	-0.53852332	0.00066508	0.00141500
105.00	0.0885	-0.076536	-0.000344	-0.51449201	0.00754855	0.00116564
110.00	0.1482	-0.108625	-0.000536	-0.34371769	-0.04505831	0.00200425
115.00	0.0542	-0.056500	-0.000178	-0.56667902	0.07528769	0.00261750
120.00	-0.0189	-0.022442	0.000062	-0.41553970	0.00869327	0.00003762
125.00	-0.0046	-0.029720	0.000018	-0.34612139	-0.00745193	0.00004920
130.00	-0.0021	-0.029693	0.000007	-0.32616656	-0.00188764	0.00023704
135.00	-0.0077	-0.032156	0.000024	-0.29956479	-0.00800199	0.00045357
140.00	-0.0013	-0.020261	0.000005	-0.27357468	0.00676547	0.00030575
145.00	-0.0027	-0.021641	0.000008	-0.25293688	-0.00069523	0.00024423
150.00	-0.0023	-0.020775	0.000005	-0.2324159	-0.0008580	0.00022954
155.00	-0.0022	-0.020647	0.000003	-0.21629738	-0.00034758	0.00021958
160.00	-0.0022	-0.020282	0.000002	-0.20209075	-0.00008781	0.00020722
160.82	-0.0096	-0.020233	0.000006	-0.19701491	-0.00004141	0.00033730
161.59	-0.0314	-0.020203	0.000097	-0.18134203	-0.00034408	0.00063060

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TABLE 2.2 (BT0)

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABOVE SPHERICAL EARTH (FT)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (FT/S**2)
0.00	7653855.50	6398534.88	182.7	1340.67	90.00	34.800
0.40	7689701.31	6386953.94	182.5	1340.68	89.92	38.754
5.00	7719888.31	6253044.88	259.8	1341.07	88.49	39.793
10.00	7719132.81	6107740.81	528.4	1343.92	86.76	40.726
15.00	7748803.56	5962426.44	1012.5	1346.96	84.82	41.860
20.00	7782196.31	5816993.00	1739.9	1353.85	82.65	43.062
25.00	7822259.88	5671466.81	2740.7	1366.90	80.27	44.352
30.00	7875915.00	5525833.94	4047.3	1388.47	77.71	45.774
35.00	7939708.75	538080.38	5693.3	1421.25	75.03	47.316
40.00	8010903.38	5234232.69	7714.0	1468.12	72.33	48.970
45.00	8090909.06	5088287.75	10145.5	1531.72	69.71	50.753
50.00	8177452.75	4942250.81	13021.2	1614.18	67.29	52.659
55.00	8267383.50	4796130.81	16372.1	1716.95	65.16	54.701
60.00	8358557.94	4649554.75	20226.5	1840.87	63.40	56.769
65.00	8446828.38	4503661.25	24605.2	1985.56	62.05	58.634
65.75	8459630.50	4481667.25	25307.9	2008.91	61.89	58.840
70.00	8532699.25	4357032.81	29515.3	2149.19	61.16	60.632
75.00	8620108.25	4210311.25	34961.7	2334.21	60.61	63.510
80.00	8704335.88	4063397.41	40959.1	2542.53	60.36	66.844
81.00	8719688.50	4034065.59	42226.1	2587.03	60.34	67.503
85.00	8777592.63	3916662.78	47523.8	2774.53	60.36	70.271
90.00	8839944.00	3769707.53	54666.8	3030.59	60.58	73.868
95.00	8890007.50	3622499.44	62394.4	3310.86	60.98	77.698
100.00	8928626.75	3475066.38	70715.1	3615.51	61.49	81.702
105.00	8958928.13	3327408.56	79634.3	3945.47	62.11	85.913
110.00	8984225.75	3175540.94	89152.8	4301.37	62.80	90.387
115.00	9005550.75	3031439.91	99284.4	4683.60	63.49	95.196
120.00	9025131.50	2883116.13	110031.5	5095.46	64.27	100.432
125.00	9046037.63	2734511.34	121386.7	5538.40	65.07	106.225
130.00	9067563.50	2585612.50	133364.8	6014.87	65.84	112.699
135.00	9088448.13	2436348.31	145988.6	6528.28	66.59	119.926
140.00	9144021.88	2316199.69	159151.9	6957.63	67.40	126.189
145.00	9153378.38	2198779.50	172701.5	7404.94	68.18	132.665
150.00	9163222.19	2081205.81	186654.6	7883.40	68.92	139.762
155.00	9178344.63	1963413.80	201037.8	8396.48	69.62	147.677
160.00	9174979.00	1846159.30	215886.4	8947.77	70.26	156.107
160.82	7005089.13	1826952.89	218365.3	9040.72	70.37	163.440
161.59	386280.61	1818793.52	220702.5	9074.66	70.48	170.928

TIME BASE 1

MACH 1

MAX Q

S-IC CEC0

S-IC 0EC0
S-IC/S-11 SEP

TABLE 2-2 (810)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	*** RADIAL DISTANCE OF THE C G (FT)	MAGNITUDE OF THE EARTH FIXED VELOCITY (FT/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (NM)
0.00	20909952.50	0.000	28.6084	-80.60	0.00
0.40	20909953.00	1.934	28.6084	-80.60	0.00
5.00	20910037.25	35.327	28.6084	-80.60	0.00
10.00	20910313.00	76.059	28.6084	-80.60	0.00
15.00	20910804.75	121.806	28.6083	-80.60	0.01
20.00	20911539.50	173.376	28.6082	-80.60	0.01
25.00	20912548.00	231.222	28.6082	-80.60	0.01
30.00	20913862.00	295.985	28.6081	-80.60	0.02
35.00	20915515.50	368.486	28.6082	-80.60	0.03
40.00	20917544.00	449.599	28.6083	-80.60	0.07
45.00	20919982.50	540.386	28.6087	-80.60	0.13
50.00	20922865.75	642.057	28.6092	-80.60	0.23
55.00	20926224.50	755.898	28.6099	-80.60	0.39
60.00	20930086.50	883.225	28.6111	-80.59	0.61
65.00	20934473.00	1024.362	28.6127	-80.59	0.92
65.75	20935177.00	1046.617	28.6129	-80.59	0.97
70.00	20939391.00	1178.460	28.6147	-80.58	1.32
75.00	20944845.50	1350.073	28.6174	-80.57	1.84
80.00	20950850.75	1542.710	28.6208	-80.56	2.50
81.00	20952119.50	1583.895	28.6216	-80.56	2.65
85.00	20957423.75	1757.658	28.6250	-80.54	3.32
90.00	20964575.25	1995.809	28.6302	-80.53	4.31
95.00	20972311.50	2258.197	28.6363	-80.50	5.51
100.00	20980641.00	2545.655	28.6436	-80.48	6.92
105.00	20989569.75	2858.994	28.6522	-80.45	8.58
110.00	20999097.50	3199.147	28.6621	-80.41	10.51
115.00	21009239.00	3567.288	28.6736	-80.37	12.73
120.00	21019995.75	3965.073	28.6866	-80.33	15.26
125.00	21031362.00	4394.832	28.7013	-80.28	18.12
130.00	21043351.00	4859.378	28.7178	-80.22	21.36
135.00	21055986.25	5361.981	28.7363	-80.15	24.98
140.00	21069161.50	5780.909	28.7567	-80.08	29.00
145.00	21082723.50	6218.646	28.7788	-80.00	33.36
150.00	21096689.75	6688.412	28.8027	-79.92	38.11
155.00	21111086.50	7193.602	28.8286	-79.82	43.25
160.00	21125949.25	7737.758	28.8565	-79.72	48.83
160.82	21128430.50	7829.602	28.8613	-79.71	49.79
161.59	21130770.25	7862.398	28.8658	-79.69	50.69

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Y COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Z COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	X COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	Y COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	Z COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)
0.00	193.980	-0.183	C.019	0.000	-0.000	-0.000
0.40	194.390	-0.167	C.015	1.934	0.001	0.003
5.00	278.666	-0.477	-0.271	35.324	0.420	-0.219
10.00	554.667	14.757	-2.462	75.859	5.485	-0.649
15.00	1045.973	47.252	-6.971	121.648	6.088	-1.224
20.00	1780.881	75.410	-11.084	173.290	5.427	0.267
25.00	2789.124	100.799	0.877	231.116	4.706	5.177
30.00	4103.100	122.730	49.294	295.566	4.032	15.215
35.00	5756.644	141.048	165.162	367.030	3.297	32.557
40.00	7785.023	155.502	391.305	445.604	2.516	59.745
45.00	10224.327	166.260	783.366	531.168	1.802	99.366
50.00	13107.981	173.632	1409.878	623.331	1.147	153.929
55.00	16467.766	177.934	2351.277	721.424	0.569	225.674
60.00	20330.544	179.360	3698.582	824.552	0.052	316.546
65.00	24718.053	178.573	5550.803	930.750	-0.377	427.809
65.75	25422.203	178.281	5878.544	946.712	-0.437	446.255
70.00	29637.672	175.772	8010.504	1037.095	-0.762	559.645
75.00	35093.094	171.081	11185.435	1145.625	-1.104	714.311
80.00	41099.696	164.591	15194.780	1257.533	-1.484	893.624
81.00	42368.561	163.159	16107.735	1280.281	-1.570	932.524
85.00	47672.859	156.113	20164.234	1372.094	-1.914	1098.507
90.00	54822.629	145.655	26224.337	1488.030	-2.225	1330.044
95.00	62555.159	133.589	33509.284	1605.169	-2.690	1588.358
100.00	70876.599	117.862	42152.304	1723.597	-3.669	1873.382
105.00	79791.051	97.281	52285.858	1842.105	-4.486	2186.430
110.00	89296.874	73.515	64064.653	1960.687	-5.307	2527.889
115.00	99404.322	41.539	77613.023	2082.187	-7.280	2896.548
120.00	110111.958	6.180	93086.522	2200.408	-6.426	3298.480
125.00	121408.505	-21.158	110652.737	2318.463	-4.530	3733.531
130.00	133301.139	-39.046	130478.781	2439.205	-2.598	4202.836
135.00	145806.061	-46.862	152743.455	2563.293	-0.599	4709.604
140.00	158810.080	-45.360	177385.018	2634.592	1.322	5145.661
145.00	172152.230	-32.635	204242.475	2702.679	3.834	5600.631
150.00	185841.125	-6.446	233446.514	2773.300	6.679	6086.347
155.00	199891.320	34.780	265162.461	2847.573	9.900	6605.993
160.00	214324.557	93.044	299568.863	2926.498	13.521	7162.983
160.82	216726.541	104.405	305473.840	2939.462	14.152	7256.861
161.59	218989.295	115.456	311082.695	2931.570	14.586	7295.409

TABLE 2-2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT**2)	TOTAL ANGLE OF ATTACK (DEG)	DRAW ALONG VELOCITY VECTØR (LB)	AERODYNAMIC LOAD (LB/FT**2)	MAGNITUDE OF THE AERODYNAMIC FORCE VECTOR (FT)
TIME BASE 1						
0.00	0.00	0.00	89.9999	0.5466	0.4257	5.7455
0.40	0.00	0.01	46.4985	1.0424	0.4193	10.9551
5.00	0.03	1.43	2.8933	1263.4190	4.1270	283.4010
10.00	0.07	6.55	3.5136	2958.9938	23.0197	1585.1154
15.00	0.11	16.56	3.4406	6116.7952	56.9791	3939.3289
20.00	0.15	32.87	2.0342	11054.5742	66.8528	4689.2994
25.00	0.21	56.80	1.3926	18123.0352	79.0926	5411.8203
30.00	0.26	89.57	0.9761	27687.9819	87.4291	5982.2369
35.00	0.33	132.15	0.6836	40131.7251	90.3293	6180.6790
40.00	0.41	184.96	0.4568	55814.5020	84.4872	5780.9413
45.00	0.49	247.81	0.2959	75054.4727	73.3373	5018.0206
50.00	0.59	319.69	0.1789	98101.2031	57.1919	3913.2944
55.00	0.70	398.84	0.1084	121512.4219	43.2533	2959.5577
60.00	0.83	482.15	0.0950	161050.9570	45.8191	3527.0108
65.00	0.98	564.28	0.1301	245289.4590	73.4320	6280.6305
65.75	1.01	575.91	0.1364	269205.1406	78.5498	6718.3502
70.00	1.16	636.16	0.1630	326553.6445	103.6902	8425.1779
75.00	1.36	692.66	0.2102	312877.8320	145.6159	11209.0568
80.00	1.60	723.78	0.3024	265027.6094	218.8390	15909.6776
81.00	1.66	725.25	0.3106	258544.4980	225.2719	16377.3521
85.00	1.87	712.06	0.3251	225200.2891	231.4734	15838.3096
90.00	2.14	650.81	0.2930	186347.6699	190.6881	13047.6223
95.00	2.41	553.05	0.5300	142679.5020	293.1124	20055.8867
100.00	2.68	456.43	0.6696	104535.8359	305.6216	20911.8210
105.00	2.97	367.25	0.5622	74062.3545	206.4821	14128.3053
110.00	3.29	288.96	0.8322	52012.4346	240.4681	16453.7642
115.00	3.62	221.02	0.5853	36277.5859	129.3640	6638.6934
120.00	3.95	164.80	0.0897	25425.5430	14.7849	758.7293
125.00	4.31	121.18	0.1320	17883.1953	15.9988	821.0226
130.00	4.68	87.71	0.1798	10694.5294	15.7720	809.3876
135.00	5.08	62.98	0.2542	7138.9274	16.0099	684.6642
140.00	5.41	43.24	0.1859	3458.7999	8.0389	343.7833
145.00	5.88	30.49	0.1710	555.5775	5.2132	222.9402
150.00	6.45	21.16	0.1550	-1481.2990	3.2807	140.3003
155.00	7.11	14.24	0.1457	-2868.9391	2.0745	79.8461
160.00	7.87	9.17	0.1449	-3715.8886	1.3287	51.1377
160.82	8.00	8.48	0.1456	-3824.2760	1.2349	47.5308
161.59	8.07	7.77	0.1517	-3915.7950	1.1783	45.3505

S-IC CECØ

S-IC ØECØ
S-IC/S-II SEP

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MØTTØN (SEC)	X PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Y PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Z PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)
0.00	20909857.3	63175.2	3853.7	-1.481	412.209	1275.730
0.40	20909857.0	63339.3	4361.4	0.420	412.193	1275.741
5.00	20909933.3	65235.4	10232.3	33.415	412.423	1275.636
10.00	20910198.5	67310.2	16609.8	73.518	417.296	1275.378
15.00	20910677.3	69401.3	22986.0	118.878	417.724	1275.019
20.00	20911397.0	71487.3	29363.7	170.087	416.902	1276.782
25.00	20912388.3	73569.6	35759.1	227.469	416.035	1282.026
30.00	20913682.8	75647.8	42192.7	291.456	415.227	1292.471
35.00	20915314.5	77721.8	48696.1	362.422	414.365	1310.302
40.00	20917318.8	79791.3	55312.3	440.448	413.458	1338.068
45.00	20919730.5	81856.5	62097.6	525.389	412.606	1378.368
50.00	20922583.5	83917.5	69121.1	616.829	411.790	1433.713
55.00	20925908.8	85974.5	76463.6	714.070	411.013	1506.352
60.00	20929732.5	88027.6	84216.7	816.188	410.243	1598.231
65.00	20934075.3	90076.9	92480.2	921.187	409.488	1710.610
65.75	20934772.0	90384.0	93770.1	936.951	409.371	1729.232
70.00	20938943.5	92122.4	101356.9	1026.118	408.685	1843.659
75.00	20944340.0	94163.7	110955.1	1132.981	407.815	1999.644
80.00	20950278.5	96200.3	121394.7	1242.931	406.778	2180.396
81.00	20951532.5	96606.9	123594.6	1265.249	406.543	2219.598
85.00	20956773.3	98231.1	132801.8	1355.206	405.540	2386.834
90.00	20963832.0	100255.6	145307.7	1468.490	404.249	2620.036
95.00	20971459.5	102273.0	159047.0	1582.577	402.609	2880.126
100.00	20979659.5	104280.4	174153.5	1697.524	400.245	3167.037
105.00	20988433.8	106275.4	190764.3	1812.078	397.806	3482.076
110.00	20997778.5	108258.5	209022.5	1926.192	395.102	3825.630
115.00	21007701.3	110223.4	229065.0	2042.694	390.981	4196.506
120.00	21018197.8	112173.3	251044.0	2155.287	389.356	4600.743
125.00	21029253.0	114117.9	275127.6	2267.052	388.426	5038.206
130.00	21040871.5	116057.0	301483.3	2380.793	387.166	5510.048
135.00	21053065.3	117985.1	330290.7	2497.097	385.567	6019.487
140.00	21065719.3	119912.5	361491.4	2560.836	383.889	6457.814
145.00	21078673.0	121828.5	394911.3	2620.774	382.473	6915.081
150.00	21091930.3	123737.1	430693.5	2682.509	380.995	7403.177
155.00	21105502.8	125638.3	468599.7	2747.089	379.461	7925.298
160.00	21119407.5	127531.6	510009.0	2815.437	377.855	8484.875
160.82	21121717.8	127841.0	516996.8	2826.590	377.588	8579.184
161.59	21123893.3	128131.7	523624.0	2817.527	377.412	8617.922

MACH 1

MAX Q

S-IC CECØ

S-IC ØECØ
S-IC/S-II SEP

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
S-IC/S-II SEP	411783.57	659530.11	67270.1	2765.96	70.48	0.620
S-II ENGINE START	415439.86	655446.64	67850.2	2764.33	70.58	0.626
S-II 90 THRUST	3742629.69	658353.04	70567.9	2760.10	71.07	5.682
	4963369.63	653340.27	74773.5	2777.47	71.82	7.616
	5068814.00	647343.84	79034.4	2801.37	72.56	7.852
	5103496.31	641243.48	83160.4	2826.64	73.29	7.981
	5129079.81	635102.30	87153.7	2853.04	74.00	8.099
	5143899.56	628934.80	91017.3	2880.49	74.68	8.202
	5151635.50	618783.38	94753.4	2909.07	75.35	8.338
	5153035.50	616283.48	96226.6	2920.89	75.62	8.374
	5154314.00	608544.84	98365.0	2938.77	76.00	8.483
	5154523.75	602788.55	99367.7	2947.38	76.18	8.508
	5154554.63	602356.28	101854.7	2969.45	76.63	8.570
	5154931.25	596168.56	105230.0	3000.23	77.17	8.659
	5152863.06	585982.83	108510.1	3031.04	77.66	8.747
	5152872.50	582805.84	111697.8	3062.92	78.14	8.840
	5149509.88	577629.26	114792.4	3095.61	78.62	8.928
	5148156.63	571452.67	117795.0	3129.10	79.09	9.022
	5147447.06	565276.08	120706.4	3163.39	79.54	9.120
	5146752.00	559222.90	123527.6	3198.52	80.41	9.219
	5146066.81	552922.90	126259.6	3234.48	80.98	9.321
	5146080.50	546751.81	128903.2	3271.25	80.83	9.426
	5146177.31	540581.16	131459.8	3308.86	81.24	9.534
	5146263.19	534411.00	133930.6	3347.30	81.64	9.644
	5152267.19	528235.51	136316.8	3386.62	82.02	9.768
	5154301.00	522052.73	138619.7	3426.80	82.39	9.888
	5156109.94	515869.63	140840.4	3467.86	82.76	10.010
	5157729.94	505686.55	142980.3	3509.79	83.11	10.135
	5159174.81	503503.45	145040.8	3552.60	83.45	10.262
	5159767.00	497320.36	147023.1	3596.30	83.77	10.391
	5160158.94	491137.27	148928.7	3640.87	84.09	10.522
	5160414.56	484951.18	150759.1	3686.34	84.40	10.657
	5160582.19	478764.77	152515.4	3732.71	84.69	10.795
	5161036.00	472578.05	154199.5	3779.99	84.97	10.937
	5161453.81	466390.87	155812.7	3828.20	85.25	11.083
	5161890.25	460203.37	157356.5	3877.35	85.51	11.233
	5162286.69	454015.40	158832.6	3927.46	85.76	11.387
	5162411.31	447827.30	160242.4	3978.54	86.01	11.545
	5162498.94	441638.93	161587.6	4030.61	86.24	11.707
	5162174.00	435450.82	162869.6	4083.67	86.46	11.873
	5161833.88	429263.30	164090.3	4137.76	86.68	12.043
	5161953.25	423075.88	165251.4	4192.88	86.88	12.219
	5162162.00	416888.38	166354.5	4249.07	87.08	12.401
	5162291.50	416700.59	167401.2	4306.33	87.26	12.588
	5162404.75	404512.80	168394.1	4364.69	87.44	12.781
	5162101.88	398325.09	169334.4	4424.19	87.61	12.979
	5161811.31	392137.85	170224.1	4484.85	87.77	13.183
	5161476.50	385951.11	171065.2	4546.70	87.92	13.394
	5161230.19	375764.48	171859.4	4609.76	88.06	13.611

INITIATE IGM

TABLE 2-2 (BT0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
390.00	5160762.19	373578.42	172609.0	4674.07	88.19	13.835
395.00	5160347.44	367394.05	173315.7	4739.65	88.32	14.067
400.00	5159910.31	361209.77	173982.0	4806.55	88.43	14.307
405.00	5159485.06	355025.48	174609.8	4874.79	88.54	14.555
410.00	5159193.50	348841.20	175201.6	4944.41	88.64	14.812
415.00	5158906.75	342656.91	175759.8	5015.46	88.74	15.078
420.00	5158573.25	336472.63	176286.3	5087.99	88.82	15.354
425.00	5158305.63	330289.67	176784.3	5162.03	88.90	15.641
430.00	5157801.00	324107.16	177255.9	5237.64	88.97	15.938
435.00	5157327.31	317925.21	177703.9	5314.86	89.03	16.246
440.00	5156813.25	311744.05	178131.2	5393.76	89.09	16.567
445.00	5156285.69	305563.30	178540.7	5474.38	89.13	16.900
450.00	5155749.06	299383.21	178935.5	5556.79	89.17	17.247
455.00	5155245.63	293203.82	179318.5	5641.07	89.20	17.609
460.00	5154635.31	287025.01	179693.0	5727.26	89.23	17.986
465.00	4103357.72	281897.52	180054.5	5799.29	89.28	14.584
470.00	4102928.75	276972.60	180395.7	5870.23	89.31	14.841
475.00	4102392.53	272047.33	180737.0	5941.50	89.30	15.108
480.00	4101581.88	267117.72	181090.2	6014.27	89.29	15.384
485.00	4100735.88	262188.86	181454.6	6088.74	89.27	15.670
490.00	3086127.97	257712.39	181827.7	6158.17	89.28	11.997
495.00	3080208.78	254057.10	182190.0	6216.47	89.32	12.139
500.00	3078457.59	250404.50	182536.5	6275.51	89.35	12.309
505.00	3076726.34	246754.82	182873.3	6335.29	89.36	12.484
510.00	3074915.63	243107.71	183207.3	6395.90	89.37	12.664
515.00	3073084.72	239463.34	183539.7	6457.64	89.38	12.849
520.00	3071198.94	235821.91	183870.2	6520.44	89.39	13.039
525.00	3069165.03	232183.41	184200.2	6584.29	89.39	13.235
530.00	3067030.44	228547.95	184531.8	6649.18	89.39	13.436
535.00	3064651.41	224915.91	184868.0	6715.14	89.39	13.643
540.00	3061921.63	221287.49	185211.3	6782.17	89.38	13.854
545.00	3058420.22	217663.45	185564.9	6850.31	89.36	14.068
550.00	3052677.69	214045.13	185931.7	6919.54	89.35	14.279
551.38	3040184.78	213043.95	186036.1	6938.88	89.34	14.288
552.08	122878.88	212963.27	186088.9	6941.82	89.35	0.577
552.26	103329.44	212943.09	186102.0	6941.90	89.35	0.485

S-II 0EC0
S-IVB ULL IGN
S-II/S-IVB SEP

TABLE 2.2 (BT0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	**** RADIAL DISTANCE OF THE C G (M)	MAGNITUDE OF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE SPHERICAL EARTH (DEG)	LONGITUDE ON EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (KM)
S-IC/S-II SEP	6440658.81	2396.459	28.8658	-79.69	93.89
S-II ENGINE START	6441239.50	2394.523	28.8695	-79.68	95.26
S-II 90 THRUST	6443959.75	2388.857	28.8872	-79.61	101.83
	6448169.69	2404.084	28.9154	-79.51	112.36
	6452435.13	2425.901	28.9453	-79.40	123.52
	6456565.75	2449.216	28.9754	-79.29	134.84
	6460563.88	2473.786	29.0059	-79.18	146.33
	6464432.19	2499.523	29.0367	-79.06	157.98
	6468173.19	2526.500	29.0678	-78.95	169.80
	6469648.31	2537.702	29.0804	-78.90	174.62
	6471789.69	2554.698	29.0992	-78.83	181.79
INITIATE IGM	6472793.75	2562.911	29.1082	-78.79	185.23
	6475284.25	2583.986	29.1310	-78.71	193.96
	6478664.63	2613.574	29.1631	-78.59	206.31
	6481949.88	2643.341	29.1954	-78.46	218.82
	6485142.81	2674.206	29.2281	-78.34	231.51
	6488242.81	2705.938	29.2611	-78.21	244.37
	6491250.69	2738.529	29.2943	-78.08	257.40
	6494167.44	2771.977	29.3279	-77.95	270.62
	6496994.25	2806.307	29.3617	-77.82	284.01
	6499731.75	2841.509	29.3958	-77.69	297.60
	6502381.25	2877.580	29.4303	-77.55	311.37
	6504943.69	2914.525	29.4650	-77.41	325.34
	6507420.38	2952.344	29.5000	-77.27	339.50
	6509812.56	2991.076	29.5354	-77.13	353.86
	6512121.63	3030.716	29.5710	-76.98	368.42
	6514348.63	3071.268	29.6069	-76.84	383.18
	6516494.88	3112.728	29.6431	-76.69	398.16
	6518561.81	3155.097	29.6796	-76.54	413.34
	6520550.69	3198.378	29.7164	-76.38	428.75
	6522463.06	3242.571	29.7535	-76.23	444.37
	6524300.19	3287.682	29.7909	-76.07	460.22
	6526083.63	3333.719	29.8286	-75.91	476.29
	6527754.75	3380.694	29.8665	-75.75	492.60
	6529375.25	3428.620	29.9048	-75.58	509.14
	6530926.44	3477.509	29.9433	-75.41	525.92
	6532410.06	3527.375	29.9822	-75.24	542.94
	6533827.56	3578.231	30.0213	-75.07	560.22
	6535180.56	3630.091	30.0607	-74.89	577.74
	6536470.69	3682.968	30.1004	-74.71	595.52
	6537699.56	3736.876	30.1403	-74.53	613.57
	6538868.88	3791.837	30.1806	-74.35	631.87
	6539980.50	3847.874	30.2211	-74.16	650.45
	6541036.00	3905.004	30.2619	-73.97	669.31
	6542037.63	3963.234	30.3030	-73.78	688.45
	6542987.13	4022.623	30.3443	-73.58	707.87
	6543886.19	4083.183	30.3859	-73.38	727.59
	6544736.63	4144.934	30.4277	-73.18	747.60
	6545540.63	4207.907	30.4699	-72.97	767.91

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST NOTION (SEC)	*** RADIUS DISTANCE OF THE C G (M)	MAGNITUDE OF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (KM)
390.00	6546299.94	4272.133	30.5122	-72.76	788.54
395.00	6547016.88	4337.639	30.5548	-72.55	809.48
400.00	6547693.50	4404.462	30.5976	-72.33	830.74
405.00	6548332.00	4472.638	30.6407	-72.11	852.33
410.00	6548934.69	4542.204	30.6839	-71.89	874.25
415.00	6549503.81	4613.203	30.7274	-71.66	896.51
420.00	6550041.94	4685.678	30.7711	-71.43	919.12
425.00	6550551.44	4759.676	30.8150	-71.20	942.09
430.00	6551035.00	4835.242	30.8591	-70.96	965.42
435.00	6551495.38	4912.427	30.9034	-70.72	989.12
440.00	6551935.19	4991.285	30.9478	-70.47	1013.19
445.00	6552357.50	5071.873	30.9924	-70.22	1037.66
450.00	6552765.38	5154.254	31.0371	-69.97	1062.51
455.00	6553161.88	5238.495	31.0820	-69.71	1087.78
460.00	6553550.44	5324.663	31.1270	-69.44	1113.45
465.00	6553926.06	5396.656	31.1720	-69.17	1139.51
470.00	6554281.75	5467.561	31.2170	-68.90	1165.92
475.00	6554637.94	5538.808	31.2619	-68.63	1192.67
480.00	6555006.44	5611.556	31.3067	-68.35	1219.76
485.00	6555386.38	5686.009	31.3514	-68.06	1247.21
490.00	6555775.25	5755.409	31.3960	-67.78	1275.02
495.00	6556153.94	5813.676	31.4403	-67.48	1303.13
500.00	6556517.13	5872.684	31.4843	-67.19	1331.52
505.00	6556870.88	5932.437	31.5280	-66.89	1360.20
510.00	6557222.44	5993.019	31.5714	-66.59	1389.17
515.00	6557572.50	6054.730	31.6144	-66.29	1418.44
520.00	6557921.00	6117.505	31.6570	-65.98	1448.01
525.00	6558269.44	6181.326	31.6992	-65.67	1477.88
530.00	6558620.00	6246.193	31.7410	-65.35	1508.06
535.00	6558975.38	6312.120	31.7824	-65.04	1538.56
540.00	6559338.44	6379.128	31.8233	-64.71	1569.38
545.00	6559712.19	6447.234	31.8637	-64.39	1600.53
550.00	6560099.56	6516.438	31.9036	-64.06	1632.01
551.38	6560209.69	6535.774	31.9146	-63.97	1640.78
552.08	6560265.50	6538.710	31.9201	-63.92	1645.22
552.26	6560279.31	6538.782	31.9215	-63.91	1646.33

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM MØTØN (SEC)	X CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Y CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Z CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	X CØMPØNØT ØF VEL. IN THE XXE SYSTEM (M/S)	Y CØMPØNØT ØF VEL. IN THE XXE SYSTEM (M/S)	Z CØMPØNØT ØF VEL. IN THE XXE SYSTEM (M/S)
S-IC/S-II SEP	66747.938	35.191	94818.007	893.543	4.446	2223.641
S-II ENGINE START	67309.188	38.057	96218.953	887.819	4.523	2223.848
S-II 90 THRUST	69932.188	52.145	102893.397	861.761	4.903	2228.000
	170.00	73971.938	77.135	829.023	5.526	2256.615
	175.00	78037.438	106.333	797.245	6.166	2291.148
	180.00	81944.688	138.853	765.722	6.839	2326.431
	185.00	85695.188	174.770	734.547	7.533	2362.203
	190.00	89290.625	214.234	703.658	8.252	2398.419
	195.00	92732.438	257.353	673.104	8.997	2435.170
	197.02	94079.000	275.812	660.849	9.307	2450.127
	200.00	96022.188	304.216	642.892	9.770	2472.464
	201.42	96928.438	318.270	634.377	9.992	2483.139
	205.00	99161.563	355.060	612.926	10.562	2510.218
	210.00	102157.188	410.141	586.125	11.475	2546.977
	215.00	105027.250	469.765	561.790	12.390	2582.923
	220.00	107773.563	534.207	536.683	13.395	2619.765
	225.00	110394.125	603.798	511.530	14.460	2657.109
	230.00	112888.688	678.901	486.291	15.569	2694.962
	235.00	115257.000	759.333	461.028	16.613	2733.319
	240.00	117498.750	845.195	435.666	17.755	2772.227
	245.00	119613.563	937.019	410.234	18.979	2811.676
	250.00	121601.000	1035.013	384.747	20.237	2851.671
	255.00	123460.938	1139.500	359.202	21.546	2892.225
	260.00	125193.000	1250.597	333.598	22.900	2933.346
	265.00	126796.875	1368.564	307.949	24.295	2975.082
	270.00	128272.313	1493.604	282.233	25.731	3017.437
	275.00	129619.000	1625.946	256.421	27.208	3060.424
	280.00	130836.375	1765.754	230.526	28.725	3104.047
	285.00	131924.063	1913.245	204.550	30.281	3148.314
	290.00	132881.688	2068.609	178.483	31.874	3193.235
	295.00	133708.750	2232.039	152.318	33.506	3238.818
	300.00	134404.750	2403.741	126.050	35.176	3285.076
	305.00	134969.063	2583.867	99.671	36.882	3332.025
	310.00	135401.188	2772.611	73.170	38.628	3379.681
	315.00	135700.563	2970.227	46.537	40.416	3428.066
	320.00	135866.313	3176.851	19.760	42.245	3477.196
	325.00	135897.875	3392.742	-7.174	44.114	3527.092
	330.00	135794.375	3618.096	-34.276	46.027	3577.771
	335.00	135554.813	3853.076	-61.557	47.982	3629.252
	340.00	135178.438	4097.985	-89.029	49.983	3681.552
	345.00	134664.188	4352.954	-116.704	52.031	3734.691
	350.00	134011.063	4618.380	-144.591	54.123	3788.693
	355.00	133217.938	4894.292	-172.704	56.261	3843.584
	360.00	132283.625	5181.042	-201.043	58.433	3899.388
	365.00	131207.250	5478.313	-229.573	60.468	3956.117
	370.00	129987.313	5785.991	-258.431	62.634	4013.824
	375.00	128622.375	6104.902	-287.598	64.947	4072.524
	380.00	127110.875	6435.523	-317.061	67.305	4132.241
	385.00	125451.188	6778.088	-346.852	69.736	4193.008
			786663.414			

INITIATE IGM

TABLE 2-2 (BYØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MØTION (SEC)	X CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Y CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Z CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	X CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)	Y CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)	Z CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)
390.00	123641.688	7132.996	8C7782.602	-376.983	72.245	4254.855
395.00	121680.750	7500.627	829213.766	-407.464	74.818	4317.811
400.00	119566.500	7881.305	850962.570	-438.319	77.459	4381.913
405.00	117296.938	8275.345	873034.844	-469.568	80.166	4447.198
410.00	114870.125	8683.072	895436.555	-501.227	82.940	4513.702
415.00	112283.938	9104.840	918173.914	-533.316	85.781	4581.469
420.00	109536.250	9541.023	941253.344	-565.857	88.691	4650.540
425.00	106624.625	9991.873	964681.531	-598.871	91.672	4720.960
430.00	103546.688	10457.849	988465.242	-632.381	94.726	4792.774
435.00	100300.000	10939.259	1012611.625	-666.411	97.852	4866.031
440.00	96881.625	11436.458	1037128.000	-700.984	101.051	4940.783
445.00	93289.125	11549.889	1062021.938	-736.125	104.329	5017.084
450.00	89519.375	12479.906	1087301.422	-771.862	107.687	5094.995
455.00	85569.375	13026.889	1112974.594	-808.221	111.128	5174.578
460.00	81436.125	13591.345	1139050.016	-845.233	114.656	5255.898
465.00	77114.063	14172.987	1165498.922	-883.762	118.015	5322.494
470.00	72600.438	14771.568	1192275.922	-921.051	121.367	5388.057
475.00	67910.375	15286.179	1219381.734	-954.806	124.488	5454.470
480.00	63050.563	16016.548	1246822.719	-989.391	127.678	5522.170
485.00	58014.125	16663.160	1274605.641	-1025.448	130.977	5591.243
490.00	52793.500	17326.404	1302730.547	-1063.188	134.271	5654.762
495.00	47377.688	18005.732	1331133.031	-1103.402	137.463	5706.350
500.00	41760.688	18700.900	1359795.063	-1143.368	140.618	5758.590
505.00	35944.188	19411.977	1388720.000	-1183.019	143.811	5811.505
510.00	29930.500	20139.081	1417911.219	-1222.661	147.047	5865.131
515.00	23715.438	20682.624	1447372.484	-1263.528	150.385	5919.513
520.00	17293.750	21643.090	1477107.500	-1305.273	153.807	5974.652
525.00	10661.563	22420.829	1507120.203	-1347.721	157.303	6030.564
530.00	3815.500	23216.197	1537414.391	-1390.773	160.868	6087.266
535.00	-3247.125	24029.629	1567994.125	-1434.403	164.500	6144.778
540.00	-10529.375	24861.365	1598863.484	-1478.619	168.198	6203.118
545.00	-18034.375	25711.704	1630026.641	-1523.440	171.966	6262.299
550.00	-25764.875	26581.107	1661487.813	-1568.872	175.802	6322.317
551.38	-27944.938	26825.137	1670249.469	-1581.554	176.875	6339.065
552.08	-29054.125	26949.140	1674687.953	-1587.483	177.271	6340.600
552.26	-29332.125	26580.152	1675797.531	-1588.916	177.355	6340.312

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	COMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN PITCH (DEG)	COMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN YAW (DEG)	COMMANDED ROLL ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN ROLL (DEG)
S-IC/S-II SEP						
S-II ENGINE START						
S-II 90 THRUST						
161.59	-68.873	-68.9424	0.000	-0.058897	0.000	0.00024871
162.22	-68.873	-69.0571	0.000	-0.059554	0.000	0.00060583
165.22	-68.873	-69.5579	0.000	-0.078274	0.000	0.01103689
170.00	-68.873	-68.7375	0.000	-0.154215	0.000	-0.01109149
175.00	-68.873	-68.7542	0.000	-0.158255	0.000	0.00105932
180.00	-68.873	-68.7856	0.000	-0.156730	0.000	-0.00008016
185.00	-68.873	-68.7809	0.000	-0.157589	0.000	0.00029827
190.00	-68.873	-68.7888	0.000	-0.157408	0.000	0.00057237
195.00	-68.873	-68.7876	0.000	-0.152115	0.000	0.00051951
197.02	-68.873	-68.7880	0.000	-0.151532	0.000	0.00059130
200.00	-68.873	-68.7862	0.000	-0.156902	0.000	0.00075286
201.42	-68.873	-68.7856	0.000	-0.158401	0.000	0.00075727
205.00	-65.373	-67.4543	0.134	-0.074252	0.000	0.04719649
210.00	-62.021	-62.0950	0.158	-0.005687	0.000	-0.02809792
215.00	-62.808	-62.2716	0.221	0.035871	0.000	-0.00097442
220.00	-63.468	-63.2496	0.269	0.096556	0.000	0.00292005
225.00	-64.025	-63.7387	0.303	0.132082	0.000	-0.00095535
230.00	-64.552	-64.2250	0.327	0.069353	0.000	-0.00144901
235.00	-65.081	-64.7469	0.378	0.044087	0.000	0.00005880
240.00	-65.592	-65.2880	0.430	0.148450	0.000	-0.00056938
245.00	-66.084	-65.7765	0.454	0.169325	0.000	-0.00071783
250.00	-66.567	-66.2620	0.475	0.182909	0.000	-0.00067253
255.00	-67.045	-66.7437	0.493	0.205649	0.000	-0.00081106
260.00	-67.519	-67.2193	0.504	0.216527	0.000	-0.00083077
265.00	-68.009	-67.6956	0.514	0.224626	0.000	-0.00110829
270.00	-68.515	-68.2046	0.522	0.233696	0.000	-0.00095776
275.00	-69.007	-68.7086	0.528	0.239867	0.000	-0.00077269
280.00	-69.490	-69.1910	0.532	0.243292	0.000	-0.00093353
285.00	-69.972	-69.6736	0.535	0.245670	0.000	-0.00093298
290.00	-70.454	-70.1572	0.537	0.247145	0.000	-0.00091880
295.00	-70.936	-70.6395	0.538	0.247760	0.000	-0.00090900
300.00	-71.418	-71.1218	0.539	0.246579	0.000	-0.00093297
305.00	-71.906	-71.6083	0.540	0.245255	0.000	-0.00091380
310.00	-72.398	-72.0997	0.541	0.246951	0.000	-0.00093648
315.00	-72.896	-72.5971	0.541	0.247295	0.000	-0.00094467
320.00	-73.400	-73.1006	0.541	0.245954	0.000	-0.00092568
325.00	-73.908	-73.6082	0.542	0.244640	0.000	-0.00090163
330.00	-74.419	-74.1197	0.543	0.244779	0.000	-0.00091091
335.00	-74.933	-74.6337	0.543	0.243567	0.000	-0.00088128
340.00	-75.449	-75.1504	0.543	0.244880	0.000	-0.00089901
345.00	-75.968	-75.6697	0.543	0.243509	0.000	-0.00088073
350.00	-76.491	-76.1917	0.543	0.240623	0.000	-0.00087629
355.00	-77.021	-76.7196	0.543	0.240648	0.000	-0.00091766
360.00	-77.555	-77.1926	0.543	0.135362	0.000	-0.00204849
365.00	-78.115	-77.7347	0.584	0.065823	0.000	-0.00086568
370.00	-78.673	-78.3250	0.637	0.187759	0.000	-0.00128913
375.00	-79.219	-78.8660	0.655	0.209465	0.000	-0.00160237
380.00	-79.770	-79.4144	0.671	0.207817	0.000	-0.00151558
385.00	-80.324	-79.9723	0.685	0.229732	0.000	-0.00158334

INITIATE IGM

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	COMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN PITCH (DEG)	COMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN YAW (DEG)	COMMANDED ROLL ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN ROLL (DEG)
390.00	-80.878	-80.5254	0.694	0.238616	0.000	-0.00163327
395.00	-81.435	-81.0811	0.700	0.242469	0.000	-0.00169604
400.00	-81.995	-81.6436	0.705	0.246878	0.000	-0.00165059
405.00	-82.556	-82.2063	0.707	0.248920	0.000	-0.00165515
410.00	-83.119	-82.7709	0.709	0.249794	0.000	-0.00164998
415.00	-83.685	-83.3382	0.711	0.250061	0.000	-0.00163354
420.00	-84.253	-83.9076	0.711	0.249644	0.000	-0.00161610
425.00	-84.823	-84.4790	0.711	0.249814	0.000	-0.00159798
430.00	-85.394	-85.0521	0.710	0.249049	0.000	-0.00157926
435.00	-85.966	-85.6262	0.709	0.245636	0.000	-0.00153654
440.00	-86.539	-86.2010	0.709	0.243766	0.000	-0.00151565
445.00	-87.113	-86.7770	0.708	0.242191	0.000	-0.00148668
450.00	-87.687	-87.3539	0.707	0.240692	0.000	-0.00145907
455.00	-88.262	-87.9312	0.706	0.239304	0.000	-0.00142970
460.00	-88.837	-88.5087	0.704	0.236660	0.000	-0.00139499
465.00	-88.368	-88.8010	0.662	0.324879	0.000	0.01430113
470.00	-85.750	-86.1829	0.475	0.141715	0.000	-0.00323004
475.00	-86.087	-85.4922	0.446	0.054260	0.000	-0.00943866
480.00	-87.315	-86.7818	0.466	0.077919	0.000	-0.00174625
485.00	-88.472	-87.9904	0.485	0.097036	0.000	-0.00159390
490.00	-90.647	-89.1614	0.599	0.142540	0.000	-0.01454529
495.00	-90.178	-90.3870	0.502	0.121080	0.000	0.01621805
500.00	-90.004	-90.2475	0.484	0.092672	0.000	-0.00377900
505.00	-90.069	-89.6656	0.477	0.064947	0.000	-0.01817474
510.00	-91.143	-90.7252	0.508	0.095824	0.000	-0.00074113
515.00	-92.076	-91.7668	0.531	0.123516	0.000	-0.00051921
520.00	-92.805	-92.5443	0.544	0.138872	0.000	-0.00077517
525.00	-93.417	-93.1863	0.552	0.147831	0.000	-0.00090477
530.00	-93.980	-93.7591	0.556	0.153367	0.000	-0.00107135
535.00	-94.534	-94.3139	0.560	0.156491	0.000	-0.00113062
540.00	-95.086	-94.8675	0.563	0.159971	0.000	-0.00113041
545.00	-95.633	-95.4178	0.565	0.163860	0.000	-0.00108673
550.00	-96.159	-95.9537	0.565	0.164604	0.000	-0.00099512
551.38	-96.134	-96.0971	0.565	0.164120	0.000	-0.00095668
552.08	-96.134	-96.1674	0.565	0.163928	0.000	-0.00074446
552.26	-96.134	-96.1845	0.565	0.163917	0.000	-0.00063474

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE OF ATTACK (DEG)	DRAW ALONG VELOCITY VECTOR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE OF THE AERODYNAMIC FORCE VECTOR (KG)
S-IC/S-II SEP	7.98	363.29	0.2303	308.5919	8.5304	42.3639
S-II ENGINE START	8.00	335.15	0.2149	283.8172	7.3456	36.4800
S-II 90 THRUST	8.13	228.35	0.0984	188.8433	2.2906	11.3756
	8.42	124.75	1.7438	138.0910	22.1821	110.1617
	8.75	65.35	2.7304	83.3107	18.1947	80.6906
	9.09	33.59	3.6843	48.2703	12.6185	51.2071
	9.18	16.40	4.6514	25.3157	7.7786	30.9042
	9.18	8.06	5.5837	13.0568	4.5913	18.2412
	9.01	4.04	6.5019	6.8465	2.6770	10.7590
	8.95	3.11	6.8653	5.3701	2.1774	8.8143
	8.87	2.15	7.3968	3.8153	1.6242	6.6357
	8.83	1.82	7.6467	3.2637	1.4193	5.8209
	8.66	1.19	9.6002	2.3692	1.1683	5.0577
	8.44	0.70	15.7393	1.8902	1.1182	5.7510
	8.26	0.43	16.2693	1.1789	0.7116	3.6186
	7.96	0.27	16.0030	0.7350	0.4367	2.2551
	7.64	0.17	16.2123	0.4688	0.2887	1.4598
	7.38	0.12	16.4108	0.3158	0.2015	0.9986
	7.07	0.08	16.5609	0.2151	0.1425	0.6932
	6.68	0.06	16.6812	0.1501	0.1021	0.4999
	6.37	0.05	16.8406	0.1115	0.0774	0.3812
	6.13	0.04	16.9898	0.0865	0.0612	0.3020
	5.94	0.03	17.1302	0.0692	0.0499	0.2464
	5.78	0.02	17.2637	0.0565	0.0418	0.2056
	5.65	0.02	17.3835	0.0473	0.0358	0.1753
	5.55	0.02	17.4580	0.0404	0.0312	0.1522
	5.46	0.02	17.5257	0.0352	0.0276	0.1342
	5.39	0.01	17.6028	0.0312	0.0247	0.1200
	5.33	0.01	17.6676	0.0280	0.0224	0.1086
	5.28	0.01	17.7197	0.0254	0.0205	0.0993
	5.24	0.01	17.7614	0.0233	0.0190	0.0917
	5.24	0.01	17.7917	0.0218	0.0178	0.0859
	5.24	0.01	17.8067	0.0206	0.0168	0.0811
	5.25	0.01	17.8058	0.0196	0.0160	0.0770
	5.26	0.01	17.7884	0.0188	0.0153	0.0735
	5.27	0.01	17.7544	0.0180	0.0146	0.0706
	5.30	0.01	17.7064	0.0174	0.0141	0.0680
	5.33	0.01	17.6447	0.0170	0.0136	0.0661
	5.33	0.01	17.5710	0.0167	0.0133	0.0645
	5.42	0.01	17.4853	0.0165	0.0129	0.0632
	5.47	0.01	17.3880	0.0163	0.0127	0.0621
	5.53	0.01	17.2792	0.0162	0.0124	0.0612
	5.58	0.01	17.1560	0.0161	0.0122	0.0604
	5.64	0.01	17.0784	0.0161	0.0120	0.0598
	5.70	0.01	16.9235	0.0161	0.0118	0.0594
	5.76	0.01	16.7138	0.0162	0.0116	0.0591
	5.83	0.01	16.5460	0.0163	0.0115	0.0590
	5.91	0.01	16.3632	0.0165	0.0114	0.0590
	5.98	0.01	16.1641	0.0167	0.0113	0.0591

INITIATE IGM

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE OF ATTACK (DEG)	DRAG ALONG VELOCITY VECTOR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE OF THE AERODYNAMIC FORCE VECTOR (KG)
390.00	6.06	0.01	15.9629	0.0168	0.0113	0.0591
395.00	6.14	0.01	15.7525	0.0168	0.0112	0.0583
400.00	6.22	0.01	15.5288	0.0169	0.0111	0.0575
405.00	6.31	0.01	15.2987	0.0169	0.0111	0.0568
410.00	6.40	0.01	15.0608	0.0170	0.0110	0.0560
415.00	6.49	0.01	14.8142	0.0171	0.0110	0.0553
420.00	6.58	0.01	14.5599	0.0172	0.0110	0.0545
425.00	6.67	0.01	14.2982	0.0173	0.0109	0.0537
430.00	6.77	0.01	14.0294	0.0174	0.0109	0.0529
435.00	6.87	0.01	13.7546	0.0173	0.0109	0.0521
440.00	6.97	0.01	13.4741	0.0174	0.0108	0.0513
445.00	7.07	0.01	13.1876	0.0178	0.0108	0.0507
450.00	7.18	0.01	12.8956	0.0180	0.0108	0.0503
455.00	7.29	0.01	12.5986	0.0183	0.0107	0.0498
460.00	7.40	0.01	12.2972	0.0185	0.0107	0.0493
465.00	7.49	0.01	12.3145	0.0190	0.0109	0.0502
470.00	7.58	0.01	15.2228	0.0225	0.0136	0.0684
475.00	7.67	0.01	16.1602	0.0240	0.0147	0.0746
480.00	7.77	0.01	15.1166	0.0234	0.0139	0.0702
485.00	7.86	0.01	14.1604	0.0229	0.0132	0.0656
490.00	7.95	0.01	13.2621	0.0222	0.0126	0.0607
495.00	8.02	0.01	12.3499	0.0213	0.0118	0.0554
500.00	8.09	0.01	12.7933	0.0221	0.0123	0.0587
505.00	8.17	0.01	13.6692	0.0233	0.0133	0.0651
510.00	8.24	0.01	12.8961	0.0226	0.0127	0.0606
515.00	8.32	0.01	12.1453	0.0219	0.0120	0.0561
520.00	8.40	0.01	11.6598	0.0216	0.0117	0.0536
525.00	8.48	0.01	11.3092	0.0214	0.0114	0.0520
530.00	8.56	0.01	11.0260	0.0214	0.0113	0.0508
535.00	8.64	0.01	10.7586	0.0213	0.0111	0.0497
540.00	8.72	0.01	10.4902	0.0212	0.0109	0.0486
545.00	8.81	0.01	10.2229	0.0211	0.0108	0.0474
550.00	8.89	0.01	9.9681	0.0211	0.0106	0.0462
551.38	8.92	0.01	9.9021	0.0210	0.0105	0.0460
552.08	8.92	0.01	9.8815	0.0210	0.0105	0.0458
552.26	8.92	0.01	9.8777	0.0210	0.0105	0.0458

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

TABLE 2-2 (BT0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Y POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Z POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)
S-IC/S-II SEP	6438562.7	39054.5	159600.6	858.782	115.035	2626.743
S-II ENGINE START	6439101.9	39127.0	161255.5	852.869	115.006	2626.956
S-II 90 THRUST	6441618.8	39471.8	169139.3	825.870	114.858	2631.143
	6445482.7	40020.2	181783.1	791.375	114.513	2659.883
	6449355.2	40591.7	195168.8	757.676	114.094	2694.544
	6453059.8	41161.1	208729.7	724.193	113.676	2729.928
	6456597.6	41728.4	222468.8	691.019	113.248	2765.775
	6459970.3	42293.6	236388.2	658.093	112.815	2802.041
	6463179.1	42856.6	250490.1	625.462	112.378	2838.816
	6464428.6	43083.3	256236.7	612.356	112.202	2853.777
	6466225.4	43417.3	264777.1	593.132	111.936	2876.111
	6467060.6	43576.1	268865.9	584.008	111.808	2886.782
INITIATE IGM	6469110.6	43975.9	279251.9	561.007	111.483	2913.841
	6471841.2	44532.4	293913.7	532.027	111.144	2950.591
	6474435.3	45087.3	308756.2	505.490	110.798	2986.539
	6476894.4	45640.5	323780.7	478.135	110.503	3023.349
	6479216.6	46192.4	338990.5	450.693	110.238	3060.636
	6481401.1	46742.9	354387.9	423.123	109.984	3098.407
	6483447.7	47292.0	369975.3	395.488	109.633	3136.658
	6485355.8	47839.4	385755.3	361.708	109.346	3175.433
	6487124.6	48385.5	401730.5	339.815	109.108	3214.725
	6488753.8	48930.5	417903.4	311.822	108.871	3254.537
	6490242.6	49474.3	434276.7	283.724	108.649	3294.884
	6491590.8	50017.0	450853.1	255.521	108.438	3335.773
	6492797.7	50558.7	467635.4	227.225	108.231	3377.250
	6493862.9	51099.3	484626.6	198.813	108.030	3419.321
	6494785.6	51639.0	501829.6	170.254	107.833	3461.999
	6495565.1	52175.3	519247.6	141.562	107.637	3505.286
	6496200.9	52715.3	536883.5	112.737	107.442	3549.190
	6496692.3	53252.1	554740.5	83.769	107.246	3593.721
	6497038.4	53787.8	572821.7	54.649	107.049	3638.888
	6497238.5	54322.5	591130.4	25.371	106.849	3684.703
	6497291.8	54856.3	609669.9	-4.074	106.645	3731.180
	6497197.4	55389.0	628443.4	-33.698	106.440	3778.338
	6496954.5	55920.7	647454.4	-63.512	106.234	3826.195
	6496561.9	56451.3	666706.5	-93.529	106.025	3874.770
	6496018.8	56980.9	686203.3	-123.764	105.812	3924.081
	6495323.9	57509.4	705948.5	-154.230	105.598	3974.145
	649476.1	58036.9	725946.0	-184.938	105.379	4024.981
	6493474.1	58563.2	746193.6	-215.901	105.160	4076.606
	6492316.7	59088.5	766713.3	-247.134	104.940	4129.038
	649102.3	59612.6	787491.3	-278.646	104.714	4182.302
	6489529.8	60135.6	808537.7	-310.454	104.485	4236.423
	6487897.3	60657.5	829857.0	-342.560	104.237	4291.423
	6486103.8	61177.6	851453.4	-374.926	103.802	4347.314
	6484147.3	61695.6	873331.6	-407.696	103.442	4404.148
	6482026.1	62212.2	895496.4	-440.852	103.175	4461.942
	6479738.2	62727.4	917952.7	-474.383	102.896	4520.716
	6477281.6	63241.2	940705.3	-508.320	102.632	4580.503

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MØTØN (SEC)	X PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Y PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Z PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)
390.00	6474654.3	63753.7	963759.4	-542.681	102.387	4641.333
395.00	6471854.1	64265.0	987120.3	-577.476	102.146	4703.234
400.00	6468878.8	64775.2	1010793.5	-612.732	101.909	4766.242
405.00	6465725.9	65284.1	1034784.6	-648.470	101.676	4830.391
410.00	6462393.3	65791.9	1059099.4	-684.709	101.443	4895.720
415.00	6458878.0	66298.6	1083743.8	-721.472	101.210	4962.268
420.00	6455177.6	66804.0	1108724.1	-758.784	100.976	5030.077
425.00	6451289.3	67308.3	1134046.7	-796.668	100.742	5099.190
430.00	6447209.9	67811.5	1159718.2	-835.150	100.508	5169.651
435.00	6442936.7	68313.4	1185745.5	-874.256	100.269	5241.507
440.00	6438466.3	68814.1	1212135.6	-914.014	100.027	5314.810
445.00	6433795.4	69313.7	1238896.0	-954.453	99.782	5389.612
450.00	6428920.5	69812.0	1266034.3	-995.601	99.535	5465.971
455.00	6423838.1	70309.0	1293558.3	-1037.492	99.286	5543.949
460.00	6418544.3	70804.8	1321476.5	-1080.158	99.035	5623.610
465.00	6413034.4	71299.3	1349760.6	-1123.978	98.799	5688.469
470.00	6407306.3	71792.8	1378362.1	-1166.588	98.539	5752.296
475.00	6401375.2	72284.2	1407285.0	-1205.748	98.019	5817.038
480.00	6395246.9	72773.0	1436534.5	-1245.837	97.494	5883.008
485.00	6388914.3	73259.2	1466117.1	-1287.500	96.993	5950.272
490.00	6382369.4	73743.0	1496032.5	-1330.737	96.523	6011.897
495.00	6375602.9	74224.5	1526215.7	-1376.129	96.098	6061.482
500.00	6368609.2	74703.8	1556648.3	-1421.339	95.587	6111.685
505.00	6361389.6	75180.4	1587333.5	-1466.301	95.066	6162.535
510.00	6353946.1	75654.4	1618274.7	-1511.326	94.535	6214.057
515.00	6346274.1	76125.8	1649475.1	-1557.648	94.040	6266.258
520.00	6338368.0	76594.8	1680938.3	-1604.922	93.564	6319.148
525.00	6330223.6	77061.5	1712667.7	-1652.973	93.099	6372.746
530.00	6321837.1	77525.8	1744666.9	-1701.705	92.639	6427.073
535.00	6313205.4	77987.9	1776939.6	-1751.093	92.181	6482.149
540.00	6304325.1	78447.7	1809489.6	-1801.146	91.724	6537.991
545.00	6295192.8	78905.1	1842320.8	-1851.885	91.269	6594.610
550.00	6285805.1	79360.3	1875436.9	-1903.318	90.815	6652.002
551.38	6283161.0	79485.9	1884654.3	-1917.675	90.689	6668.011
552.08	6281816.3	79549.4	1889323.0	-1924.211	90.621	6669.191
552.26	6281479.4	79565.3	1890490.0	-1925.773	90.604	6668.816

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED PITCH CHANL (DEG)	GIMBAL DEFLECTION ANG REQUIRED YAW CHANNEL (DEG)	GIMBAL DEFLECTION ANG REQUIRED ROLL CHANNEL (DEG)	EULERIAN ATTITUDE RATE IN PITCH (DEG/S)	EULERIAN ATTITUDE RATE IN YAW (DEG/S)	EULERIAN ATTITUDE RATE IN ROLL (DEG/S)
S-IC/S-II SEP	161.59	0.0000	0.00000	-0.18134203	-0.00034408	0.00063060
S-II ENGINE START	162.22	-0.3382	-0.046743	-0.18277502	-0.00175370	0.00087263
S-II 90 THRUST	165.22	-0.6843	-0.075824	-0.06371170	-0.01446111	0.01476796
	170.00	0.3107	-0.136319	-0.002692	-0.00379261	-0.00468804
	175.00	0.0376	-0.123472	0.000466	0.00254326	0.00221993
	180.00	0.0822	-0.126453	-0.000113	-0.00085263	-0.00056891
	185.00	0.0697	-0.125745	0.000060	0.00035073	0.00022817
	190.00	0.0676	-0.126311	-0.000081	-0.00029454	-0.00000438
	195.00	0.0689	-0.120174	0.000065	0.00107297	-0.00003988
	197.02	0.0679	-0.121456	-0.000094	-0.00026875	0.00006932
	200.00	0.0708	-0.127827	0.000115	-0.00168152	-0.0001015
	201.42	0.0707	-0.127691	0.000113	-0.00047171	-0.00000518
	205.00	-0.4920	-0.112439	0.006999	0.04474987	-0.01357170
	210.00	0.9081	-0.134880	-0.007786	-0.00427220	-0.02370935
	215.00	-0.0168	-0.127266	0.001353	0.01578708	0.00996401
	220.00	0.0465	-0.126561	0.000281	0.00836343	-0.00286778
	225.00	0.0743	-0.128491	0.000042	0.00641096	0.00022840
	230.00	0.0813	-0.140574	-0.000027	-0.03070620	0.00048981
	235.00	0.0670	-0.141755	0.000216	0.01991293	-0.00011690
	240.00	0.0612	-0.120377	0.000120	0.01263926	-0.00037408
	245.00	0.0665	-0.131520	0.000132	0.00046730	-0.00026026
	250.00	0.0652	-0.132157	0.000146	0.00488875	-0.00033045
	255.00	0.0648	-0.130576	0.000132	0.00330638	-0.00035949
	260.00	0.0647	-0.132234	0.000137	0.00156398	-0.00035600
	265.00	0.0685	-0.132941	0.000088	0.00177949	-0.00052621
	270.00	0.0630	-0.132836	0.000147	0.00163718	-0.00032284
	275.00	0.0616	-0.133246	0.000158	0.00086641	-0.00043507
	280.00	0.0633	-0.133722	0.000134	0.00055874	-0.00042549
	285.00	0.0626	-0.134038	0.000138	0.00038825	-0.00040802
	290.00	0.0620	-0.134371	0.000140	0.00020022	-0.00041236
	295.00	0.0617	-0.134714	0.000140	0.0005064	-0.00042040
	300.00	0.0616	-0.135985	0.000137	-0.00056588	-0.00041478
	305.00	0.0611	-0.136613	0.000140	0.00023187	-0.00042256
	310.00	0.0607	-0.136303	0.000137	0.00022735	-0.00043297
	315.00	0.0602	-0.136538	0.000138	-0.00001800	-0.00042918
	320.00	0.0596	-0.137601	0.000141	-0.00057418	-0.00042670
	325.00	0.0591	-0.137988	0.000144	0.00021628	-0.00043869
	330.00	0.0584	-0.138548	0.000143	-0.00041195	-0.00042983
	335.00	0.0579	-0.138988	0.000146	0.00016884	-0.00044127
	340.00	0.0574	-0.138487	0.000144	0.00015360	-0.00044116
	345.00	0.0568	-0.139427	0.000146	-0.00066362	-0.00043653
	350.00	0.0565	-0.140508	0.000145	-0.00038257	-0.00044735
	355.00	0.0561	-0.140225	0.000139	0.00035740	-0.00044705
	360.00	0.0654	-0.148349	0.000008	-0.04383323	0.00009568
	365.00	0.0566	-0.154903	0.000216	0.01867747	-0.00010386
	370.00	0.0499	-0.132213	0.000148	0.01690924	-0.00048972
	375.00	0.0542	-0.141175	0.000134	-0.00291093	-0.00038928
	380.00	0.0530	-0.143855	0.000159	0.00346514	-0.00040101
	385.00	0.0516	-0.141415	0.000152	0.00349001	-0.00046515

INITIATE IGM

TABLE 2-2 (BT0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED PITCH CHANL (DEG)	GIMBAL DEFLECTION ANG REQUIRED YAW CHANNEL (DEG)	GIMBAL DEFLECTION ANG REQUIRED ROLL CHANNEL (DEG)	EULERIAN ATTITUDE RATE IN PITCH (DEG/S)	EULERIAN ATTITUDE RATE IN YAW (DEG/S)	EULERIAN ATTITUDE RATE IN ROLL (DEG/S)
390.00	0.0523	-0.142721	0.000153	-0.11034727	0.00070383	-0.00047437
395.00	0.0520	-0.143345	0.000150	-0.11213500	0.00096084	-0.00046695
400.00	0.0508	-0.143529	0.000157	-0.11256897	0.00065099	-0.00048092
405.00	0.0503	-0.144021	0.000155	-0.11264289	0.00023426	-0.00049128
410.00	0.0497	-0.144414	0.000156	-0.11319355	0.00011987	-0.00049288
415.00	0.0489	-0.144775	0.000157	-0.11370611	-0.00002556	-0.00049295
420.00	0.0483	-0.145147	0.000158	-0.11406409	-0.00013191	-0.00049469
425.00	0.0476	-0.144980	0.000159	-0.11449290	0.00021912	-0.00049590
430.00	0.0468	-0.145361	0.000159	-0.11473768	-0.00066026	-0.00048811
435.00	0.0461	-0.146081	0.000162	-0.11489086	-0.00052648	-0.00048589
440.00	0.0454	-0.146358	0.000162	-0.11509703	-0.00031416	-0.00048399
445.00	0.0446	-0.146620	0.000163	-0.11529699	-0.00032195	-0.00048089
450.00	0.0438	-0.146759	0.000164	-0.11543504	-0.00028453	-0.00047811
455.00	0.0430	-0.146766	0.000164	-0.11550358	-0.00028719	-0.00047459
460.00	0.0422	-0.147214	0.000166	-0.11550246	-0.00058668	-0.00045899
465.00	-0.0660	-0.105790	0.002551	0.15673845	-0.00228017	0.00369483
470.00	0.2014	-0.129730	-0.003707	0.66314520	-0.04866845	-0.02066302
475.00	0.0783	-0.121969	-0.000690	-0.22143361	0.00315706	0.00436341
480.00	0.0337	-0.119738	0.00074	-0.25279788	0.00424576	-0.00050957
485.00	0.0308	-0.120223	0.000123	-0.22897345	0.00335001	-0.00019823
490.00	0.2408	-0.131884	-0.003240	-0.40336674	0.01468645	-0.02100065
495.00	0.0010	-0.131978	-0.000728	0.16231329	-0.02469529	-0.02701504
500.00	-0.0855	-0.122179	0.000216	-0.06989873	0.00448353	0.01133563
505.00	0.1145	-0.130369	-0.003177	-0.03776510	-0.00070750	-0.00045426
510.00	-0.0022	-0.125749	0.000417	-0.25479939	0.00765640	0.00112275
515.00	0.0053	-0.126149	0.000137	-0.17293176	0.00390538	-0.00064887
520.00	0.0073	-0.126257	0.000095	-0.14019175	0.00233604	-0.00031351
525.00	0.0091	-0.126273	0.000041	-0.11905068	0.00134936	-0.00034499
530.00	0.0104	-0.126207	0.000008	-0.11176832	0.00092726	-0.00031922
535.00	0.0106	-0.126568	-0.000000	-0.11066964	0.00040979	-0.00030693
540.00	0.0103	-0.126332	0.000001	-0.11069164	0.00088928	-0.00030432
545.00	0.0099	-0.125984	0.000006	-0.10909419	0.00059460	-0.00030720
550.00	0.0094	-0.126102	0.000017	-0.10460972	-0.00026111	-0.00028060
551.38	0.0091	-0.126158	0.000021	-0.10247153	-0.00045730	-0.00025450
552.08	-0.0538	-0.126327	-0.000143	-0.09836495	-0.00009571	0.00029364
552.26	-0.0597	-0.126299	-0.000129	-0.09754047	-0.00003261	0.00039809

S-II 0EC0
S-IVB ULL IGN
S-II/S-IVB SEP

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABOVE SPHERICAL EARTH (FT)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (FT/S**2)
S-IC/S-II SEP	161.59	92572.63	1454014.97	220702.5	70.48	2.033
S-II ENGINE START	162.22	93394.59	222605.5	9074.66	70.58	2.053
S-II 90 THRUST	165.22	841376.61	231522.0	9055.45	71.07	18.641
	170.00	1115809.86	1440368.70	9112.45	71.82	24.988
	175.00	1139514.70	1427148.84	259299.1	72.56	25.761
	180.00	1147311.59	1413699.84	9273.75	73.29	26.185
	185.00	1153062.98	1400160.86	9360.37	74.00	26.572
	190.00	1156394.61	1386563.86	9450.44	74.68	26.910
	195.00	1158133.72	1364183.81	9544.21	75.35	27.355
	197.02	1158448.44	1358672.47	9582.98	75.62	27.474
	200.00	1158735.86	1341611.70	9641.62	76.00	27.830
	201.42	1158783.02	1337739.73	9669.90	76.18	27.912
	205.00	1158789.95	1327968.27	9742.28	76.63	28.117
	210.00	1158874.63	1314326.67	9843.28	77.17	28.409
	215.00	1158409.67	1300689.47	9944.36	77.66	28.698
	220.00	1158411.80	1287071.55	10048.94	78.14	29.001
	225.00	1157655.86	1273454.50	10156.19	78.62	29.292
	230.00	1157351.63	1255837.47	10266.07	79.09	29.601
	235.00	1157192.11	1246220.41	10378.58	79.54	29.921
	240.00	1157035.86	1232603.36	10493.84	79.98	30.247
	245.00	1156881.83	1218986.31	10611.80	80.41	30.581
	250.00	1156884.89	1205381.39	10732.46	80.83	30.926
	255.00	1156906.66	1191777.44	10855.84	81.24	31.280
	260.00	1156925.97	1178174.56	10981.96	81.64	31.641
	265.00	1158275.72	1164559.92	11110.94	82.02	32.049
	270.00	1158732.94	1150929.23	11242.78	82.39	32.441
	275.00	1159139.61	1137297.84	11377.49	82.76	32.841
	280.00	1159503.80	1123666.47	11515.07	83.11	33.250
	285.00	1159828.61	1110035.08	11655.53	83.45	33.668
	290.00	1159961.75	1096403.69	11798.87	83.77	34.090
	295.00	1160049.86	1082772.33	11945.12	84.09	34.522
	300.00	1160107.33	1065134.33	12094.29	84.40	34.964
	305.00	1160145.00	105495.61	12246.41	84.69	35.417
	310.00	1160247.03	1041856.26	12401.53	84.97	35.884
	315.00	1160340.95	1028215.84	12559.70	85.25	36.363
	320.00	1160439.08	1014574.74	12720.96	85.51	36.855
	325.00	1160528.19	1000932.60	12885.36	85.76	37.360
	330.00	1160556.20	987290.19	13052.95	86.01	37.877
	335.00	1160575.91	973647.16	13223.77	86.24	38.409
	340.00	1160502.86	960004.71	13397.88	86.46	38.952
	345.00	1160426.39	946363.56	13575.32	86.68	39.511
	350.00	1160453.23	93722.65	13756.17	86.88	40.090
	355.00	1160500.16	919081.55	13940.51	87.08	40.686
	360.00	1160529.28	905439.80	14128.39	87.26	41.300
	365.00	1160554.73	891798.05	14319.83	87.44	41.933
	370.00	1160486.64	878156.48	14515.05	87.61	42.582
	375.00	1160421.33	864515.96	14714.09	87.77	43.252
	380.00	1160346.06	850876.53	14917.00	87.92	43.942
	385.00	1160290.69	837237.36	15123.90	88.06	44.656

INITIATE IGM

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABOVE SPHERICAL EARTH (FT)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (FT/S**2)
390.00	1160185.47	823599.43	566302.4	15334.88	88.19	45.391
395.00	1160092.23	805965.23	568621.1	15550.05	88.32	46.151
400.00	1159993.97	796331.23	570806.9	15769.51	88.43	46.938
405.00	1159898.36	782697.20	572866.7	15993.39	88.54	47.751
410.00	1159832.81	765063.18	574808.5	16221.82	88.64	48.595
415.00	1159768.36	755429.17	576639.6	16454.93	88.74	49.469
420.00	1159693.38	741795.16	578367.2	16692.87	88.82	50.375
425.00	1159633.22	728164.07	580000.8	16935.80	88.90	51.316
430.00	1159519.77	714533.95	581548.3	17183.86	88.97	52.290
435.00	1159413.28	700905.10	583018.1	17437.22	89.03	53.301
440.00	1159297.72	687277.97	584420.1	17696.06	89.09	54.353
445.00	1159179.11	672651.74	585763.6	17960.57	89.13	55.447
450.00	1159058.48	660026.98	587058.7	18230.95	89.17	56.585
455.00	1158945.30	646403.77	588315.2	18507.43	89.20	57.772
460.00	1158808.09	632781.81	589544.1	18790.23	89.23	59.009
465.00	922471.50	621477.65	590730.1	19026.54	89.28	47.847
470.00	922375.06	610620.05	591849.5	19259.27	89.31	48.692
475.00	922254.52	595761.68	592969.1	19493.09	89.30	49.568
480.00	922072.27	588893.76	594128.0	19731.84	89.29	50.473
485.00	921882.09	578027.48	595323.5	19976.19	89.27	51.411
490.00	693789.16	568158.55	596547.4	20203.98	89.28	39.359
495.00	692458.47	560100.02	597736.1	20395.25	89.32	39.826
500.00	692064.79	552047.41	598872.9	20588.94	89.35	40.384
505.00	691675.59	544001.25	599977.9	20785.08	89.36	40.958
510.00	691268.52	535960.75	601073.9	20983.93	89.37	41.548
515.00	690856.91	527926.30	602164.4	21186.48	89.38	42.155
520.00	690432.98	515898.32	603248.7	21392.52	89.39	42.780
525.00	689975.73	511876.80	604331.3	21602.00	89.39	43.422
530.00	689495.86	503861.98	605419.1	21814.91	89.39	44.082
535.00	688961.03	495854.70	606522.3	22031.29	89.39	44.759
540.00	688347.35	487855.40	607648.7	22251.22	89.38	45.452
545.00	687560.20	475865.75	608808.6	22474.76	89.36	46.156
550.00	686269.23	471888.72	610012.3	22701.89	89.35	46.849
551.38	683460.72	465681.51	610354.5	22765.36	89.34	46.876
552.08	27624.27	469503.63	610528.0	22775.01	89.35	1.893
552.26	23229.38	469459.16	610570.8	22775.25	89.35	1.592

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

TABLE 2-2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	***** RADIAL DISTANCE ØF THE C G (FT)	ØF THE EARTH FIXED VELOCITY (FT/SEC)	MAGNITUDE LATITUDE ØN SPHERICAL EARTH (DEG)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (NM)
S-IC/S-II SEP	21130770.25	7862.398		28.8658	-79.69	50.69
S-II ENGINE START	161.59			28.8695	-79.68	51.44
S-II 90 THRUST	162.22	21132675.25	7856.048	28.8872	-79.61	54.98
	165.22	21141600.00	7837.458	28.8872	-79.51	60.67
	170.00	21155412.25	7887.416	28.9154	-79.40	66.70
	175.00	21169406.50	7958.994	28.9453	-79.29	72.81
	180.00	21182958.25	8035.485	28.9754	-79.01	79.01
	185.00	21196075.50	8116.097	29.0059	-79.06	85.30
	190.00	21208766.75	8200.536	29.0367	-78.95	91.68
	195.00	21221040.50	8289.041	29.0678	-78.90	98.16
	197.02	21225880.25	8325.793	29.0804	-78.83	100.02
	200.00	21232905.75	8381.555	29.0992	-78.71	104.73
	201.42	21236199.75	8408.501	29.1082	-78.59	111.40
INITIATE IGM	205.00	21244370.75	8477.644	29.1310	-78.46	118.16
	210.00	21255461.25	8574.716	29.1631	-78.34	125.00
	215.00	21266239.50	8672.378	29.1954	-78.21	131.95
	220.00	21276715.00	8773.641	29.2281	-78.08	138.99
	225.00	21286885.75	8877.749	29.2611	-77.95	146.12
	230.00	21296754.00	8984.674	29.2943	-77.82	153.36
	235.00	21306323.50	9094.413	29.3279	-77.69	160.69
	240.00	21315597.75	9207.045	29.3617	-77.55	168.13
	245.00	21324579.00	9322.536	29.3958	-77.41	175.67
	250.00	21332711.75	9440.878	29.4303	-77.27	183.31
	255.00	21341678.50	9562.089	29.4650	-77.13	191.07
	260.00	21349804.25	9686.167	29.5000	-76.98	198.93
	265.00	21357652.50	9813.241	29.5354	-76.84	206.90
	270.00	21365228.25	9943.295	29.5710	-76.54	214.99
	275.00	21372534.75	10076.340	29.6069	-76.23	223.19
	280.00	21379576.25	10212.363	29.6431	-75.91	231.51
	285.00	21386357.50	10351.369	29.6796	-75.75	239.94
	290.00	21392882.75	10493.366	29.7164	-75.58	248.50
	295.00	21399156.75	10638.356	29.7535	-75.41	257.18
	300.00	21405184.25	10786.358	29.7909	-75.24	265.98
	305.00	21410969.75	10937.399	29.8286	-75.07	274.91
	310.00	21416518.00	11091.516	29.8665	-74.89	283.97
	315.00	21421834.75	11248.752	29.9048	-74.71	293.17
	320.00	21426923.75	11409.149	29.9433	-74.53	302.49
	325.00	21431791.50	11572.753	29.9822	-74.35	311.96
	330.00	21436442.00	11739.604	30.0213	-74.16	321.56
	335.00	21440881.00	11909.747	30.0607	-73.97	331.30
	340.00	21445113.75	12083.227	30.1004	-73.78	341.18
	345.00	21449145.50	12260.092	30.1403	-73.58	351.22
	350.00	21452981.75	12440.410	30.1806	-73.38	361.40
	355.00	21456628.75	12624.257	30.2211	-73.18	371.73
	360.00	21460091.75	12811.693	30.2619	-72.97	382.22
	365.00	21463378.00	13002.735	30.3030	-72.77	392.87
	370.00	21466493.00	13197.581	30.3443	-72.57	403.67
	375.00	21469442.75	13396.271	30.3859	-72.37	414.64
	380.00	21472232.75	13598.863	30.4277	-72.17	
	385.00	21474870.75	13805.471	30.4699	-71.97	

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	*** RADIAL DISTANCE OF THE C G (FT)	MAGNITUDE OF THE EARTH FIXED VELOCITY (FT/SEC)	SUB VEHICLE LATITUDE ON SPHERICAL EARTH (DEG)	LONGITUDE ON EARTH (DEG)	ARC RANGE ALONG SPHER- EARTH TO SUB VEHICLE PT. (NM)
390.00	21477361.75	14016.185	30.5122	-72.76	425.78
395.00	21479714.00	14231.101	30.5548	-72.55	437.08
400.00	21481934.00	14450.335	30.5976	-72.33	448.56
405.00	21484028.75	14674.007	30.6407	-72.11	460.22
410.00	21486006.00	14902.244	30.6839	-71.89	472.06
415.00	21487873.25	15135.181	30.7274	-71.66	484.08
420.00	21489638.75	15372.961	30.7711	-71.43	496.29
425.00	21491310.25	15615.733	30.8150	-71.20	508.69
430.00	21492896.75	15863.654	30.8591	-70.96	521.28
435.00	21494407.25	16116.885	30.9034	-70.72	534.08
440.00	21495850.25	16375.605	30.9478	-70.47	547.08
445.00	21497235.75	16640.004	30.9924	-70.22	560.29
450.00	21498574.00	16910.283	31.0371	-69.97	573.71
455.00	21499874.75	17186.662	31.0820	-69.71	587.35
460.00	21501149.50	17469.366	31.1270	-69.44	601.22
465.00	21502382.00	17705.566	31.1720	-69.17	615.29
470.00	21503549.00	17938.193	31.2170	-68.90	629.54
475.00	21504717.50	18171.943	31.2619	-68.63	643.99
480.00	21505926.50	18410.617	31.3067	-68.35	658.62
485.00	21507173.00	18654.885	31.3514	-68.06	673.44
490.00	21508448.75	18882.575	31.3960	-67.78	688.46
495.00	21509691.25	19073.740	31.4403	-67.48	703.63
500.00	21510882.75	19267.336	31.4843	-67.19	718.97
505.00	21512043.50	19463.376	31.5280	-66.89	734.45
510.00	21513196.75	19662.137	31.5714	-66.59	750.09
515.00	21514345.25	19864.598	31.6144	-66.29	765.90
520.00	21515488.75	20070.554	31.6570	-65.98	781.86
525.00	21516631.75	20279.942	31.6992	-65.67	797.99
530.00	21517782.00	20492.760	31.7410	-65.35	814.29
535.00	21518948.00	20709.057	31.7824	-65.04	830.76
540.00	21520139.00	20928.898	31.8233	-64.71	847.40
545.00	21521365.25	21152.342	31.8637	-64.39	864.22
550.00	21522636.25	21379.388	31.9036	-64.06	881.21
551.38	21522997.50	21442.829	31.9146	-63.97	885.95
552.08	21523180.50	21452.461	31.9201	-63.92	888.35
552.26	21523226.00	21452.696	31.9215	-63.91	888.95

S-II DECØ
S-IVB ULL IGN
S-II/S-IVB SEP

*** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X COORDINATE EARTH FIXED, PT. CENTERED PLUMB SYS, (FT)	Y COORDINATE EARTH FIXED, PT. CENTERED PLUMB SYS, (FT)	Z COORDINATE EARTH FIXED, PT. CENTERED PLUMB SYS, (FT)	X COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	Y COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	Z COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)
S-IC/S-II SEP	218989.295	115.456	311082.695	2931.570	14.586	7295.409
S-II ENGINE START	220830.666	124.859	315678.977	2912.792	14.840	7296.089
S-II 90 THRUST	229436.309	171.079	337576.762	2827.300	16.087	7309.711
	242690.082	253.069	372734.594	2719.893	18.128	7403.593
	256028.336	348.862	410034.336	2615.633	20.230	7516.891
	268847.398	455.553	447907.449	2512.212	22.437	7632.647
	281152.188	573.392	486363.402	2409.931	24.714	7750.010
	292948.242	702.866	525409.906	2308.588	27.073	7868.830
	304240.277	844.332	565054.680	2208.347	29.519	7989.401
	308658.133	904.893	581234.797	2168.138	30.536	8038.474
	315033.422	998.083	605306.633	2109.226	32.053	8111.758
	318006.684	1044.191	616842.031	2081.291	32.782	8146.781
	325333.207	1164.895	646174.680	2010.911	34.653	8235.623
	335161.375	1345.608	687656.828	1922.984	37.649	8356.224
	344577.590	1541.225	729731.711	1843.144	40.651	8474.156
	353587.801	1752.648	772403.758	1760.772	43.946	8595.030
	362185.445	1980.965	815684.578	1678.248	47.441	8717.549
	370368.707	2227.365	859582.070	1595.444	51.081	8841.737
	378139.762	2491.248	904104.633	1512.561	54.505	8967.582
	385494.586	2772.950	949260.859	1429.350	58.250	9095.232
	392432.945	3074.210	995059.828	1345.912	62.266	9224.659
	398953.410	3395.712	1041510.414	1262.295	66.395	9355.875
	405055.566	3738.518	1088621.563	1178.484	70.688	9488.926
	410738.188	4103.009	1126402.656	1094.480	75.132	9623.840
	416000.242	4490.040	1184863.281	1010.332	79.708	9760.767
	420840.918	4900.276	1234013.656	925.962	84.419	9899.727
	425259.184	5334.469	1283863.969	841.276	89.267	10040.763
	429253.195	5793.157	1334424.688	756.320	94.243	10183.882
	432821.727	6277.049	1385706.250	671.097	99.346	10329.115
	435963.539	6786.775	1437719.328	585.576	104.574	10476.493
	438677.000	7322.962	1490474.766	499.732	109.928	10626.043
	440960.465	7886.288	1543983.406	413.551	115.406	10777.810
	442811.883	8477.253	1598256.563	327.005	121.003	10931.840
	444225.613	9096.493	1653305.672	240.060	126.732	11088.194
	445211.816	9744.839	1709142.438	152.682	132.599	11246.934
	445755.617	10422.740	1765778.984	64.830	138.600	11408.123
	445859.168	11131.045	1823227.719	-23.536	144.732	11571.824
	445519.602	11870.393	1881501.328	-112.454	151.007	11738.094
	444733.633	12641.325	1940612.922	-201.959	157.421	11906.994
	443498.809	13444.832	2000575.656	-292.090	163.987	12078.583
	441811.637	14281.345	2061403.188	-382.886	170.706	12252.923
	439668.836	15152.166	2123109.438	-474.380	177.569	12430.094
	437066.723	16057.388	2185708.813	-566.613	184.585	12610.185
	434001.391	16998.171	2249216.156	-659.591	191.709	12793.266
	430465.977	17573.469	2313646.344	-753.193	198.386	12979.386
	426467.559	18582.911	2379015.219	-847.872	205.492	13168.714
	421989.418	20029.206	2445338.844	-943.564	213.080	13361.301
	417030.426	21113.919	2512633.688	-1040.228	220.816	13557.222
	411585.258	22237.822	2580916.688	-1137.966	228.792	13756.588

INITIATE IGM

TABLE 2-2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Y COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Z COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	X COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	Y COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	Z COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)
390.00	405648.578	23402.217	2650205.375	-1236.819	237.023	13959.496
395.00	399215.055	24608.357	2720517.594	-1336.823	245.467	14166.047
400.00	392278.539	25857.299	2751871.938	-1438.056	254.129	14376.356
405.00	384832.469	27150.083	2864287.531	-1540.578	263.012	14590.543
410.00	376870.488	28487.770	2937783.938	-1644.446	272.112	14808.735
415.00	368385.621	29871.521	3012381.594	-1749.725	281.434	15031.067
420.00	359370.895	31202.569	3088101.500	-1856.487	290.981	15257.677
425.00	349818.320	32781.736	3164965.625	-1964.800	300.760	15488.713
430.00	339720.102	34310.527	3242996.188	-2074.741	310.781	15724.325
435.00	329068.238	35889.957	3322216.594	-2186.387	321.037	15964.668
440.00	317853.098	37521.187	34022650.906	-2299.815	331.534	16209.917
445.00	306066.680	39205.674	3484323.906	-2415.108	342.286	16460.250
450.00	293698.734	40544.572	3567261.875	-2532.355	353.303	16715.862
455.00	280739.418	42739.137	3651491.438	-2651.645	364.594	16976.961
460.00	267178.887	44591.027	3737040.719	-2773.075	376.167	17243.761
465.00	252998.891	46499.301	3823818.625	-2899.481	387.188	17462.249
470.00	238190.410	48463.148	3911666.375	-3021.822	398.187	17677.353
475.00	222803.066	50479.591	4000596.219	-3132.566	408.424	17895.244
480.00	206858.799	52547.730	4090625.688	-3246.034	418.890	18117.357
485.00	190335.055	54669.159	4181777.000	-3364.330	429.716	18343.974
490.00	173207.020	56845.158	4274050.313	-3488.151	440.522	18552.369
495.00	155438.605	59073.923	4367234.313	-3620.085	450.994	18721.622
500.00	137010.129	61354.658	4461269.875	-3751.208	461.345	18893.010
505.00	117927.124	63687.588	4556167.938	-3881.295	471.821	19066.619
510.00	98197.178	66073.099	4651939.625	-4011.356	482.437	19242.555
515.00	77806.553	68512.546	4748597.375	-4145.433	493.390	19420.974
520.00	56738.024	71007.514	4846153.188	-4282.391	504.618	19601.878
525.00	34978.879	73559.148	4944620.063	-4421.656	516.087	19785.314
530.00	12518.045	76168.624	5044010.438	-4562.904	527.783	19971.344
535.00	-10653.297	78837.366	5144337.625	-4706.045	539.698	20160.032
540.00	-34545.193	81566.158	5245615.063	-4851.112	551.831	20351.437
545.00	-59167.896	84355.982	5347856.375	-4998.162	564.192	20545.599
550.00	-84530.429	87208.355	5451075.438	-5147.218	576.779	20742.510
551.38	-91682.865	88008.979	5479821.063	-5188.825	580.300	20797.458
552.08	-95321.932	88415.813	5494383.000	-5208.276	581.597	20802.493
552.26	-96234.005	88517.558	5498023.375	-5212.980	581.874	20801.548

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT**2)	TOTAL ANGLE OF ATTACK (DEG)	DRAG ALONG VELOCITY VECTOR (LB)	AERODYNAMIC LOAD (LB/FT**2)	MAGNITUDE OF THE AERODYNAMIC FORCE VECTOR (FT)
S-IC/S-II SEP	7.98	7.59	0.2303	680.3286	1.7472	93.3963
S-II ENGINE START	8.00	7.00	0.2149	625.7098	1.5045	80.4247
S-II 90 THRUST	8.13	4.77	0.0984	416.3281	0.4692	25.0790
	8.42	2.61	1.7438	304.4385	4.5433	242.8650
	8.75	1.36	2.7304	183.6687	3.7266	177.8924
	9.09	0.70	3.6843	106.4177	2.5845	112.8922
	9.18	0.34	4.6514	55.8116	1.5932	68.1321
	9.18	0.17	5.5837	28.7853	0.9404	40.2149
	9.01	0.08	6.5019	15.0939	0.5483	23.7196
	8.95	0.06	6.8653	11.8390	0.4460	19.4321
	8.87	0.04	7.3968	8.4113	0.3327	14.6293
	8.83	0.04	7.6467	7.1951	0.2907	12.8330
	8.66	0.02	9.6002	5.2233	0.2393	11.1502
	8.44	0.01	15.7393	4.1672	0.2290	12.6789
	8.26	0.01	16.2693	2.5990	0.1457	7.9777
	7.96	0.01	16.0030	1.6204	0.0894	4.9717
	7.64	0.00	16.2123	1.0335	0.0591	3.2184
	7.38	0.00	16.4108	0.6963	0.0413	2.2015
	7.07	0.00	16.5609	0.4742	0.0292	1.5283
	6.68	0.00	16.6812	0.3310	0.0209	1.1022
	6.37	0.00	16.8406	0.2458	0.0159	0.8403
	6.13	0.00	16.9898	0.1907	0.0125	0.6659
	5.94	0.00	17.1302	0.1525	0.0102	0.5432
	5.78	0.00	17.2637	0.1245	0.0086	0.4532
	5.65	0.00	17.3835	0.1042	0.0073	0.3864
	5.55	0.00	17.4580	0.0891	0.0064	0.3355
	5.46	0.00	17.5257	0.0777	0.0056	0.2960
	5.39	0.00	17.6028	0.0687	0.0051	0.2646
	5.33	0.00	17.6676	0.0617	0.0046	0.2395
	5.28	0.00	17.7197	0.0560	0.0042	0.2190
	5.24	0.00	17.7614	0.0514	0.0039	0.2021
	5.24	0.00	17.7917	0.0481	0.0036	0.1893
	5.24	0.00	17.8067	0.0454	0.0034	0.1788
	5.25	0.00	17.8058	0.0432	0.0033	0.1698
	5.26	0.00	17.7884	0.0413	0.0031	0.1621
	5.27	0.00	17.7544	0.0398	0.0030	0.1556
	5.30	0.00	17.7064	0.0384	0.0029	0.1499
	5.33	0.00	17.6447	0.0375	0.0028	0.1457
	5.38	0.00	17.5710	0.0369	0.0027	0.1423
	5.42	0.00	17.4853	0.0364	0.0026	0.1394
	5.47	0.00	17.3880	0.0360	0.0026	0.1369
	5.53	0.00	17.2792	0.0357	0.0025	0.1349
	5.58	0.00	17.1560	0.0356	0.0025	0.1332
	5.64	0.00	17.0784	0.0355	0.0025	0.1319
	5.70	0.00	16.9235	0.0355	0.0024	0.1309
	5.76	0.00	16.7138	0.0357	0.0024	0.1302
	5.83	0.00	16.5460	0.0360	0.0024	0.1300
	5.91	0.00	16.3632	0.0364	0.0023	0.1301
	5.98	0.00	16.1641	0.0368	0.0023	0.1303

TABLE 2-2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT*2)	TOTAL ANGLE OF ATTACK (DEG)	DRAG ALONG VELOCITY VECTØR (LB)	AERODYNAMIC LOAD (LB/FT*2)	MAGNITUDE OF THE AERODYNAMIC FORCE VECTOR (FT)
390.00	6.06	0.00	15.9629	0.0371	0.0023	0.1303
395.00	6.14	0.00	15.7525	0.0371	0.0023	0.1285
400.00	6.22	0.00	15.5288	0.0372	0.0023	0.1268
405.00	6.31	0.00	15.2987	0.0373	0.0023	0.1251
410.00	6.40	0.00	15.0608	0.0374	0.0023	0.1235
415.00	6.49	0.00	14.8142	0.0376	0.0023	0.1218
420.00	6.58	0.00	14.5599	0.0379	0.0022	0.1201
425.00	6.67	0.00	14.2982	0.0381	0.0022	0.1184
430.00	6.77	0.00	14.0294	0.0384	0.0022	0.1166
435.00	6.87	0.00	13.7546	0.0381	0.0022	0.1149
440.00	6.97	0.00	13.4741	0.0383	0.0022	0.1131
445.00	7.07	0.00	13.1876	0.0392	0.0022	0.1118
450.00	7.18	0.00	12.8956	0.0397	0.0022	0.1108
455.00	7.29	0.00	12.5986	0.0403	0.0022	0.1098
460.00	7.40	0.00	12.2972	0.0408	0.0022	0.1086
465.00	7.49	0.00	12.3145	0.0418	0.0022	0.1108
470.00	7.58	0.00	15.2228	0.0496	0.0028	0.1509
475.00	7.67	0.00	16.1602	0.0529	0.0030	0.1644
480.00	7.77	0.00	15.1166	0.0515	0.0029	0.1548
485.00	7.86	0.00	14.1604	0.0505	0.0027	0.1445
490.00	7.95	0.00	13.2621	0.0490	0.0026	0.1339
495.00	8.02	0.00	12.3499	0.0471	0.0024	0.1220
500.00	8.09	0.00	12.7933	0.0486	0.0025	0.1295
505.00	8.17	0.00	13.6692	0.0514	0.0027	0.1434
510.00	8.24	0.00	12.8961	0.0498	0.0026	0.1335
515.00	8.32	0.00	12.1453	0.0483	0.0025	0.1237
520.00	8.40	0.00	11.6598	0.0476	0.0024	0.1182
525.00	8.48	0.00	11.3092	0.0473	0.0023	0.1147
530.00	8.56	0.00	11.0260	0.0471	0.0023	0.1121
535.00	8.64	0.00	10.7586	0.0469	0.0023	0.1096
540.00	8.72	0.00	10.4902	0.0468	0.0022	0.1070
545.00	8.81	0.00	10.2229	0.0466	0.0022	0.1044
550.00	8.89	0.00	9.9681	0.0464	0.0022	0.1019
551.38	8.92	0.00	9.9021	0.0464	0.0022	0.1013
552.08	8.92	0.00	9.8815	0.0463	0.0022	0.1010
552.26	8.92	0.00	9.8777	0.0463	0.0022	0.1009

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM MOTION (SEC)	X POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Y POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Z POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)
S-IC/S-II SEP	21123893.3	128131.7	523624.0	2817.527	377.412	8617.922
S-II ENGINE START	21125662.3	128369.4	529C53.5	2798.125	377.316	8618.623
S-II 90 THRUST	21133519.8	129500.6	554919.0	2709.548	376.830	8632.358
	21146596.5	131299.7	596401.4	2596.374	375.698	8726.650
	21159301.8	133174.8	640317.5	2485.815	374.324	8840.368
	21171455.8	135043.0	684808.8	2375.962	372.951	8956.455
	21183063.0	136904.2	729884.5	2267.123	371.549	9074.066
	21194128.3	138758.4	775551.7	2159.096	370.129	9193.048
	21204655.5	140605.5	821817.8	2052.040	368.694	9313.702
	21208755.3	141349.3	840671.7	2009.043	368.116	9362.784
	21214650.0	142445.4	868691.3	1945.970	367.245	9436.060
	21217390.3	142966.2	882105.9	1916.038	366.825	9471.069
	21224115.8	144277.8	916180.7	1840.574	365.759	9559.844
	21233074.5	146103.8	964283.8	1745.495	364.647	9680.418
	21241585.3	147924.2	1012979.7	1658.433	363.510	9798.356
	21249653.5	149739.2	1062272.6	1568.685	362.543	9919.124
	21257272.0	151549.7	1112173.4	1478.652	361.673	10041.458
	21264439.3	153356.1	1162689.8	1388.200	360.841	10165.377
	21271153.8	155157.5	1213829.7	1297.532	359.689	10290.871
	21277413.8	156953.4	1265601.3	1206.391	358.747	10418.087
	21283217.0	158745.2	1318C13.3	1114.878	357.967	10546.998
	21288562.0	160533.1	1371074.1	1023.038	357.187	10677.616
	21293446.8	162317.2	1424792.4	930.853	356.460	10809.986
	21297870.0	164097.7	1479176.9	838.322	355.768	10944.136
	21301829.5	165874.9	1534236.9	745.488	355.090	11080.217
	21305324.3	167648.6	1589982.2	652.272	354.428	11218.245
	21308351.5	169419.2	1646422.6	558.576	353.782	11358.264
	21310909.0	171186.5	1703568.1	464.443	353.140	11500.281
	21312595.0	172950.6	1761428.8	369.873	352.499	11644.323
	21314607.0	174711.5	1820C14.7	274.832	351.856	11790.424
	21315742.5	176469.1	1879336.4	179.293	351.210	11938.609
	21316399.0	178223.5	1939404.3	83.237	350.555	12088.920
	21316574.0	179974.6	2000229.2	-13.368	349.885	12241.406
	21316264.5	181722.4	2061822.0	-110.558	349.212	12396.123
	21315467.3	183466.8	2124194.2	-208.371	348.536	12553.134
	21314179.5	185207.7	2187357.2	-306.855	347.851	12712.500
	21312397.5	186945.2	2251323.1	-406.051	347.153	12874.281
	21310117.8	188679.3	2316104.0	-506.003	346.449	13038.534
	21307336.3	190409.7	2381712.5	-606.752	345.733	13205.318
	21304648.8	192136.6	2448161.4	-708.338	345.014	13374.691
	21300251.5	193859.8	2515463.7	-810.806	344.291	13546.713
	21295939.3	195579.5	2583632.9	-914.194	343.551	13721.462
	21291108.0	197279.3	2652682.8	-1018.550	342.798	13899.024
	21285752.3	199007.4	2722627.8	-1123.883	341.987	14079.470
	21279867.8	200713.9	2793482.2	-1230.072	340.557	14262.841
	21273449.0	202413.5	2865261.3	-1337.586	339.378	14449.306
	21266489.8	204108.2	2937980.4	-1446.366	338.499	14638.916
	21258983.5	205798.4	3011655.7	-1556.373	337.584	14831.744
	21250523.8	207484.2	3086303.3	-1667.718	336.719	15027.897

INITIATE IGM

TABLE 2-2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Y POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Z POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)
390.00	21242303.8	209165.7	3161940.3	-1780.448	335.915	15227.470
395.00	21233116.8	210843.3	3238583.7	-1894.605	335.124	15430.558
400.00	21223355.3	212517.0	3316251.7	-2010.274	334.348	15637.277
405.00	21213011.5	214186.8	3394962.7	-2127.526	333.581	15847.741
410.00	21202077.5	215852.8	3474735.5	-2246.421	332.817	16062.074
415.00	21190544.5	217515.0	3555589.9	-2367.036	332.053	16280.408
420.00	21178404.3	219173.3	3637546.3	-2489.448	331.286	16502.877
425.00	21165647.0	220827.9	3720625.7	-2613.739	330.517	16729.625
430.00	21152263.5	222478.5	3804849.8	-2739.992	329.750	16960.797
435.00	21138243.5	224125.3	3890241.2	-2868.295	328.968	17196.546
440.00	21123576.8	225768.2	3976822.9	-2998.734	328.173	17437.039
445.00	21108252.5	227407.0	4064619.4	-3131.406	327.370	17682.453
450.00	21092258.8	229041.9	4153655.7	-3266.409	326.559	17932.975
455.00	21075584.0	230672.6	4243957.8	-3403.845	325.743	18188.809
460.00	21058216.0	232299.3	4335552.8	-3543.825	324.918	18450.164
465.00	21040138.8	233921.7	4428348.5	-3687.592	324.143	18662.957
470.00	21021346.0	235540.7	4522185.4	-3827.390	323.289	18872.363
475.00	21001887.0	237153.0	4617076.6	-3955.866	321.585	19084.771
480.00	20981781.3	238756.6	4713039.7	-4087.390	319.862	19301.208
485.00	20961004.5	240351.7	4810095.4	-4224.082	318.220	19521.889
490.00	20939532.0	241938.9	4908243.0	-4365.936	316.676	19724.071
495.00	20917332.0	243518.8	5007269.3	-4514.857	315.282	19886.750
500.00	20894387.0	245091.1	5107113.8	-4663.184	313.607	20051.458
505.00	20870700.5	246654.9	5207787.1	-4810.699	311.898	20218.292
510.00	20846279.5	248210.0	5309300.1	-4958.420	310.155	20387.326
515.00	20821109.0	249756.7	5411663.8	-5110.394	308.531	20558.588
520.00	20795170.5	251295.4	5514889.4	-5265.491	306.970	20732.112
525.00	20768449.8	252826.4	5618988.5	-5423.141	305.443	20907.958
530.00	20740935.3	254349.9	5723972.7	-5583.023	303.933	21086.197
535.00	20712616.0	255865.8	5829854.3	-5745.055	302.431	21266.894
540.00	20683481.3	257374.2	5936645.6	-5909.273	300.930	21450.102
545.00	20653519.5	258875.1	6044359.4	-6075.740	299.439	21635.860
550.00	20622720.0	260368.6	6153008.3	-6244.482	297.950	21824.153
551.38	20614045.0	260780.7	6183249.1	-6291.584	297.536	21876.678
552.08	20609633.3	260988.9	6198566.2	-6313.028	297.313	21880.547
552.26	20608528.0	261040.9	6202395.1	-6318.154	297.256	21879.317

S-II ØECØ
S-IVB ULL IGN
S-II/S-IVB SEP

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
S-II/SIVB SEP	552.26	28057.60	186102.0	6941.90	89.35	0.169
S-IVB ENGINE START	552.38	27419.89	186111.3	6941.90	89.35	0.165
	555.00	29590.00	186299.7	6942.07	89.39	0.178
S-IVB 90 THRUST	557.88	821429.03	186492.7	6947.03	89.43	4.941
	560.00	876252.34	186505.11	6957.48	89.45	5.285
S-IVB ULLAGE JETT	564.18	900974.79	186884.5	6979.27	89.48	5.462
	565.00	902091.87	186932.8	6983.61	89.48	5.477
	570.00	907250.80	187212.2	7010.57	89.53	5.544
	575.00	910170.60	187466.0	7037.81	89.57	5.599
	580.00	909059.80	187695.3	7065.33	89.61	5.630
	585.00	909615.63	187900.0	7093.11	89.65	5.671
	590.00	909525.98	188080.7	7121.12	89.69	5.709
	595.00	909420.95	188238.7	7149.37	89.73	5.747
	600.00	909227.76	188374.4	7177.85	89.76	5.785
	605.00	909048.85	188489.2	7206.55	89.79	5.824
	610.00	908821.38	188584.2	7235.49	89.82	5.863
	615.00	911927.50	188660.5	7264.70	89.85	5.925
	620.00	909618.47	188719.5	7294.15	89.88	5.952
	625.00	909249.41	188762.2	7323.81	89.90	5.992
	630.00	908830.19	188790.2	7353.70	89.93	6.032
	635.00	908481.31	188804.5	7383.80	89.95	6.073
	640.00	908213.52	188806.7	7414.14	89.96	6.115
	645.00	907900.11	188798.2	7444.71	89.98	6.158
	650.00	907554.74	188780.2	7475.51	89.99	6.201
	655.00	907210.86	188754.4	7506.54	90.00	6.244
	660.00	906866.91	188722.1	7537.80	90.01	6.289
	665.00	906507.66	188684.7	7569.29	90.01	6.334
	670.00	906148.42	188643.5	7601.04	90.02	6.379
	675.00	905816.03	188594.7	7633.06	90.03	6.425
	680.00	905506.99	188541.2	7665.30	90.04	6.473
	685.00	905204.26	188486.5	7697.78	90.03	6.521
	690.00	904913.95	188434.2	7730.47	90.03	6.570
	695.00	904720.19	188387.5	7763.39	90.02	6.620
S-IVB GCSI	699.23	904628.78	188354.9	7791.41	90.00	6.664
	700.00	41038.01	188349.7	7792.91	90.00	0.302
	705.00	630.81	188315.9	7793.08	90.00	0.005
EPØI	709.23	630.81	188287.0	7793.10	90.00	0.005

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG
TABLE 2.2 (BTØ)

TIME FROM FIRST MOTION (SEC)	*** RADIAL DISTANCE OF THE C G (M)	MAGNITUDE OF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN EARTH (DEG)	ARC RANGE ALONG SPHER- EARTH TO SUB VEHICLE PT. (KM)
S-II/SIVB SEP	552.26	6560279.31	6538.782	31.9215	-63.91
S-IVB ENGINE START	552.38	6560289.19	6538.788	31.9225	1646.33
	555.00	6560488.63	6538.932	31.9429	1647.13
S-IVB 90 THRUST	557.88	6560694.00	6543.875	31.9651	1663.74
	560.00	6560838.19	6554.311	31.9813	1682.06
S-IVB ULLAGE JET	564.18	6561112.81	6576.069	32.0127	1695.52
	565.00	6561164.63	6580.412	32.0188	1722.19
	570.00	6561466.06	6607.338	32.0554	1727.41
	575.00	6561742.19	6634.554	32.0912	1759.42
	580.00	6561994.06	6662.049	32.1262	1791.57
	585.00	6562221.75	6689.800	32.1603	1823.85
	590.00	6562425.88	6717.791	32.1934	1856.26
	595.00	6562607.50	6746.019	32.2257	1888.80
	600.00	6562767.38	6774.478	32.2571	1921.48
	605.00	6562906.69	6803.168	32.2875	1954.30
	610.00	6563026.56	6832.087	32.3169	1987.26
	615.00	6563128.19	6861.285	32.3454	2020.35
	620.00	6563212.81	6890.725	32.3730	2053.59
	625.00	6563281.63	6920.374	32.3995	2086.96
	630.00	6563336.00	6950.248	32.4250	2120.48
	635.00	6563377.25	6980.348	32.4495	2154.15
	640.00	6563406.69	7010.678	32.4730	2187.96
	645.00	6563425.81	7041.240	32.4954	2221.91
	650.00	6563436.00	7072.033	32.5168	2256.02
	655.00	6563438.69	7103.058	32.5371	2290.27
	660.00	6563435.19	7134.315	32.5562	2324.67
	665.00	6563427.19	7165.806	32.5743	2359.22
	670.00	6563415.88	7197.548	32.5913	2393.93
	675.00	6563397.38	7229.568	32.6070	2428.79
	680.00	6563374.56	7261.810	32.6217	2463.80
	685.00	6563351.06	7294.280	32.6351	2498.97
	690.00	6563330.38	7326.973	32.6474	2534.30
	695.00	6563315.75	7359.890	32.6585	2569.78
S-IVB GCS1	699.23	6563310.56	7387.906	32.6669	2605.43
	700.00	6563310.38	7389.405	32.6683	2635.69
	705.00	6563309.50	7389.579	32.6769	2641.23
EPØI	709.23	6563308.69	7389.596	32.6831	2677.10
					2707.43

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM MOTION (SEC)	X CØRDATE EARTH FIXED, PT. CENTERED PLUMB SYS, (M)	Y CØRDATE EARTH FIXED, PT. CENTERED PLUMB SYS, (M)	Z CØRDATE EARTH FIXED, PT. CENTERED PLUMB SYS, (M)	X CØMPØNT ØF VEL. IN THE XXE SYSTEM (M/S)	Y CØMPØNT ØF VEL. IN THE XXE SYSTEM (M/S)	Z CØMPØNT ØF VEL. IN THE XXE SYSTEM (M/S)
S-II/SIVB SEP	552.26	-29332.125	26580.152	1615797.531	-1588.916	6340.312
S-IVB ENGINE START	552.38	-29538.813	27602.320	1676590.047	-1589.937	6340.061
	555.00	-33718.000	27468.071	1653168.750	-1611.293	6334.781
S-IVB 90 THRUST	557.88	-38399.125	27585.353	1711432.625	-1635.232	6333.709
	560.00	-41878.438	28367.728	1724841.063	-1653.229	6339.790
S-IVB ULLAGE JETT	564.18	-48865.375	29131.893	1751393.875	-1688.747	6352.874
	565.00	-50250.313	29282.094	1756578.875	-1695.906	6355.452
	570.00	-58841.313	30210.405	1788395.563	-1740.425	6371.251
	575.00	-67654.000	31153.204	1820291.563	-1784.712	6387.175
	580.00	-76689.438	32110.810	1852267.078	-1829.503	6403.011
	585.00	-85949.563	33683.750	1884321.531	-1874.587	6418.791
	590.00	-95435.750	34772.216	1916454.750	-1919.941	6434.504
	595.00	-105149.313	35076.535	1948666.359	-1965.535	6450.152
	600.00	-115091.438	36096.779	1980956.063	-2011.348	6465.736
	605.00	-125263.125	37133.175	2013323.531	-2057.371	6481.256
	610.00	-135665.563	38185.811	2045768.406	-2103.600	6496.712
	615.00	-146299.563	39254.804	2078290.453	-2150.040	6512.153
	620.00	-157166.313	40340.230	2110889.656	-2196.690	6527.541
	625.00	-168266.813	41442.252	2143565.563	-2243.550	6542.840
	630.00	-179602.063	42560.849	2176317.781	-2290.592	6558.075
	635.00	-191173.063	43696.138	2209146.000	-2337.827	6573.242
	640.00	-202980.750	44848.159	2242049.906	-2385.255	6588.346
	645.00	-215025.875	46017.071	2275029.188	-2432.883	6603.386
	650.00	-227309.813	47202.865	2308083.563	-2480.711	6618.359
	655.00	-239833.313	48405.637	2341212.594	-2528.740	6633.266
	660.00	-252597.563	49625.477	2374415.938	-2576.973	6648.105
	665.00	-265603.313	50862.446	2407693.344	-2625.414	6662.876
	670.00	-278852.063	52116.648	2441044.406	-2674.423	6677.452
	675.00	-292349.813	53388.353	2474466.938	-2724.524	6691.639
	680.00	-306096.375	54677.738	2507961.000	-2774.123	6706.020
	685.00	-320090.813	55984.794	2541527.063	-2823.647	6720.437
	690.00	-334332.500	57309.618	2575165.313	-2873.026	6734.917
	695.00	-348820.813	58652.230	2608876.188	-2922.300	6749.451
S-IVB GCS1	699.23	-361264.125	59601.426	2637438.250	-2963.954	6761.762
	700.00	-363555.188	60012.613	2642658.688	-2970.339	6760.580
	705.00	-378502.125	61387.868	2676420.000	-3008.435	6743.800
EPØI	709.23	-391289.375	62560.652	2704901.625	-3040.531	6729.321

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TABLE 2.2 (B10)

TIME FROM FIRST MOTION (SEC)	COMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN PITCH (DEG)	COMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN YAW (DEG)	COMMANDED ROLL ATTITUDE (DEG)	EULERIAN ANGLE OF ATTITUDE IN ROLL (DEG)
S-II/SIVB SEP	552.26	-96.134	-56.1845	0.565	0.163917	-0.00063474
S-IVB ENGINE START	552.38	-96.134	-56.1967	0.565	0.163884	-0.00055016
	555.00	-96.134	-56.4431	0.565	0.150598	0.00131241
S-IVB 90 THRUST	557.88	-96.134	-56.5138	0.565	0.169151	0.05096086
	560.00	-96.134	-55.9820	0.565	0.342893	0.23566859
S-IVB ULLAGE JETT	564.18	-95.509	-58.1983	0.101	-0.136499	-0.13249274
	565.00	-99.671	-59.1442	0.106	-0.242284	-0.21366115
	570.00	-100.199	-59.9895	0.259	0.035569	0.33424254
	575.00	-101.050	-100.6472	0.390	0.107586	0.65638945
	580.00	-101.879	-101.5819	0.508	0.252154	0.83556080
	585.00	-102.562	-102.2596	0.579	0.324083	0.87656877
	590.00	-103.184	-102.9023	0.627	0.377243	0.89657822
	595.00	-103.751	-103.4760	0.656	0.408590	0.92925724
	600.00	-104.291	-104.0207	0.673	0.427259	0.96103910
	605.00	-104.814	-104.5471	0.681	0.437774	0.94387063
	610.00	-105.325	-105.0586	0.682	0.438497	0.92864455
	615.00	-105.834	-105.5622	0.680	0.435794	0.91464245
	620.00	-106.363	-106.0990	0.678	0.433476	0.89328048
	625.00	-106.834	-106.5756	0.669	0.424847	0.88433227
	630.00	-107.305	-107.0453	0.660	0.414503	0.87657604
	635.00	-107.779	-107.5170	0.651	0.404363	0.86827458
	640.00	-108.255	-107.9915	0.642	0.394311	0.85925148
	645.00	-108.733	-108.4702	0.633	0.384760	0.84925419
	650.00	-109.207	-108.9437	0.625	0.375681	0.84036057
	655.00	-109.684	-109.4176	0.618	0.367063	0.83135838
	660.00	-110.162	-109.8972	0.611	0.359401	0.82110444
	665.00	-110.643	-110.3740	0.606	0.352988	0.81157409
	670.00	-111.125	-110.8517	0.600	0.3461369	0.79923621
	675.00	-111.603	-111.3277	0.590	0.339414	0.78868049
	680.00	-112.080	-111.8042	0.580	0.332700	0.77823537
	685.00	-112.557	-112.2817	0.570	0.326000	0.76823537
	690.00	-113.034	-112.7592	0.560	0.319300	0.75823537
	695.00	-113.511	-113.2367	0.550	0.312600	0.74823537
S-IVB GCS1	699.23	-113.988	-113.7142	0.540	0.305900	0.73823537
	700.00	-114.465	-114.1917	0.530	0.299200	0.72823537
	705.00	-114.942	-114.6692	0.520	0.292500	0.71823537
	709.23	-115.419	-115.1467	0.510	0.285800	0.70823537
EP01		-115.896	-115.6242	0.500	0.279100	0.69823537

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE OF ATTACK (DEG)	DRAG ALONG VELOCITY VECTOR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE OF THE AERODYNAMIC FORCE VECTOR (KG)
S-II/SIVB SEP	8.92	0.01	9.8777	0.0118	0.0105	0.0174
S-IVB ENGINE START	8.92	0.01	9.8752	0.0118	0.0105	0.0174
	8.92	0.01	9.8310	0.0118	0.0104	0.0172
S-IVB 90 THRUST	8.92	0.01	9.9757	0.0117	0.0105	0.0175
	8.93	0.01	10.6532	0.0118	0.0112	0.0192
S-IVB ULLAGE JETT	8.95	0.01	8.7334	0.0115	0.0092	0.0143
	8.96	0.01	7.8528	0.0113	0.0082	0.0123
	8.98	0.01	7.3488	0.0113	0.0077	0.0115
	9.02	0.01	7.0393	0.0113	0.0074	0.0110
	9.05	0.01	6.4538	0.0113	0.0068	0.0101
	9.08	0.01	6.1274	0.0113	0.0064	0.0096
	9.11	0.01	5.8362	0.0113	0.0061	0.0091
	9.15	0.01	5.6133	0.0114	0.0059	0.0088
	9.18	0.01	5.4186	0.0114	0.0057	0.0086
	9.22	0.01	5.2411	0.0114	0.0056	0.0083
	9.26	0.01	5.0774	0.0115	0.0054	0.0081
	9.29	0.01	4.9204	0.0116	0.0053	0.0079
	9.33	0.01	4.7293	0.0116	0.0051	0.0076
	9.37	0.01	4.5974	0.0117	0.0050	0.0075
	9.41	0.01	4.4712	0.0118	0.0049	0.0073
	9.45	0.01	4.3420	0.0119	0.0048	0.0071
	9.49	0.01	4.2089	0.0119	0.0047	0.0070
	9.53	0.01	4.0706	0.0120	0.0046	0.0068
	9.57	0.01	3.9364	0.0121	0.0045	0.0066
	9.62	0.01	3.8008	0.0122	0.0043	0.0065
	9.66	0.01	3.6585	0.0123	0.0042	0.0063
	9.70	0.01	3.5179	0.0124	0.0041	0.0061
	9.74	0.01	1.0913	0.0125	0.0013	0.0019
	9.79	0.01	1.7746	0.0126	0.0021	0.0031
	9.83	0.01	2.3652	0.0128	0.0028	0.0042
	9.88	0.01	2.8771	0.0129	0.0035	0.0052
	9.92	0.01	3.3744	0.0130	0.0041	0.0061
	9.97	0.01	3.7303	0.0132	0.0046	0.0068
S-IVB GCSI	10.00	0.01	4.0009	0.0133	0.0049	0.0074
	700.00	0.01	4.0552	0.0133	0.0050	0.0075
	705.00	0.01	4.4125	0.0133	0.0055	0.0081
EPØI	709.23	0.01	4.7162	0.0133	0.0058	0.0087

TABLE 2-2 (BT0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Y PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Z PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)
S-11/S1VB SEP	6281479.4	79565.3	1890490.0	-1925.773	90.604	6668.816
S-1VB ENGINE START	6281238.6	79576.6	1891323.6	-1926.884	90.591	6668.503
	6276167.4	79813.2	1908759.8	-1950.126	90.331	6661.919
S-1VB 90 THRUST	6270506.0	80073.3	1927565.1	-1976.308	90.027	6659.382
	6266303.1	80263.6	1942061.5	-1996.193	89.828	6664.361
S-1VB ULLAGE JETT	6257869.2	80638.8	1969967.7	-2035.487	89.450	6675.249
	6256205.0	80711.7	1975415.6	-2043.384	89.325	6677.386
	6245865.3	81156.5	2008835.1	-2092.432	88.636	6690.437
	6235281.3	81598.2	2042320.1	-2141.267	88.069	6703.583
	6224451.7	82037.3	2075870.6	-2190.617	87.552	6716.581
	6213374.6	82473.8	2109485.7	-2240.269	87.085	6729.471
	6202048.5	82908.2	2143165.0	-2290.204	86.645	6742.241
	6190472.1	83340.3	2176907.8	-2340.389	86.223	6754.897
	6178644.3	83770.4	2210713.7	-2390.802	85.810	6767.436
	6166563.8	84198.4	2244581.9	-2441.437	85.403	6779.860
	6154229.5	84624.4	2278511.9	-2492.286	84.995	6792.170
	6141640.5	85048.4	2312503.3	-2543.358	84.585	6804.412
	6128795.6	85470.3	2346555.7	-2594.652	84.171	6816.549
	6115693.6	85890.1	2380668.4	-2646.163	83.751	6828.545
	6102333.6	86307.8	2414840.8	-2697.866	83.324	6840.425
	6088714.6	86723.3	2449072.3	-2749.771	82.889	6852.186
	6074835.6	87136.6	2483362.3	-2801.878	82.446	6863.831
	6060695.5	87547.8	2517710.3	-2854.194	81.995	6875.359
	6046293.3	87956.6	2552115.6	-2906.720	81.536	6886.767
	6031628.0	88363.1	2586577.6	-2959.454	81.070	6898.055
	6016698.4	88767.3	2621095.8	-3012.402	80.598	6909.222
	6001503.7	89169.1	2655669.5	-3065.566	80.118	6920.267
	5986042.4	89568.5	2690298.0	-3119.301	79.650	6931.046
	5970308.5	89965.6	2724978.8	-3174.123	79.214	6941.347
	5954302.1	90360.6	2759711.7	-3228.461	78.760	6951.815
	5938024.1	90753.2	2794496.8	-3282.738	78.298	6962.274
	5921475.0	91143.5	2829334.3	-3336.883	77.825	6972.757
	5904655.4	91531.4	2864224.3	-3390.936	77.338	6983.251
S-1VB GCS1	5890222.3	91857.6	2893767.6	-3436.641	76.928	6992.110
	5887566.1	91916.9	2899165.6	-3443.616	76.842	6990.350
	5870244.1	92299.7	2934066.7	-3485.133	76.258	6969.947
EPØI	5855435.3	92621.0	2963498.0	-3520.094	75.761	6952.383

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG
TABLE 2-2 (BTØ)

TIME FROM FIRST MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED PITCH CHANL (DEG)	GIMBAL DEFLECTION ANG REQUIRED YAW CHANNEL (DEG)	GIMBAL DEFLECTION ANG REQUIRED ROLL CHANNEL (DEG)	EULERIAN ATTITUDE RATE IN PITCH (DEG/S)	EULERIAN ATTITUDE RATE IN YAW (DEG/S)	EULERIAN ATTITUDE RATE IN ROLL (DEG/S)
S-II/SIVB SEP	0.0000	0.000000	0.000000	-0.09754047	-0.00003261	0.00039809
S-IVB ENGINE START	-0.0131	-0.037970	0.000000	-0.09725539	-0.00049523	0.00039791
552.38	-0.3225	-0.343120	0.000000	-0.09077111	-0.00066290	0.00057110
555.00	-0.1864	-0.274223	0.000000	0.18654976	0.06824908	0.07199447
S-IVB 90 THRUST	0.2996	-0.127238	0.000000	0.20489661	0.06092363	0.07519590
557.88	0.0561	-0.341896	0.000000	-1.21792682	-0.17605795	-0.11470229
560.00	-0.4787	-0.364774	0.000000	-1.04163486	-0.08729824	-0.06284157
S-IVB ULLAGE JETT	0.2659	-0.140574	0.000000	0.12611097	0.03803986	0.12566721
564.18	0.0869	-0.203092	0.000000	-0.29474812	0.02927439	0.03794278
565.00	0.1046	-0.186881	0.000000	-0.13704765	0.02273922	0.02647148
570.00	0.1117	-0.194325	0.000000	-0.13628961	0.01419409	0.00161086
575.00	0.1102	-0.193969	0.000000	-0.11958167	0.01016279	0.00489106
580.00	0.1131	-0.195708	0.000000	-0.11119465	0.00655923	0.00632198
585.00	0.1140	-0.196308	0.000000	-0.10669375	0.00470421	-0.00445093
590.00	0.1141	-0.195982	0.000000	-0.10347662	0.00269825	-0.00398264
600.00	0.1157	-0.197503	0.000000	-0.10137145	0.00134176	-0.00368374
605.00	0.1192	-0.198584	0.000000	-0.10121891	0.00086484	-0.00371696
610.00	0.1098	-0.198554	0.000000	-0.10623351	0.00090105	-0.00487239
615.00	0.1187	-0.200193	0.000000	-0.09127526	-0.00077410	-0.00163520
620.00	0.1168	-0.200954	0.000000	-0.09512758	-0.00053369	-0.00250249
625.00	0.1190	-0.201765	0.000000	-0.09413963	-0.00062501	-0.00228922
630.00	0.1192	-0.202541	0.000000	-0.09533472	-0.00052016	-0.00255760
640.00	0.1187	-0.203281	0.000000	-0.09584789	-0.00044360	-0.00267057
645.00	0.1204	-0.204003	0.000000	-0.09430297	-0.00040376	-0.00230741
650.00	0.1213	-0.204733	0.000000	-0.09552383	-0.00026127	-0.00257726
655.00	0.1202	-0.205406	0.000000	-0.09537685	-0.00004812	-0.00251635
660.00	0.1234	-0.206122	0.000000	-0.09583614	0.00022305	-0.00258966
665.00	-0.3885	-0.185952	0.000000	-0.52466641	0.01627021	-0.10001982
670.00	0.2346	-0.211635	0.000000	0.18840791	-0.00754165	0.06429737
675.00	0.1054	-0.208795	0.000000	0.01098639	-0.00160352	0.02328319
680.00	0.1290	-0.209493	0.000000	0.04191452	-0.00379283	0.03030310
685.00	0.1250	-0.209849	0.000000	0.02844244	-0.000518000	0.01515109
690.00	0.1142	-0.209197	0.000000	-0.01518351	0.00295784	0.00596576
695.00	0.1282	-0.211686	0.000000	0.00376140	-0.00087952	-0.01403308
S-IVB GCS1	0.1296	-0.212113	0.000000	0.00325740	-0.00085180	-0.01414953
700.00	0.1436	-0.216890	0.000000	0.00351921	-0.00140133	-0.01417321
705.00	0.1565	-0.222699	0.000000	0.00382093	-0.00188190	-0.01417716
709.23						
EPØI						

TABLE 2-2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABOVE SPHERICAL EARTH (FT)	MAGNITUDE ØF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (FT/S**2)
S-II/SIVB SEP	552.26	6307.60	367053.34	22775.25	89.35	0.553
S-IVB ENGINE START	552.38	6164.24	367053.34	22775.27	89.35	0.540
555.00	6652.10	366971.37	611219.4	22775.81	89.39	0.583
S-IVB 90 THRUST	557.88	184664.59	611852.8	22792.10	89.43	16.211
560.00	196989.36	365537.69	612296.8	22826.39	89.45	17.338
564.18	202547.19	363623.76	613138.1	22897.86	89.48	17.921
S-IVB ULLAGE JETT	565.00	202758.32	613296.6	22912.12	89.48	17.969
570.00	203958.09	360744.00	614213.4	23000.55	89.53	18.190
575.00	204614.49	358366.97	615046.1	23089.94	89.57	18.370
580.00	204364.77	355986.44	615798.2	23180.23	89.61	18.470
585.00	204489.73	353606.72	616469.8	23271.35	89.65	18.606
590.00	204469.57	351226.85	617062.8	23363.26	89.69	18.730
595.00	204445.96	348847.23	617580.9	23455.94	89.73	18.856
600.00	204402.53	346467.78	618026.3	23549.37	89.76	18.981
605.00	204362.31	344088.51	618402.8	23643.55	89.79	19.109
610.00	204311.17	341709.16	618714.6	23738.48	89.82	19.237
615.00	205009.46	339326.87	618964.8	23834.31	89.85	19.438
620.00	204490.36	336942.97	619158.4	23930.94	89.88	19.526
625.00	204407.40	334561.97	619298.6	24028.25	89.90	19.657
630.00	204313.15	332181.75	619390.5	24126.30	89.93	19.789
635.00	204234.72	325802.30	619437.4	24225.08	89.95	19.924
640.00	204174.52	327423.50	619444.4	24324.61	89.96	20.063
645.00	204104.06	325045.31	619416.7	24424.90	89.98	20.203
650.00	204026.42	322667.66	619357.7	24525.95	89.99	20.344
655.00	203949.11	320290.54	619273.0	24627.75	90.00	20.487
660.00	203871.79	317913.89	619167.0	24730.31	90.01	20.632
665.00	203791.03	315537.80	619044.1	24833.64	90.01	20.780
670.00	203710.27	313162.13	618909.0	24937.79	90.02	20.928
675.00	203635.54	310786.84	618748.9	25042.85	90.03	21.081
680.00	203566.07	308411.93	618573.5	25148.64	90.04	21.236
685.00	203498.01	306037.29	618393.9	25255.17	90.03	21.394
690.00	203432.75	303662.98	618222.3	25362.44	90.03	21.554
695.00	203389.19	301288.86	618069.3	25470.44	90.02	21.719
S-IVB GCS1	699.23	299281.49	617962.3	25562.36	90.00	21.863
700.00	9225.71	295259.20	617945.0	25567.28	90.00	0.992
705.00	141.81	295256.92	617834.3	25567.85	90.00	0.015
709.23	141.81	295256.92	617739.4	25567.92	90.00	0.015
EPØI						

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MØIØN (SEC)	**** RADIAL DISTANCE ØF THE C G (FT)	MAGNITUDE ØF THE EARTH FIXED VELOCITY (FT/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN EARTH (DEG)	ARC RANGE ALONG SPHER- EARTH TO SUB VEHICLE PT. (NM)
S-II/SIVB SEP	552.26	21523226.00	31.9215	-63.91	888.95
S-IVB ENGINE START	552.38	21523258.25	31.9225	-63.90	889.38
	555.00	21523912.75	31.9429	-63.73	898.35
S-IVB 90 THRUST	557.88	21524586.50	31.9651	-63.53	908.24
	560.00	21525059.50	31.9813	-63.39	915.51
S-IVB ULLAGE JETT	564.18	21525960.50	32.0127	-63.11	929.91
	565.00	21526130.50	32.0188	-63.06	932.72
	570.00	21527119.50	32.0554	-62.72	950.01
	575.00	21528025.50	32.0912	-62.38	967.37
	580.00	21528851.75	32.1262	-62.04	984.80
	585.00	21529598.75	32.1603	-61.70	1002.30
	590.00	21530268.50	32.1934	-61.36	1019.87
	595.00	21530864.25	32.2257	-61.02	1037.52
	600.00	21531388.75	32.2571	-60.67	1055.24
	605.00	21531846.00	32.2875	-60.32	1073.03
	610.00	21532239.25	32.3169	-59.97	1090.90
	615.00	21532572.75	32.3454	-59.62	1108.85
	620.00	21532850.25	32.3730	-59.27	1126.87
	625.00	21533076.00	32.3995	-58.91	1144.97
	630.00	21533254.50	32.4250	-58.56	1163.15
	635.00	21533389.75	32.4495	-58.20	1181.40
	640.00	21533486.25	32.4730	-57.84	1199.74
	645.00	21533549.00	32.4954	-57.47	1218.15
	650.00	21533582.50	32.5168	-57.11	1236.65
	655.00	21533591.25	32.5371	-56.75	1255.22
	660.00	21533579.75	32.5562	-56.38	1273.88
	665.00	21533553.50	32.5743	-56.01	1292.62
	670.00	21533516.50	32.5913	-55.64	1311.44
	675.00	21533455.75	32.6070	-55.27	1330.35
	680.00	21533381.00	32.6217	-54.89	1349.34
	685.00	21533303.75	32.6351	-54.52	1368.41
	690.00	21533236.00	32.6474	-54.14	1387.57
	695.00	21533188.00	32.6585	-53.76	1406.82
S-IVB GCS1	699.23	21533171.00	32.6669	-53.44	1423.16
	700.00	21533170.25	32.6683	-53.38	1426.15
	705.00	21533167.50	32.6769	-52.99	1445.52
EPØI	709.23	21533164.75	32.6831	-52.67	1461.90

**** MEASURED FROM THE CENTER ØF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (BT0)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X COORDINATE EARTH FIXED, PT. CENTERED PLUMB SYS, (FT)	Y COORDINATE EARTH FIXED, PT. CENTERED PLUMB SYS, (FT)	Z COORDINATE EARTH FIXED, PT. CENTERED PLUMB SYS, (FT)	X COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	Y COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	Z COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)
S-II/SIVB SEP						
S-IVB ENGINE START	552.26	-96234.005	88517.558	5498023.375	-5212.980	20801.548
	552.38	-96885.867	88590.287	5500623.500	-5216.329	20800.724
	555.00	-110623.359	90118.342	5555015.563	-5286.393	20783.400
S-IVB 90 THRUST	557.88	-125981.381	91815.462	5614936.375	-5364.934	20779.886
	560.00	-137396.447	93669.972	5658927.313	-5423.981	20799.834
S-IVB ULLAGE JETT	564.18	-160332.594	95577.076	5746042.875	-5540.507	20842.763
	565.00	-164863.229	96669.862	5763054.000	-5563.996	20851.219
	570.00	-193048.924	99115.502	5867439.500	-5710.055	20903.053
	575.00	-221961.941	102208.674	5972085.125	-5855.355	20955.298
	580.00	-251605.766	105250.426	6076991.688	-6002.308	21007.254
	585.00	-281986.750	108542.486	6182157.188	-6150.218	21059.026
	590.00	-313108.414	111785.486	6287581.188	-6299.018	21110.576
	595.00	-344978.059	115080.495	6393262.313	-6448.605	21161.917
	600.00	-377596.578	118427.751	6459199.625	-6598.910	21213.043
	605.00	-410968.254	121828.001	6605392.125	-6749.906	21263.963
	610.00	-445096.988	125281.531	6711838.563	-6901.573	21314.672
	615.00	-479985.438	128788.726	6818538.188	-7053.936	21365.333
	620.00	-515637.504	132349.836	6925490.938	-7206.990	21415.816
	625.00	-552056.469	135865.393	7032695.375	-7360.728	21466.011
	630.00	-589245.609	139835.330	7140150.188	-7515.067	21515.992
	635.00	-627208.203	143360.031	7247854.313	-7670.036	21565.754
	640.00	-665947.336	147139.627	7355806.750	-7825.641	21615.308
	645.00	-705465.461	150574.641	7464006.500	-7981.900	21664.651
	650.00	-745767.094	154865.043	7572452.563	-8138.817	21713.776
	655.00	-786854.695	158811.145	7681143.625	-8296.390	21762.683
	660.00	-828732.156	162813.242	7750078.500	-8454.636	21811.368
	665.00	-871401.938	166871.541	7859256.313	-8613.563	21859.830
	670.00	-914868.969	170586.379	8008675.813	-8774.353	21907.850
	675.00	-959152.922	175158.635	8118329.813	-8938.727	21954.196
	680.00	-1004253.195	179388.904	8228218.438	-9101.453	22001.379
	685.00	-1050166.703	183677.145	8338343.313	-9263.934	22048.677
	690.00	-1096891.391	188023.680	8448705.000	-9425.939	22096.184
	695.00	-1144425.234	192428.576	8559305.000	-9587.598	22143.866
S-IVB GCS1	699.23	-1185249.750	196198.902	8653012.500	-9724.259	22184.258
	700.00	-1192766.359	196891.775	8670140.000	-9745.205	22180.381
	705.00	-1241804.859	201403.766	8780905.375	-9870.195	22125.327
EP01	709.23	-1283757.781	205251.482	8874349.125	-9975.497	22077.825

TABLE 2-2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTØR (LB)	AERODYNAMIC LOAD (LB/FT**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTOR (FT)
S-II/SIVB SEP	8.92	0.00	9.8777	0.0261	0.0022	0.0384
S-IVB ENGINE START	8.92	0.00	9.8752	0.0261	0.0022	0.0384
	8.92	0.00	9.8310	0.0259	0.0021	0.0379
S-IVB 90 THRUST	8.92	0.00	9.9757	0.0258	0.0021	0.0386
	8.93	0.00	10.6532	0.0261	0.0023	0.0423
S-IVB ULLAGE JETT	8.95	0.00	8.7334	0.0253	0.0019	0.0316
	8.96	0.00	7.8528	0.0250	0.0017	0.0270
	8.98	0.00	7.3488	0.0249	0.0016	0.0253
	9.02	0.00	7.0393	0.0249	0.0015	0.0242
	9.05	0.00	6.4538	0.0249	0.0014	0.0222
	9.08	0.00	6.1274	0.0249	0.0013	0.0211
	9.11	0.00	5.8362	0.0250	0.0013	0.0202
	9.15	0.00	5.6133	0.0250	0.0012	0.0195
	9.18	0.00	5.4186	0.0251	0.0012	0.0189
	9.22	0.00	5.2411	0.0252	0.0011	0.0183
	9.26	0.00	5.0774	0.0253	0.0011	0.0178
	9.29	0.00	4.9204	0.0255	0.0011	0.0174
	9.33	0.00	4.7293	0.0256	0.0010	0.0168
	9.37	0.00	4.5974	0.0258	0.0010	0.0164
	9.41	0.00	4.4712	0.0259	0.0010	0.0161
	9.45	0.00	4.3420	0.0261	0.0010	0.0157
	9.49	0.00	4.2089	0.0263	0.0010	0.0154
	9.53	0.00	4.0706	0.0265	0.0009	0.0150
	9.57	0.00	3.9364	0.0267	0.0009	0.0146
	9.62	0.00	3.8008	0.0270	0.0009	0.0142
	9.66	0.00	3.6585	0.0272	0.0009	0.0138
	9.70	0.00	3.5179	0.0274	0.0008	0.0134
	9.74	0.00	1.0913	0.0275	0.0003	0.0042
	9.79	0.00	1.7746	0.0278	0.0004	0.0069
	9.83	0.00	2.3652	0.0281	0.0006	0.0093
	9.88	0.00	2.8771	0.0284	0.0007	0.0114
	9.92	0.00	3.3744	0.0287	0.0008	0.0135
	9.97	0.00	3.7303	0.0290	0.0009	0.0150
S-IVB GCSI	10.00	0.00	4.0009	0.0292	0.0010	0.0162
	10.01	0.00	4.0552	0.0293	0.0010	0.0165
	10.01	0.00	4.4125	0.0293	0.0011	0.0179
EPØI	10.01	0.00	4.7162	0.0293	0.0012	0.0192

TABLE 2.2 (BTØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG

TIME FROM FIRST MOTION (SEC)	X POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Y POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Z POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)
S-IVB SEP						
552.26	20608528.0	261040.9	6202395.1	-6318.154	297.256	21879.317
S-IVB ENGINE START						
552.38	20607738.0	261078.0	6205129.9	-6321.799	297.215	21878.289
555.00	20591100.3	261854.4	6262335.4	-6398.051	296.361	21856.688
S-IVB 90 THRUST						
557.88	20572525.0	262707.7	6325344.9	-6483.949	295.363	21848.366
560.00	20558737.0	263332.0	6371592.7	-6549.190	294.711	21864.701
564.18	20531066.8	264562.8	6463148.6	-6678.106	293.473	21900.424
S-IVB ULLAGE JETT						
565.00	20525606.8	264802.2	6481022.3	-6704.016	293.061	21907.433
570.00	20491684.0	266261.4	6590666.3	-6864.934	290.801	21950.253
575.00	20456959.3	267710.8	6700525.3	-7025.155	288.940	21993.382
580.00	20421429.3	269151.1	6810598.9	-7187.063	287.244	22036.026
585.00	20385087.0	270583.5	6920884.8	-7349.965	285.713	22078.317
590.00	20347928.0	272008.4	7031381.1	-7513.792	284.268	22120.214
595.00	20309947.8	273426.3	7142085.9	-7678.440	282.883	22161.734
600.00	20271142.5	274837.3	7252997.5	-7843.839	281.529	22202.872
605.00	20231508.3	276241.6	7364113.8	-8009.963	280.192	22243.635
610.00	20191041.5	277639.2	7475432.8	-8176.791	278.856	22284.021
615.00	20149739.0	279030.1	7586953.1	-8344.352	277.510	22324.187
620.00	20107596.8	280414.3	7698673.5	-8512.638	276.150	22364.005
625.00	20064611.3	281791.6	7810591.8	-8681.836	274.775	22403.364
630.00	20020779.5	283162.0	7922705.9	-8851.267	273.373	22442.340
635.00	19976097.8	284525.3	8035014.0	-9021.557	271.945	22480.925
640.00	19930562.8	285881.4	8147514.0	-9192.514	270.491	22519.130
645.00	19884171.5	287230.2	8260204.2	-9364.155	269.012	22556.951
650.00	19836920.3	288571.5	8373082.5	-9536.483	267.507	22594.378
655.00	19788805.5	289905.2	8486146.9	-9709.495	265.979	22631.414
660.00	19739824.0	291231.2	8599395.4	-9883.209	264.428	22668.050
665.00	19689972.5	292549.4	8712826.3	-10057.632	262.855	22704.286
670.00	19639246.5	293859.8	8826437.0	-10233.927	261.318	22739.653
675.00	19587626.3	295162.8	8940219.1	-10413.788	259.890	22773.448
680.00	19535111.5	296458.6	9054172.0	-10592.064	258.398	22807.792
685.00	19481706.0	297746.8	9168296.5	-10770.136	256.883	22842.108
690.00	19427411.3	299027.3	9282592.9	-10947.778	255.332	22876.500
695.00	19372229.0	300300.0	9397061.1	-11125.117	253.735	22910.928
S-IVB GCS1						
699.23	19324876.0	301369.9	9493988.0	-11275.068	252.388	22939.994
700.00	19316161.5	301564.7	9511698.1	-11297.954	252.107	22934.219
705.00	19259331.0	302820.4	9626203.0	-11434.163	250.189	22867.280
709.23	19210745.5	303874.8	9722762.3	-11548.866	248.560	22809.655
EPØ1						

TABLE 2.2 (EPØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MØTION (SEC)	TOTAL VEHICLE ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE ØF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
709.00	630.81	135740.66	188287.0	7793.10	90.00	0.005
712.00	630.81	135739.98	188267.7	7793.11	90.00	0.005
717.00	630.81	135738.75	188233.0	7793.14	90.00	0.005
721.00	630.81	135737.77	188204.8	7793.16	90.00	0.005
726.00	630.81	135736.54	188169.1	7793.18	90.00	0.005
731.00	630.81	135735.31	188133.2	7793.20	90.00	0.005
736.00	630.81	135734.08	188096.8	7793.23	90.00	0.005
741.00	630.81	135732.86	188060.2	7793.25	90.00	0.005
746.00	630.81	135731.63	188023.2	7793.28	90.00	0.005
751.00	630.81	135730.40	187985.9	7793.30	90.00	0.005
756.00	630.81	135729.17	187948.1	7793.33	90.00	0.005
761.00	830.98	135727.94	187910.0	7793.36	90.00	0.006
766.00	830.98	135726.71	187871.6	7793.39	90.00	0.006
771.00	830.98	135725.48	187832.8	7793.43	90.00	0.006
776.00	830.98	135724.26	187793.8	7793.46	90.00	0.006
781.00	830.98	135723.03	187754.2	7793.49	90.00	0.006
786.00	830.98	135721.80	187714.5	7793.53	90.00	0.006
791.00	825.14	135720.57	187674.5	7793.56	90.00	0.006
796.00	817.18	135719.34	187634.0	7793.60	90.00	0.006
927.00	608.66	135687.16	186475.4	7794.49	90.00	0.004
1247.00	190.77	135608.56	183197.0	7796.33	90.00	0.001
1567.00	187.96	135529.95	180343.6	7797.90	90.00	0.001
1887.00	185.15	135447.32	179231.1	7798.96	89.99	0.001
2207.00	158.61	135371.13	180688.2	7799.02	89.99	0.001
2527.00	126.35	135319.02	184622.3	7797.97	89.99	0.001
2847.00	104.38	135266.91	190013.4	7796.15	89.98	0.001
3167.00	94.42	135214.80	195363.0	7794.16	89.97	0.001
3487.00	90.85	135162.70	199355.4	7792.55	89.96	0.001
3807.00	86.47	135110.59	201381.7	7791.55	89.96	0.001
4127.00	79.07	135067.73	201676.5	7791.07	89.97	0.001
4447.00	71.67	135029.51	201024.4	7790.79	89.98	0.001
4767.00	65.59	134991.29	200219.4	7790.46	89.99	0.000
5087.00	64.45	134954.30	199588.1	7790.02	90.01	0.000
5407.00	63.32	134921.46	198848.9	7789.70	90.02	0.000
5727.00	62.28	134888.63	197379.8	7789.85	90.02	0.000
6047.00	62.28	134855.74	194726.9	7790.75	90.02	0.000
6367.00	62.28	134822.43	191049.5	7792.36	90.01	0.000
6687.00	62.28	134789.13	187235.9	7794.29	90.01	0.000
7007.00	60.55	134755.82	184599.2	7795.91	90.00	0.000
7327.00	58.76	134722.32	184288.4	7796.65	90.00	0.000
7647.00	56.96	134688.82	186713.5	7796.23	90.00	0.000
7967.00	55.56	134655.32	191300.1	7794.81	89.99	0.000
8287.00	55.38	134625.05	196719.5	7792.88	89.98	0.000
8383.00	55.33	134616.13	198276.1	7792.30	89.98	0.000
8423.00	55.31	134612.42	198897.3	7792.06	89.98	0.000
8463.00	55.29	134608.70	199498.8	7791.83	89.97	0.000
8503.00	55.27	134604.99	200078.6	7791.60	89.97	0.000
8543.00	55.24	134601.27	200634.5	7791.38	89.97	0.000
8583.00	55.22	134597.56	201165.3	7791.17	89.97	0.000

TABLE 2.2 (EPØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE ØF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
8623.00	55.20	134593.84	201668.9	7790.97	89.97	0.000
8663.00	55.18	134590.13	202144.2	7790.77	89.97	0.000
8703.00	55.13	134586.41	202590.2	7790.58	89.97	0.000
8743.00	54.79	134582.70	203005.8	7790.40	89.97	0.000
8783.00	54.45	134578.98	203390.3	7790.23	89.97	0.000
8823.00	54.11	134575.27	203743.8	7790.08	89.96	0.000
8863.00	53.77	134571.55	204065.4	7789.93	89.96	0.000
8903.00	53.44	134567.83	204355.4	7789.79	89.96	0.000
8943.00	53.10	134564.12	204614.0	7789.66	89.96	0.000
8983.00	52.76	134560.40	204841.7	7789.54	89.96	0.000
9023.00	52.42	134554.68	205039.1	7789.43	89.96	0.000
9063.00	52.08	134542.06	205207.0	7789.32	89.96	0.000
9103.00	51.75	134529.44	205346.4	7789.23	89.96	0.000
9143.00	51.41	134516.82	205458.3	7789.14	89.96	0.000
9183.00	51.07	134504.20	205544.1	7789.07	89.96	0.000
9223.00	50.73	134491.58	205605.0	7789.00	89.96	0.000
9263.00	50.39	134478.96	205642.9	7788.93	89.96	0.000
9267.00	50.36	134477.70	205645.2	7788.93	89.96	0.000
9272.00	50.32	134476.12	205648.0	7788.92	89.96	0.000
9277.00	50.27	134474.54	205650.4	7788.91	89.96	0.000

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AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	*** RADIAL DISTANCE ØF THE C G (M)	MAGNITUDE ØF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN EARTH (DEG)	ARC RANGE ALONG SPHER- EARTH TO SUB VEHICLE PT. (KM)
709.00	6563308.69	0.000	32.6831	-52.69	1461.02
712.00	6563308.00	0.000	32.6867	-52.46	1472.63
717.00	6563306.75	0.000	32.6922	-52.08	1492.00
721.00	6563305.81	0.000	32.6956	-51.77	1507.49
726.00	6563304.44	0.000	32.6987	-51.39	1526.86
731.00	6563303.13	0.000	32.7006	-51.01	1546.23
736.00	6563301.81	0.000	32.7011	-50.62	1565.60
741.00	6563300.38	0.000	32.7003	-50.24	1584.97
746.00	6563298.94	0.000	32.6983	-49.86	1604.33
751.00	6563297.50	0.000	32.6950	-49.48	1623.70
756.00	6563296.00	0.000	32.6903	-49.09	1643.07
761.00	6563294.50	0.000	32.6844	-48.71	1662.44
766.00	6563292.94	0.000	32.6772	-48.33	1681.81
771.00	6563291.38	0.000	32.6687	-47.95	1701.17
776.00	6563289.75	0.000	32.6589	-47.56	1720.54
781.00	6563288.13	0.000	32.6478	-47.18	1739.91
786.00	6563286.50	0.000	32.6354	-46.80	1759.28
791.00	6563284.75	0.000	32.6217	-46.42	1778.65
796.00	6563283.13	0.000	32.6067	-46.04	1798.02
927.00	6563240.88	0.000	31.7613	-36.11	2305.51
1247.00	6563214.19	0.000	26.3404	-13.29	3545.60
1567.00	6563347.31	0.000	17.3049	6.77	4786.15
1887.00	6563586.00	0.000	6.2205	24.66	6026.57
2207.00	6563896.94	0.000	-5.5325	41.75	7265.83
2527.00	6564330.81	0.000	-16.6916	59.55	8502.70
2847.00	6565006.50	0.000	-25.8889	79.45	9734.16
3167.00	6566039.75	0.000	-31.5659	102.11	10583.80
3487.00	6567452.88	0.000	-32.3924	126.42	9395.13
3807.00	6569114.06	0.000	-28.1466	149.80	8163.88
4127.00	6570749.00	0.000	-19.9072	170.54	6931.35
4447.00	6572026.94	0.000	-9.2378	-171.16	5698.41
4767.00	6572685.50	0.000	2.4346	-154.05	4465.25
5087.00	6572636.06	0.000	13.8488	-136.64	3232.75
5407.00	6571996.44	0.000	23.6965	-117.45	2003.21
5727.00	6571033.88	0.000	30.4664	-95.56	790.05
6047.00	6570050.06	0.000	32.6879	-71.50	530.65
6367.00	6569266.69	0.000	29.7589	-47.62	1727.80
6687.00	6568768.31	0.000	22.4510	-26.10	2956.32
7007.00	6568523.88	0.000	12.2803	-7.23	4189.15
7327.00	6568466.25	0.000	0.7357	10.08	5422.97
7647.00	6568576.63	0.000	-10.8868	27.29	6656.02
7967.00	6568920.81	0.000	-21.3057	45.90	7886.83
8287.00	6569613.06	0.000	-29.0579	67.07	9112.61
8383.00	6569902.19	0.000	-30.6255	73.96	9477.70
8423.00	6570034.25	0.000	-31.1565	76.89	9628.95
8463.00	6570173.00	0.000	-31.6122	79.86	9779.35
8503.00	6570318.31	0.000	-31.9905	82.85	9928.48
8543.00	6570470.13	0.000	-32.2896	85.87	10075.54
8583.00	6570628.31	0.000	-32.5081	88.91	10218.99

**** MEASURED FROM THE CENTER ØF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (EPØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	*** RADIAL DISTANCE ØF THE C G (M)	MAGNITUDE ØF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN ØBLATE EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (KM)
8623.00	6570792.69	0.000	-32.6450	91.96	10355.38
8663.00	6570962.81	0.000	-32.6996	95.01	10475.95
8703.00	6571138.69	0.000	-32.6715	98.06	10556.95
8743.00	6571319.63	0.000	-32.5611	101.11	10556.98
8783.00	6571505.38	0.000	-32.3689	104.15	10476.06
8823.00	6571695.56	0.000	-32.0957	107.18	10355.58
8863.00	6571889.69	0.000	-31.7429	110.18	10219.28
8903.00	6572087.31	0.000	-31.3123	113.15	10075.93
8943.00	6572287.69	0.000	-30.8057	116.10	9928.99
8983.00	6572490.38	0.000	-30.2254	119.00	9779.99
9023.00	6572694.94	0.000	-29.5740	121.87	9629.72
9063.00	6572900.56	0.000	-28.8542	124.70	9478.61
9103.00	6573106.69	0.000	-28.0688	127.49	9326.92
9143.00	6573312.50	0.000	-27.2209	130.23	9174.82
9183.00	6573517.56	0.000	-26.3136	132.93	9022.41
9223.00	6573720.81	0.000	-25.3501	135.58	8869.76
9263.00	6573921.94	0.000	-24.3334	138.19	8716.92
9267.00	6573941.75	0.000	-24.2290	138.45	8701.63
9272.00	6573966.50	0.000	-24.0977	138.77	8682.51
9277.00	6573991.19	0.000	-23.9656	139.09	8663.39

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**** MEASURED FROM THE CENTER ØF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	X CØØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Y CØØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Z CØØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	COMPONENT ØF VEL. IN THE XXE SYSTEM (M/S)	COMPONENT ØF VEL. IN THE XXE SYSTEM (M/S)	COMPONENT ØF VEL. IN THE XXE SYSTEM (M/S)	Z
709.00	5982673.813	79577.352	2697840.844	-3038.794	278.332	6730.107	
712.00	5973521.875	80421.017	2718010.219	-3061.538	279.867	6719.745	
717.00	5958119.438	81826.658	2751565.375	-3099.352	282.391	6702.306	
721.00	5945661.438	82560.229	2778346.406	-3129.532	284.405	6688.202	
726.00	5929919.375	84388.533	2811742.844	-3167.170	286.914	6670.379	
731.00	5913989.500	85829.382	2845049.719	-3204.707	289.414	6652.344	
736.00	5897872.188	87282.659	2878265.875	-3242.142	291.905	6634.098	
741.00	5881567.875	88748.404	2911390.188	-3279.474	294.386	6615.640	
746.00	5865077.250	90226.499	2944421.781	-3316.702	296.859	6596.973	
751.00	5848400.750	91716.969	2977359.500	-3353.825	299.321	6578.095	
756.00	5831538.875	93219.718	3010202.250	-3390.841	301.774	6559.009	
761.00	5814492.188	94734.696	3042949.094	-3427.752	304.217	6539.717	
766.00	5797261.188	96261.853	3075598.938	-3464.556	306.650	6520.221	
771.00	5779846.438	97801.139	3108150.781	-3501.249	309.072	6500.517	
776.00	5762248.563	99352.533	3140603.625	-3537.832	311.484	6480.606	
781.00	5744468.063	100915.975	3172956.406	-3574.302	313.885	6460.489	
786.00	5726505.438	102491.359	3205208.063	-3610.659	316.275	6440.167	
791.00	5708361.375	104078.740	3237357.625	-3646.901	318.654	6419.640	
796.00	5690036.250	105677.894	3269404.000	-3683.027	321.022	6398.908	
927.00	5147525.438	151595.191	4068922.563	-4584.539	378.422	5785.239	
1247.00	3381560.906	289091.332	5617583.938	-6332.984	466.579	3788.405	
1567.00	1181564.625	439312.668	6441151.750	-7267.878	453.260	1302.932	
1887.00	-1169820.984	566411.211	6433611.500	-7268.808	321.077	-1349.078	
2207.00	-3370399.781	632749.914	5556853.938	-6335.693	76.752	-3822.695	
2527.00	-5137617.938	606854.133	4040674.063	-4590.096	-248.230	-5796.211	
2847.00	-6245320.750	470782.488	1968156.891	-2258.818	-601.863	-7014.915	
3167.00	-6552748.813	225441.758	-351476.199	357.366	-920.021	-7324.571	
3487.00	-6021956.313	-107246.084	-2618391.875	2923.678	-1137.432	-6689.815	
3807.00	-4721911.000	-485857.332	-4541008.375	5113.664	-1199.403	-5196.749	
4127.00	-2819309.688	-854612.648	-5873318.563	6649.283	-1072.512	-3041.010	
4447.00	-557376.711	-115153.172	-6446300.688	7335.380	-752.569	-503.000	
4767.00	1774998.375	-1318608.563	-6189575.500	7084.916	-268.233	2087.898	
5087.00	3880166.594	-1311560.922	-5140292.688	5930.861	320.876	4396.549	
5407.00	5489874.625	-1110892.266	-3437775.906	4021.853	931.678	6126.623	
5727.00	6399441.250	-723469.961	-1304694.422	1602.082	1469.613	7059.241	
6047.00	6493460.688	-187549.248	982445.734	-1020.719	1843.058	7080.011	
6367.00	5760061.750	430539.488	3129151.219	-3513.920	1978.032	6191.957	
6687.00	4292368.063	1047787.672	4860702.875	-5560.706	1831.209	4514.263	
7007.00	2277061.125	1571553.188	5957303.063	-6899.409	1399.387	2267.420	
7327.00	-28925.053	1917211.500	6282373.000	-7357.623	723.497	-254.593	
7647.00	-2331132.375	2017601.375	5800112.375	-6876.056	-114.366	-2723.040	
7967.00	-4335606.563	1838742.453	4579548.125	-5517.131	-1001.169	-4819.026	
8287.00	-5786886.563	1385302.688	2784366.063	-3456.123	-1808.889	-6276.428	
8383.00	-6084519.375	1201510.188	2167627.906	-2738.611	-2016.785	-6559.611	
8423.00	-6187905.938	1119217.094	1903292.375	-2429.859	-2097.226	-6654.905	
8463.00	-6278843.125	1033784.695	1635417.063	-2116.255	-2173.719	-6736.580	
8503.00	-6357149.688	945374.938	1364549.500	-1798.428	-2246.059	-6804.490	
8543.00	-6422669.188	854157.914	1051242.844	-1477.015	-2314.051	-6858.520	
8583.00	-6475271.250	760311.305	816054.125	-1152.661	-2377.508	-6898.581	

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST OPP

TIME FROM FIRST MOTION (SEC)	X COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Y COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Z COORDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	X COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)	Y COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)	Z COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)
8623.00	-6514851.125	664020.102	539543.438	-826.015	-2436.252	-6924.614
8663.00	-6541330.188	565476.227	262272.453	-497.732	-2490.114	-6936.587
8703.00	-6554656.250	464878.176	-15196.050	-168.470	-2538.935	-6934.497
8743.00	-6554803.125	362430.586	-252300.148	161.114	-2582.566	-6918.370
8783.00	-6541771.250	258243.857	-568479.242	490.359	-2620.869	-6888.258
8823.00	-6515587.313	152833.732	-843175.492	818.607	-2653.716	-6844.245
8863.00	-6476304.313	46120.878	-1115834.906	1145.201	-2680.991	-6786.438
8903.00	-6424001.625	-61569.798	-1385908.359	1469.489	-2702.590	-6714.977
8943.00	-6358784.250	-170009.395	-1652852.969	1790.820	-2718.421	-6630.023
8983.00	-6280783.500	-278965.457	-1916132.813	2108.552	-2728.402	-6531.771
9023.00	-6190156.063	-388202.652	-2175220.063	2422.051	-2732.468	-6420.436
9063.00	-6087084.063	-497483.273	-2429596.375	2730.886	-2730.563	-6296.264
9103.00	-5971774.063	-606567.547	-2678753.500	3033.842	-2722.646	-6159.523
9143.00	-5844457.563	-715214.367	-2922194.469	3330.910	-2708.687	-6010.509
9183.00	-5705390.000	-823181.672	-3159434.656	3621.296	-2688.672	-5849.540
9223.00	-5554850.438	-930227.281	-3350002.656	3904.415	-2662.600	-5676.960
9263.00	-5393140.875	-1036108.977	-3613441.156	4179.702	-2630.483	-5493.133
9267.00	-5376367.750	-1046623.852	-3635375.656	4206.779	-2626.939	-5474.146
9272.00	-5355249.375	-1059747.266	-3662686.594	4240.505	-2622.425	-5450.261
9277.00	-5333962.625	-1072647.906	-3689877.906	4274.099	-2617.817	-5426.208

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AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	COMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE ØF IN PITCH (DEG)	COMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE ØF IN YAW (DEG)	COMMANDED ROLL ATTITUDE (DEG)	EULERIAN ANGLE ØF IN ROLL (DEG)
709.00	-112.139	-112.1385	0.361	0.361217	0.775	0.77519783
712.00	-112.139	-112.1385	0.361	0.361217	0.775	0.77519783
717.00	-112.139	-112.1385	0.361	0.361217	0.775	0.77519783
721.00	-117.653	-117.6531	0.544	0.544211	0.000	0.00000000
726.00	-117.993	-117.9933	0.539	0.539355	0.000	0.00000000
731.00	-118.333	-118.3334	0.534	0.534481	0.000	0.00000000
736.00	-118.674	-118.6736	0.530	0.529588	0.000	0.00000000
741.00	-119.014	-119.0137	0.525	0.524677	0.000	0.00000000
746.00	-119.354	-119.3538	0.520	0.519746	0.000	0.00000000
751.00	-119.694	-119.6940	0.515	0.514798	0.000	0.00000000
756.00	-120.034	-120.0341	0.510	0.509831	0.000	0.00000000
761.00	-120.374	-120.3743	0.505	0.504846	0.000	0.00000000
766.00	-120.714	-120.7144	0.500	0.499844	0.000	0.00000000
771.00	-121.055	-121.0546	0.495	0.494824	0.000	0.00000000
776.00	-121.395	-121.3947	0.490	0.489786	0.000	0.00000000
781.00	-121.735	-121.7349	0.485	0.484731	0.000	0.00000000
786.00	-122.075	-122.0750	0.480	0.479659	0.000	0.00000000
791.00	-122.415	-122.4152	0.475	0.474571	0.000	0.00000000
796.00	-122.755	-122.7553	0.469	0.469465	0.000	0.00000000
927.00	-131.668	-131.6677	0.330	0.330342	0.000	0.00000000
1247.00	-153.442	-153.4417	-0.036	-0.035900	0.000	0.00000000
1567.00	-175.220	-175.2204	-0.397	-0.397090	0.000	0.00000000
1887.00	162.996	162.9960	-0.702	-0.701630	0.000	0.00000000
2207.00	141.209	141.2094	-0.906	-0.905970	0.000	0.00000000
2527.00	119.424	119.4238	-0.981	-0.980896	0.000	0.00000000
2847.00	97.644	97.6441	-0.916	-0.915770	0.000	0.00000000
3167.00	75.874	75.8742	-0.720	-0.720030	0.000	0.00000000
3487.00	54.116	54.1158	-0.422	-0.421747	0.000	0.00000000
3807.00	32.368	32.3680	-0.064	-0.063528	0.000	0.00000000
4127.00	10.628	10.6281	0.304	0.303611	0.000	0.00000000
4447.00	-11.107	-11.1074	0.627	0.627481	0.000	0.00000000
4767.00	-32.841	-32.8411	0.862	0.862092	0.000	0.00000000
5087.00	-54.575	-54.5747	0.974	0.974141	0.000	0.00000000
5407.00	-76.309	-76.3089	0.948	0.947720	0.000	0.00000000
5727.00	-98.044	-98.0441	0.787	0.786562	0.000	0.00000000
6047.00	-119.782	-119.7818	0.514	0.513517	0.000	0.00000000
6367.00	-141.524	-141.5240	0.167	0.167338	0.000	0.00000000
6687.00	-163.273	-163.2726	-0.203	-0.202769	0.000	0.00000000
7007.00	174.971	174.9714	-0.544	-0.544096	0.000	0.00000000
7327.00	153.209	153.2091	-0.808	-0.807947	0.000	0.00000000
7647.00	131.444	131.4442	-0.957	-0.956640	0.000	0.00000000
7967.00	109.682	109.6819	-0.969	-0.968971	0.000	0.00000000
8287.00	87.927	87.9270	-0.843	-0.843273	0.000	0.00000000
8383.00	81.402	81.4025	-0.781	-0.780863	0.000	0.00000000
8423.00	78.684	78.6843	-0.752	-0.751826	0.000	0.00000000
8463.00	75.966	75.9662	-0.721	-0.721098	0.000	0.00000000
8503.00	73.248	73.2484	-0.689	-0.688750	0.000	0.00000000
8543.00	70.531	70.5307	-0.655	-0.654855	0.000	0.00000000
8583.00	67.813	67.8132	-0.619	-0.619488	0.000	0.00000000

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	COMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN PITCH (DEG)	COMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN YAW (DEG)	COMMANDED ROLL ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN ROLL (DEG)
8623.00	65.096	65.0958	-0.583	-0.582730	0.000	0.00000000
8663.00	62.379	62.3787	-0.545	-0.544664	0.000	0.00000000
8703.00	59.662	59.6617	-0.505	-0.505375	0.000	0.00000000
8743.00	56.945	56.9449	-0.465	-0.464951	0.000	0.00000000
8783.00	54.228	54.2283	-0.423	-0.423485	0.000	0.00000000
8823.00	51.512	51.5118	-0.381	-0.381069	0.000	0.00000000
8863.00	48.795	48.7955	-0.338	-0.337799	0.000	0.00000000
8903.00	46.079	46.0793	-0.294	-0.293772	0.000	0.00000000
8943.00	43.363	43.3633	-0.249	-0.249087	0.000	0.00000000
8983.00	40.647	40.6474	-0.204	-0.203844	0.000	0.00000000
9023.00	37.932	37.9317	-0.158	-0.158146	0.000	0.00000000
9063.00	35.216	35.2161	-0.112	-0.112095	0.000	0.00000000
9103.00	32.501	32.5006	-0.066	-0.065793	0.000	0.00000000
9143.00	29.785	29.7852	-0.019	-0.019346	0.000	0.00000000
9183.00	27.070	27.0700	0.027	0.027142	0.000	0.00000000
9223.00	24.355	24.3548	0.074	0.073568	0.000	0.00000000
9263.00	21.640	21.6398	0.120	0.119826	0.000	0.00000000
9267.00	21.368	21.3683	0.124	0.124439	0.000	0.00000000
9272.00	21.029	21.0289	0.130	0.130201	0.000	0.00000000
9277.00	20.690	20.6895	0.136	0.135958	0.000	0.00000000

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TABLE 2.2 (EPØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MØTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTØR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTOR (KG)
709.00	10.01	0.01	4.7162	1.0953	0.0000	0.0000
712.00	10.01	0.01	4.9049	1.0953	0.0000	0.0000
717.00	10.01	0.01	5.2454	1.0953	0.0000	0.0000
721.00	10.01	0.01	0.0321	1.0953	0.0000	0.0000
726.00	10.01	0.01	0.0198	1.0953	0.0000	0.0000
731.00	10.01	0.01	0.0076	1.0953	0.0000	0.0000
736.00	10.01	0.01	0.0053	1.0953	0.0000	0.0000
741.00	10.01	0.01	0.0174	1.0954	0.0000	0.0000
746.00	10.01	0.01	0.0297	1.0955	0.0000	0.0000
751.00	10.01	0.01	0.0420	1.0956	0.0000	0.0000
756.00	10.01	0.01	0.0544	1.0957	0.0000	0.0000
761.00	10.01	0.01	0.0667	1.0958	0.0000	0.0000
766.00	10.01	0.01	0.0790	1.0959	0.0000	0.0000
771.00	10.01	0.01	0.0914	1.0961	0.0000	0.0000
776.00	10.01	0.01	0.1037	1.0962	0.0000	0.0000
781.00	10.01	0.01	0.1160	1.0964	0.0000	0.0000
786.00	10.01	0.01	0.1284	1.0966	0.0000	0.0000
791.00	10.01	0.01	0.1407	1.0968	0.0000	0.0000
796.00	10.01	0.01	0.1530	1.0970	0.0000	0.0000
927.00	10.01	0.01	0.4722	1.1071	0.0000	0.0000
1247.00	10.06	0.01	1.1864	1.1636	0.0000	0.0000
1567.00	10.12	0.01	1.7240	1.2410	0.0000	0.0000
1887.00	10.16	0.01	2.0056	1.2939	0.0000	0.0000
2207.00	10.15	0.01	1.9895	1.2840	0.0000	0.0000
2527.00	10.10	0.01	1.6781	1.2102	0.0000	0.0000
2847.00	10.02	0.01	1.1173	1.1103	0.0000	0.0000
3167.00	9.96	0.01	0.3906	1.0275	0.0000	0.0000
3487.00	9.93	0.01	0.3989	0.9811	0.0000	0.0000
3807.00	9.93	0.01	1.1265	0.9735	0.0000	0.0000
4127.00	9.94	0.01	1.6888	0.9906	0.0000	0.0000
4447.00	9.95	0.01	2.0018	1.0080	0.0000	0.0000
4767.00	9.95	0.01	2.0191	1.0040	0.0000	0.0000
5087.00	9.93	0.01	1.7378	0.9735	0.0000	0.0000
5407.00	9.90	0.01	1.1989	0.9328	0.0000	0.0000
5727.00	9.87	0.01	0.4817	0.9062	0.0000	0.0000
6047.00	9.88	0.01	0.3100	0.9123	0.0000	0.0000
6367.00	9.91	0.01	1.0550	0.9565	0.0000	0.0000
6687.00	9.97	0.01	1.6453	1.0281	0.0000	0.0000
7007.00	10.01	0.01	1.9926	1.0979	0.0000	0.0000
7327.00	10.04	0.01	2.0454	1.1293	0.0000	0.0000
7647.00	10.02	0.01	1.7953	1.1026	0.0000	0.0000
7967.00	9.97	0.01	1.2787	1.0322	0.0000	0.0000
8287.00	9.91	0.01	0.5715	0.9535	0.0000	0.0000
8383.00	9.90	0.01	0.3376	0.9333	0.0000	0.0000
8423.00	9.89	0.01	0.2388	0.9256	0.0000	0.0000
8463.00	9.89	0.01	0.1402	0.9184	0.0000	0.0000
8503.00	9.88	0.01	0.0471	0.9117	0.0000	0.0000
8543.00	9.88	0.01	0.0699	0.9055	0.0000	0.0000
8583.00	9.87	0.01	0.1663	0.8999	0.0000	0.0000

TABLE 2.2 (EPØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST NOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTØR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTØR (KG)
8623.00	9.87	0.01	0.2653	0.8948	0.0000	0.0000
8663.00	9.86	0.01	0.3642	0.8903	0.0000	0.0000
8703.00	9.86	0.01	0.4624	0.8863	0.0000	0.0000
8743.00	9.86	0.01	0.5598	0.8829	0.0000	0.0000
8783.00	9.85	0.01	0.6559	0.8800	0.0000	0.0000
8823.00	9.85	0.01	0.7505	0.8777	0.0000	0.0000
8863.00	9.85	0.01	0.8434	0.8759	0.0000	0.0000
8903.00	9.85	0.01	0.9345	0.8746	0.0000	0.0000
8943.00	9.85	0.01	1.0233	0.8738	0.0000	0.0000
8983.00	9.85	0.01	1.1099	0.8735	0.0000	0.0000
9023.00	9.85	0.01	1.1938	0.8736	0.0000	0.0000
9063.00	9.85	0.01	1.2751	0.8742	0.0000	0.0000
9103.00	9.85	0.01	1.3533	0.8751	0.0000	0.0000
9143.00	9.85	0.01	1.4285	0.8763	0.0000	0.0000
9183.00	9.85	0.01	1.5003	0.8779	0.0000	0.0000
9223.00	9.86	0.01	1.5687	0.8797	0.0000	0.0000
9263.00	9.86	0.01	1.6334	0.8818	0.0000	0.0000
9267.00	9.86	0.01	1.6396	0.8820	0.0000	0.0000
9272.00	9.86	0.01	1.6474	0.8823	0.0000	0.0000
9277.00	9.86	0.01	1.6551	0.8825	0.0000	0.0000

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TABLE 2-2 (EP0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST 0PP

TIME FROM MOTION (SEC)	X POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Y POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Z POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)
709.00	5856238.6	92597.3	2961918.5	-3518.203	75.771	6953.336
712.00	5845645.6	92830.6	2982754.3	-3542.969	75.434	6940.772
717.00	5827827.6	93206.3	3017405.4	-3584.131	74.843	6919.638
721.00	5813425.2	93504.7	3045049.8	-3616.970	74.368	6902.556
726.00	5795237.7	93875.1	3079508.7	-3657.906	73.772	6880.984
731.00	5776845.9	94242.4	3113859.1	-3698.713	73.174	6859.169
736.00	5758250.4	94606.8	3148099.8	-3739.389	72.573	6837.112
741.00	5739451.9	94968.1	3182229.7	-3779.935	71.969	6814.815
746.00	5720451.0	95326.5	3216247.4	-3820.347	71.363	6792.277
751.00	5701248.3	95681.8	3250151.9	-3860.625	70.754	6769.500
756.00	5681844.6	96034.0	3283941.9	-3900.767	70.143	6746.485
761.00	5662240.6	96383.2	3317616.3	-3940.774	69.529	6723.236
766.00	5642436.8	96729.3	3351173.7	-3980.644	68.913	6699.752
771.00	5622434.1	97072.3	3384613.2	-4020.374	68.294	6676.032
776.00	5602233.0	97412.2	3417933.5	-4059.962	67.673	6652.077
781.00	5581834.4	97749.0	3451133.5	-4099.408	67.050	6627.888
786.00	5561238.9	98082.7	3484211.9	-4138.709	66.424	6603.465
791.00	5540447.2	98413.3	3517167.6	-4177.865	65.796	6578.809
796.00	5519460.1	98740.7	3549999.4	-4216.874	65.166	6553.921
927.00	4902589.9	106160.5	4362278.6	-5182.017	47.878	5822.243
1247.00	2933954.3	114151.1	5869809.4	-6973.629	1.349	3485.863
1567.00	546066.8	106905.2	6539718.0	-7770.605	-46.282	650.220
1887.00	-1920275.2	85098.5	6275823.8	-7457.766	-88.609	-2279.828
2207.00	-4112627.8	51390.1	5115503.4	-6078.842	-119.783	-4884.471
2527.00	-5717674.9	10111.1	3224675.3	-3831.748	-135.336	-6790.265
2847.00	-6506512.6	-33292.6	873783.2	-1039.442	-132.855	-7725.400
3167.00	-6367326.4	-72983.1	-1601470.3	1898.410	-112.337	-7558.597
3487.00	-5320756.2	-103516.3	-3848412.2	4563.635	-76.219	-6315.950
3807.00	-3516331.4	-120581.9	-5547443.9	6578.257	-29.079	-4175.397
4127.00	-1211023.3	-121604.9	-6457041.0	7656.568	22.929	-1441.231
4447.00	1267066.3	-106130.5	-6447853.8	7645.084	72.850	1497.949
4767.00	3565165.5	-75936.4	-5521233.9	6545.160	113.848	4223.645
5087.00	5356126.9	-34842.1	-3809230.4	4513.747	140.149	6347.506
5407.00	6385201.9	11768.3	-1555698.0	1840.772	147.870	7567.632
5727.00	6506238.8	57651.0	918705.6	-1093.320	135.586	7711.553
6047.00	5702111.5	56534.9	3262233.0	-3872.219	104.560	6759.498
6367.00	4086879.8	122970.5	5141746.6	-6101.891	58.616	4846.074
6687.00	1889776.1	133086.1	6289654.4	-7464.980	3.691	2241.659
7007.00	-576835.7	125155.1	6541949.5	-7765.652	-52.886	-684.167
7327.00	-2961559.4	59892.5	5862076.2	-6959.109	-103.359	-3513.947
7647.00	-4924267.8	60420.9	4346738.6	-5159.973	-140.619	-5842.606
7967.00	-6185204.2	11893.1	2212200.2	-2626.236	-159.245	-7337.338
8287.00	-6565217.6	-39185.6	-237064.6	279.278	-156.311	-7786.309
8383.00	-6495966.5	-53959.1	-981388.9	1161.851	-151.148	-7703.712
8423.00	-6442204.4	-59951.8	-1288316.3	1525.735	-148.430	-7639.786
8463.00	-6373953.6	-65829.2	-1592342.8	1886.152	-145.386	-7558.697
8503.00	-6291368.9	-71578.4	-1892785.5	2242.294	-142.021	-7460.631
8543.00	-6194637.4	-77186.7	-2188969.8	2593.362	-138.342	-7345.814
8583.00	-6083977.7	-82641.6	-2480230.6	2938.572	-134.356	-7214.504

TABLE 2.2 (EPØ)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG IST ØPP

TIME FROM FIRST MØTION (SEC)	X POSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Y POSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Z POSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)
8623.00	-5959639.7	-87931.2	-2765914.0	3277.151	-130.071	-7067.001
8663.00	-5821903.8	-93043.5	-3045379.0	3608.341	-125.496	-6903.637
8703.00	-5671080.5	-57967.1	-3317598.3	3931.401	-120.640	-6724.782
8743.00	-5507509.4	-102691.0	-3583160.2	4245.609	-115.512	-6530.838
8783.00	-5331558.8	-107204.6	-3840269.8	4550.263	-110.122	-6322.242
8823.00	-5143624.5	-111497.5	-4088750.3	4844.680	-104.483	-6099.463
8863.00	-4944129.3	-115560.0	-4328044.2	5128.203	-98.605	-5863.000
8903.00	-4733521.8	-119382.8	-4557614.7	5400.197	-92.500	-5613.386
8943.00	-4512275.1	-122957.2	-4776946.8	5660.054	-86.181	-5351.179
8983.00	-4280886.6	-126274.7	-4985548.6	5907.193	-79.662	-5076.969
9023.00	-4039876.4	-129327.6	-5182952.1	6141.060	-72.955	-4791.368
9063.00	-3789785.6	-132108.7	-5368714.5	6361.133	-66.074	-4495.019
9103.00	-3531176.2	-134611.4	-5542418.9	6566.916	-59.034	-4188.584
9143.00	-3264628.8	-136829.5	-5703675.6	6757.951	-51.850	-3872.750
9183.00	-2990742.3	-138757.7	-5852122.7	6933.807	-44.537	-3548.226
9223.00	-2710131.8	-140391.0	-5987427.1	7094.091	-37.110	-3215.738
9263.00	-2423427.4	-141725.2	-6109285.1	7238.442	-29.585	-2876.032
9267.00	-2394446.4	-141842.0	-6120720.4	7251.988	-28.828	-2841.694
9272.00	-2358144.6	-141983.8	-6134821.2	7268.691	-27.880	-2798.681
9277.00	-2321759.8	-142120.8	-6148706.8	7285.139	-26.931	-2755.570

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AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED PITCH CHANL (DEG)	GIMBAL DEFLECTION ANG REQUIRED YAW CHANNEL (DEG)	GIMBAL DEFLECTION ANG REQUIRED ROLL CHANNEL (DEG)	EULERIAN ATTITUDE RATE IN PITCH (DEG/S)	EULERIAN ATTITUDE RATE IN YAW (DEG/S)	EULERIAN ATTITUDE RATE IN ROLL (DEG/S)
709.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
712.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
717.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
721.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
726.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
731.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
736.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
741.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
746.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
751.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
756.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
761.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
766.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
771.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
776.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
781.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
786.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
791.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
796.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
927.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
1247.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
1567.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
1887.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
2207.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
2527.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
2847.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
3167.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
3487.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
3807.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
4127.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
4447.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
4767.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
5087.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
5407.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
5727.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
6047.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
6367.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
6687.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
7007.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
7327.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
7647.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
7967.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8287.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8383.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8423.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8463.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8503.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8543.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8583.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000

TABLE 2.2 (EP0)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST 0PP

TIME FROM FIRST MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED PITCH CHANL (DEG)	GIMBAL DEFLECTION ANG REQUIRED YAW CHANNEL (DEG)	GIMBAL DEFLECTION ANG REQUIRED ROLL CHANNEL (DEG)	EULERIAN ATTITUDE RATE IN PITCH (DEG/S)	EULERIAN ATTITUDE RATE IN YAW (DEG/S)	EULERIAN ATTITUDE RATE IN ROLL (DEG/S)
8623.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8663.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8703.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8743.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8783.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8823.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8863.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8903.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8943.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
8983.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9023.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9063.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9103.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9143.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9183.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9223.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9263.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9267.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9272.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
9277.00	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000

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AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TABLE 2-2 (EPØ)

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABOVE SPHERICAL EARTH (FT)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (FT/S**2)
709.00	141.81	295256.92	617739.6	25567.92	90.00	0.015
712.00	141.81	295255.42	617676.2	25567.96	90.00	0.015
717.00	141.81	295252.71	617562.2	25568.04	90.00	0.015
721.00	141.81	295250.55	617469.7	25568.10	90.00	0.015
726.00	141.81	295247.84	617352.6	25568.17	90.00	0.015
731.00	141.81	295245.13	617234.7	25568.25	90.00	0.015
736.00	141.81	295242.42	617115.6	25568.33	90.00	0.015
741.00	141.81	295239.72	616995.4	25568.41	90.00	0.015
746.00	141.81	295237.01	616873.9	25568.50	90.00	0.015
751.00	141.81	295234.30	616751.6	25568.58	90.00	0.015
756.00	141.81	295231.59	616627.6	25568.66	90.00	0.015
761.00	186.81	295228.88	616502.5	25568.76	90.00	0.020
766.00	186.81	295226.18	616376.6	25568.87	90.00	0.020
771.00	186.81	295223.47	616249.5	25568.98	90.00	0.020
776.00	186.81	295220.77	616121.3	25569.09	90.00	0.020
781.00	186.81	295218.05	615991.5	25569.20	90.00	0.020
786.00	186.81	295215.35	615861.1	25569.32	90.00	0.020
791.00	185.50	295212.64	615729.9	25569.43	90.00	0.020
796.00	183.71	295209.93	615597.0	25569.54	90.00	0.020
927.00	136.83	299138.98	611795.8	25572.48	90.00	0.015
1247.00	42.89	298965.69	601040.0	25578.51	90.00	0.005
1567.00	42.25	298792.39	591678.5	25583.66	90.00	0.005
1887.00	41.62	298610.21	588028.5	25587.13	89.99	0.004
2207.00	35.66	298442.25	592808.9	25587.34	89.99	0.004
2527.00	28.40	298327.37	605716.3	25583.89	89.99	0.003
2847.00	23.47	298212.49	623403.6	25577.91	89.98	0.003
3167.00	21.23	298097.61	640954.8	25571.40	89.97	0.002
3487.00	20.42	297982.73	654053.2	25566.10	89.96	0.002
3807.00	19.44	297867.85	660701.0	25562.83	89.96	0.002
4127.00	17.78	297773.36	661668.3	25561.24	89.97	0.002
4447.00	16.11	297689.11	659528.8	25560.35	89.98	0.002
4767.00	14.75	297604.85	656887.7	25559.25	89.99	0.002
5087.00	14.49	297523.30	654816.7	25557.80	90.01	0.002
5407.00	14.23	297450.91	652391.5	25556.75	90.02	0.002
5727.00	14.00	297378.52	647571.5	25557.26	90.02	0.001
6047.00	14.00	297306.02	638867.7	25560.21	90.02	0.001
6367.00	14.00	297232.58	626802.7	25565.50	90.01	0.001
6687.00	14.00	297159.15	614290.9	25571.82	90.01	0.001
7007.00	13.61	297085.72	605640.4	25577.14	90.00	0.001
7327.00	13.21	297011.88	604620.9	25579.55	90.00	0.001
7647.00	12.81	296938.02	612577.3	25578.17	90.00	0.001
7967.00	12.49	296864.17	627624.9	25573.51	89.99	0.001
8287.00	12.45	296797.42	645405.3	25567.20	89.98	0.001
8383.00	12.44	296777.76	650512.1	25565.29	89.98	0.001
8423.00	12.43	296769.57	652550.3	25564.51	89.98	0.001
8463.00	12.43	296761.38	654523.5	25563.74	89.97	0.001
8503.00	12.42	296753.19	656425.9	25563.00	89.97	0.001
8543.00	12.42	296745.00	658249.8	25562.28	89.97	0.001
8583.00	12.41	296736.81	659991.1	25561.58	89.97	0.001

TABLE 2.2 (EPØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG IST ØPP

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABOVE SPHERICAL EARTH (FT)	MAGNITUDE ØF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (FT/S**2)
8623.00	12.41	296728.62	661643.4	25560.91	89.97	0.001
8663.00	12.40	296720.43	663202.8	25560.27	89.97	0.001
8703.00	12.39	296712.24	664666.0	25559.65	89.97	0.001
8743.00	12.32	296704.05	666029.5	25559.07	89.97	0.001
8783.00	12.24	296695.86	667290.9	25558.51	89.97	0.001
8823.00	12.16	296687.67	668450.7	25557.99	89.96	0.001
8863.00	12.09	296679.48	669505.8	25557.50	89.96	0.001
8903.00	12.01	296671.29	670457.5	25557.04	89.96	0.001
8943.00	11.94	296663.10	671305.8	25556.62	89.96	0.001
8983.00	11.86	296654.91	672052.9	25556.22	89.96	0.001
9023.00	11.78	296642.29	672700.3	25555.86	89.96	0.001
9063.00	11.71	296614.47	673251.4	25555.52	89.96	0.001
9103.00	11.63	296586.64	673708.5	25555.22	89.96	0.001
9143.00	11.56	296558.82	674075.9	25554.94	89.96	0.001
9183.00	11.48	296531.00	674357.3	25554.68	89.96	0.001
9223.00	11.40	296503.17	674557.1	25554.45	89.96	0.001
9263.00	11.33	296475.34	674681.3	25554.24	89.96	0.001
9267.00	11.32	296472.56	674688.9	25554.22	89.96	0.001
9272.00	11.31	296469.09	674698.2	25554.20	89.96	0.001
9277.00	11.30	296465.61	674706.2	25554.17	89.96	0.001

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TABLE 2.2 (EP0)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST 0PP

TIME FROM FIRST MOTION (SEC)	*** RADIAL DISTANCE OF THE C G (FT)	MAGNITUDE OF THE EARTH FIXED VELOCITY (FT/SEC)	SUB VEHICLE LATITUDE 0N SPHERICAL EARTH (DEG)	LONGITUDE 0N 0BLATE EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (NM)
709.00	21533164.75	0.000	32.6831	-52.69	788.89
712.00	21533162.50	0.000	32.6867	-52.46	795.16
717.00	21533158.50	0.000	32.6922	-52.08	805.62
721.00	21533155.50	0.000	32.6956	-51.77	813.98
726.00	21533151.00	0.000	32.6987	-51.39	824.44
731.00	21533146.50	0.000	32.7006	-51.01	834.90
736.00	21533142.25	0.000	32.7011	-50.62	845.36
741.00	21533137.50	0.000	32.7003	-50.24	855.81
746.00	21533132.75	0.000	32.6983	-49.86	866.27
751.00	21533128.00	0.000	32.6950	-49.48	876.73
756.00	21533123.25	0.000	32.6903	-49.09	887.19
761.00	21533118.25	0.000	32.6844	-48.71	897.64
766.00	21533113.25	0.000	32.6772	-48.33	908.10
771.00	21533108.00	0.000	32.6687	-47.95	918.56
776.00	21533102.75	0.000	32.6589	-47.56	929.02
781.00	21533097.25	0.000	32.6478	-47.18	939.48
786.00	21533092.00	0.000	32.6354	-46.80	949.93
791.00	21533086.25	0.000	32.6217	-46.42	960.39
796.00	21533081.00	0.000	32.6067	-46.04	970.85
927.00	21532942.25	0.000	31.7613	-36.11	1244.88
1247.00	21532854.75	0.000	26.3404	-13.29	1914.47
1567.00	21533291.50	0.000	17.3049	6.77	2584.31
1887.00	21534074.75	0.000	6.2205	24.66	3254.09
2207.00	21535094.75	0.000	-5.5325	41.75	3923.23
2527.00	21536518.25	0.000	-16.6916	59.55	4591.09
2847.00	21538735.00	0.000	-25.8889	79.45	5256.03
3167.00	21542125.00	0.000	-31.5659	102.11	5714.80
3487.00	21546761.25	0.000	-32.3924	126.42	5072.96
3807.00	21552211.25	0.000	-28.1466	149.80	4408.14
4127.00	21557575.25	0.000	-19.9072	170.54	3742.63
4447.00	21561768.00	0.000	-9.2378	-171.16	3076.89
4767.00	21563928.75	0.000	2.4346	-154.05	2411.04
5087.00	21563766.50	0.000	13.8488	-136.64	1745.54
5407.00	21561668.00	0.000	23.6965	-117.45	1081.65
5727.00	21558510.00	0.000	30.4664	-95.56	426.59
6047.00	21555282.25	0.000	32.6879	-71.50	286.53
6367.00	21552712.00	0.000	29.7589	-47.62	932.94
6687.00	21551077.00	0.000	22.4510	-26.10	1596.28
7007.00	21550275.00	0.000	12.2803	-7.23	2261.96
7327.00	21550086.00	0.000	0.7357	10.08	2928.17
7647.00	21550448.00	0.000	-10.8868	27.29	3593.96
7967.00	21551577.25	0.000	-21.3057	45.90	4258.55
8287.00	21553848.50	0.000	-29.0579	67.07	4920.42
8383.00	21554797.00	0.000	-30.6255	73.96	5117.55
8423.00	21555230.25	0.000	-31.1565	76.89	5199.22
8463.00	21555685.50	0.000	-31.6122	79.86	5280.43
8503.00	21556162.25	0.000	-31.9905	82.85	5360.95
8543.00	21556660.25	0.000	-32.2896	85.87	5440.35
8583.00	21557179.25	0.000	-32.5081	88.91	5517.81

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2-2 (EPØ)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	**** RADIAL DISTANCE ØF THE C G (FT)	MAGNITUDE ØF THE EARTH FIXED VELOCITY (FT/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN ØBLATE (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (NM)
8623.00	21557718.75	0.000	-32.6450	91.96	5591.46
8663.00	21558276.75	0.000	-32.6996	95.01	5656.56
8703.00	21558853.75	0.000	-32.6715	98.06	5700.30
8743.00	21559447.50	0.000	-32.5611	101.11	5700.31
8783.00	21560056.75	0.000	-32.3689	104.15	5656.62
8823.00	21560680.75	0.000	-32.0957	107.18	5591.56
8863.00	21561317.75	0.000	-31.7429	110.18	5517.97
8903.00	21561966.00	0.000	-31.3123	113.15	5440.57
8943.00	21562623.50	0.000	-30.8057	116.10	5361.22
8983.00	21563288.50	0.000	-30.2254	119.00	5280.77
9023.00	21563959.50	0.000	-29.5740	121.87	5199.63
9063.00	21564634.25	0.000	-28.8542	124.70	5118.04
9103.00	21565310.50	0.000	-28.0688	127.49	5036.14
9143.00	21565985.75	0.000	-27.2209	130.23	4954.01
9183.00	21566658.50	0.000	-26.3136	132.93	4871.71
9223.00	21567325.25	0.000	-25.3501	135.58	4789.29
9263.00	21567985.25	0.000	-24.3334	138.19	4706.76
9267.00	21568050.25	0.000	-24.2290	138.45	4698.50
9272.00	21568131.50	0.000	-24.0977	138.77	4688.18
9277.00	21568212.50	0.000	-23.9656	139.09	4677.86

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*** MEASURED FROM THE CENTER ØF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (EP0)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST 0PP

TIME FROM FIRST MOTION (SEC)	X CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Y CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Z CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	X CØMPONENT 0F VEL. IN THE XXE SYSTEM (FT/S)	Y CØMPONENT 0F VEL. IN THE XXE SYSTEM (FT/S)	Z CØMPONENT 0F VEL. IN THE XXE SYSTEM (FT/S)
709.00	19628194.750	261C80.549	8851183.750	-9969.797	913.162	22080.405
712.00	19598168.750	263848.477	8917356.250	-10044.415	918.199	22046.406
717.00	19547635.750	268460.164	9027445.375	-10168.476	926.481	21989.193
721.00	19506763.000	272179.227	9115308.625	-10267.494	933.087	21942.918
726.00	19455116.000	276865.266	9224878.000	-10390.977	941.318	21884.445
731.00	19402852.500	281592.457	9334152.500	-10514.130	949.520	21825.276
736.00	19349974.250	286360.430	9443129.375	-10636.948	957.693	21765.412
741.00	19296482.500	291169.305	9551805.000	-10759.429	965.835	21704.857
746.00	19242379.250	296C18.695	9660176.375	-10881.569	973.946	21643.611
751.00	19187666.250	300508.688	9768239.750	-11003.363	982.025	21581.677
756.00	19132345.250	305838.965	9875991.500	-11124.808	990.072	21519.057
761.00	19076417.750	310809.367	9983428.750	-11245.906	998.087	21455.765
766.00	19019885.750	315819.723	10090547.625	-11366.652	1006.068	21391.800
771.00	1896275C.750	320669.875	10157345.000	-11487.038	1014.016	21327.155
776.00	18905014.750	325559.754	103C3817.625	-11607.059	1021.928	21261.831
781.00	18846679.750	331C89.152	104C9961.875	-11726.712	1029.806	21195.830
786.00	18787747.250	336257.738	10515774.375	-11845.993	1037.648	21129.156
791.00	18728219.500	341465.680	10621252.000	-11964.898	1045.453	21061.810
796.00	18668097.750	346712.246	10726391.000	-12083.423	1053.221	20993.792
927.00	16888206.750	497359.551	13349483.375	-15041.139	1241.542	18980.444
1247.00	11094359.875	948462.367	18430393.250	-20777.506	1530.770	12429.151
1567.00	3876524.344	1441314.516	21132387.500	-23844.744	1487.074	4274.711
1887.00	-3837995.344	1858304.484	211C7649.000	-23847.796	1053.402	-4426.108
2207.00	-11057742.000	2075951.156	18362381.500	-20786.395	251.810	-12541.649
2527.00	-16855701.750	1990591.234	13256804.625	-15059.371	-814.403	-19016.442
2847.00	-20489897.250	1544561.953	6457207.625	-7410.819	-1974.615	-23014.814
3167.00	-2149819.500	739638.305	-1153137.125	1172.461	-3018.440	-24030.745
3487.00	-19757074.250	-351857.227	-8550524.500	9592.119	-3731.732	-21948.211
3807.00	-15491833.875	-1594C20.109	-14858321.375	16777.113	-3935.049	-17049.702
4127.00	-9249703.625	-2803847.250	-19269417.750	21815.233	-3518.739	-9977.067
4447.00	-1828663.734	-3778C61.563	-2114928C.250	24066.207	-2469.057	-1650.262
4767.00	5823485.438	-4326143.563	-203C7006.000	23244.473	-880.028	6850.061
5087.00	12730205.250	-4304233.688	-1686477.250	19458.205	1052.741	14424.374
5407.00	18011399.500	-3644659.656	-11278805.375	13195.054	3056.687	20100.470
5727.00	20995942.000	-2373589.094	-4280493.500	5256.176	4821.565	23160.238
6047.00	21304004.750	-616631.383	3223247.125	-3348.816	6046.779	23228.383
6367.00	1889784C.250	1412531.125	10266244.125	-11528.609	6489.607	20314.820
6687.00	14082572.250	3437623.563	15947187.875	-18243.787	6007.905	14810.574
7007.00	7470672.938	5157226.688	19544957.500	-22635.857	4591.164	7439.041
7327.00	-94898.468	6290C63.938	20611459.750	-24139.182	2373.677	-835.280
7647.00	-7648072.063	6619427.063	1902924C.000	-22559.239	-375.216	-8933.858
7967.00	-1422443C.875	6032619.563	15024764.125	-18100.823	-3284.675	-15810.454
8287.00	-18985848.250	4544556.313	9135059.125	-11338.988	-5934.674	-20591.955
8383.00	-19962333.750	3941962.531	7111640.063	-8984.943	-6616.749	-21521.033
8423.00	-20301528.500	3611572.063	6244391.500	-7971.979	-688C.664	-21833.678
8463.00	-20599878.750	3391682.063	5365541.500	-6943.095	-7131.623	-22101.639
8503.00	-2085679C.250	3101623.781	4476868.375	-5900.354	-7368.959	-22324.442
8543.00	-21071745.250	2802355.344	3580193.031	-4845.849	-7592.030	-22501.706
8583.00	-21244328.000	2494459.656	2677342.906	-3781.695	-7800.223	-22633.140

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG IST 0PP

TIME FROM FIRST MOTION (SEC)	X CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Y CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Z CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	COMPONENT OF VEL. IN THE XXE SYSTEM (FT/S)	COMPONENT OF VEL. IN THE YYE SYSTEM (FT/S)	COMPONENT OF VEL. IN THE ZZE SYSTEM (FT/S)
8623.00	-21374183.250	2178543.625	1770155.625	-2710.023	-7992.953	-22718.550
8663.00	-21461057.000	1855236.953	860473.922	-1632.980	-8169.665	-22757.832
8703.00	-21504777.500	1525190.859	-49855.807	-552.722	-8329.839	-22750.975
8743.00	-21505259.500	1189076.719	-958989.984	528.590	-8472.986	-22698.063
8783.00	-21462504.000	847584.828	-1865089.359	1608.790	-8598.651	-22599.273
8823.00	-21376598.750	501423.004	-2766323.781	2685.719	-8706.417	-22454.871
8863.00	-21247717.500	151315.217	-3660875.656	3757.222	-8795.902	-22265.218
8903.00	-21076120.750	-202000.648	-4546943.375	4821.157	-8866.765	-22030.763
8943.00	-20862153.000	-557173.602	-5422745.938	5875.393	-8918.703	-21752.045
8983.00	-20606245.000	-915240.992	-6286524.938	6917.823	-8951.452	-21429.694
9023.00	-20308910.750	-1273630.734	-7136548.750	7946.360	-8964.791	-21064.423
9063.00	-19970748.000	-1632162.953	-7971116.688	8958.945	-8958.541	-20657.034
9103.00	-19592434.500	-1990051.000	-8788561.250	9953.550	-8932.564	-20208.410
9143.00	-19174725.500	-2346503.813	-9587252.125	10928.183	-8886.768	-19719.519
9183.00	-18718471.000	-2700727.250	-10365599.250	11880.891	-8821.104	-19191.406
9223.00	-18224574.750	-3051926.750	-11122055.875	12809.762	-8735.565	-18625.196
9263.00	-17694031.500	-3399307.656	-11855121.875	13712.934	-8630.192	-18022.090
9267.00	-17639001.750	-3433605.281	-11927085.375	13801.767	-8618.567	-17959.797
9272.00	-17569715.750	-3476861.094	-12016688.250	13912.418	-8603.757	-17881.434
9277.00	-17499877.250	-3519842.188	-12105898.500	14022.634	-8588.639	-17802.518

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TABLE 2-2 (EP0)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST 0PP

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT**2)	TOTAL ANGLE OF ATTACK (DEG)	DRAG ALONG VELOCITY VECTØR (LB)	AERODYNAMIC LOAD (LB/FT**2)	MAGNITUDE OF THE AERODYNAMIC FORCE VECTOR (FT)
709.00	10.01	0.00	4.7162	2.4147	0.0000	0.0000
712.00	10.01	0.00	4.9049	2.4147	0.0000	0.0000
717.00	10.01	0.00	5.2454	2.4146	0.0000	0.0000
721.00	10.01	0.00	0.0321	2.4146	0.0000	0.0000
726.00	10.01	0.00	0.0198	2.4147	0.0000	0.0000
731.00	10.01	0.00	0.0076	2.4147	0.0000	0.0000
736.00	10.01	0.00	0.0053	2.4148	0.0000	0.0000
741.00	10.01	0.00	0.0174	2.4150	0.0000	0.0000
746.00	10.01	0.00	0.0297	2.4151	0.0000	0.0000
751.00	10.01	0.00	0.0420	2.4153	0.0000	0.0000
756.00	10.01	0.00	0.0544	2.4155	0.0000	0.0000
761.00	10.01	0.00	0.0667	2.4158	0.0000	0.0000
766.00	10.01	0.00	0.0790	2.4161	0.0000	0.0000
771.00	10.01	0.00	0.0914	2.4164	0.0000	0.0000
776.00	10.01	0.00	0.1037	2.4167	0.0000	0.0000
781.00	10.01	0.00	0.1160	2.4171	0.0000	0.0000
786.00	10.01	0.00	0.1284	2.4175	0.0000	0.0000
791.00	10.01	0.00	0.1407	2.4180	0.0000	0.0000
796.00	10.01	0.00	0.1530	2.4184	0.0000	0.0000
927.00	10.01	0.00	0.4722	2.4408	0.0000	0.0000
1247.00	10.06	0.00	1.1864	2.5652	0.0000	0.0000
1567.00	10.12	0.00	1.7240	2.7360	0.0000	0.0000
1887.00	10.16	0.00	2.0056	2.8526	0.0000	0.0000
2207.00	10.15	0.00	1.9895	2.8308	0.0000	0.0000
2527.00	10.10	0.00	1.6781	2.6681	0.0000	0.0000
2847.00	10.02	0.00	1.1173	2.4478	0.0000	0.0000
3167.00	9.96	0.00	0.3906	2.2653	0.0000	0.0000
3487.00	9.93	0.00	0.3989	2.1631	0.0000	0.0000
3807.00	9.93	0.00	1.1265	2.1463	0.0000	0.0000
4127.00	9.94	0.00	1.6888	2.1838	0.0000	0.0000
4447.00	9.95	0.00	2.0018	2.2224	0.0000	0.0000
4767.00	9.95	0.00	2.0191	2.2134	0.0000	0.0000
5087.00	9.93	0.00	1.7378	2.1463	0.0000	0.0000
5407.00	9.90	0.00	1.1989	2.0564	0.0000	0.0000
5727.00	9.87	0.00	0.4817	1.9978	0.0000	0.0000
6047.00	9.88	0.00	0.3100	2.0112	0.0000	0.0000
6367.00	9.91	0.00	1.0550	2.1088	0.0000	0.0000
6687.00	9.97	0.00	1.6453	2.2665	0.0000	0.0000
7007.00	10.01	0.00	1.9926	2.4205	0.0000	0.0000
7327.00	10.04	0.00	2.0454	2.4898	0.0000	0.0000
7647.00	10.02	0.00	1.7953	2.4308	0.0000	0.0000
7967.00	9.97	0.00	1.2787	2.2756	0.0000	0.0000
8287.00	9.91	0.00	0.5715	2.1022	0.0000	0.0000
8383.00	9.90	0.00	0.3376	2.0575	0.0000	0.0000
8423.00	9.89	0.00	0.2388	2.0406	0.0000	0.0000
8463.00	9.89	0.00	0.1402	2.0247	0.0000	0.0000
8503.00	9.88	0.00	0.0471	2.0099	0.0000	0.0000
8543.00	9.88	0.00	0.0699	1.9963	0.0000	0.0000
8583.00	9.87	0.00	0.1663	1.9839	0.0000	0.0000

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTØR (LB)	AERODYNAMIC LOAD (LB/FT**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTOR (FT)
8623.00	9.87	0.00	0.2653	1.9727	0.0000	0.0000
8663.00	9.86	0.00	0.3642	1.9627	0.0000	0.0000
8703.00	9.86	0.00	0.4624	1.9539	0.0000	0.0000
8743.00	9.86	0.00	0.5598	1.9464	0.0000	0.0000
8783.00	9.85	0.00	0.6559	1.9401	0.0000	0.0000
8823.00	9.85	0.00	0.7505	1.9350	0.0000	0.0000
8863.00	9.85	0.00	0.8434	1.9310	0.0000	0.0000
8903.00	9.85	0.00	0.9345	1.9282	0.0000	0.0000
8943.00	9.85	0.00	1.0233	1.9265	0.0000	0.0000
8983.00	9.85	0.00	1.1099	1.9258	0.0000	0.0000
9023.00	9.85	0.00	1.1938	1.9260	0.0000	0.0000
9063.00	9.85	0.00	1.2751	1.9272	0.0000	0.0000
9103.00	9.85	0.00	1.3533	1.9292	0.0000	0.0000
9143.00	9.85	0.00	1.4285	1.9320	0.0000	0.0000
9183.00	9.85	0.00	1.5003	1.9354	0.0000	0.0000
9223.00	9.86	0.00	1.5687	1.9394	0.0000	0.0000
9263.00	9.86	0.00	1.6334	1.9440	0.0000	0.0000
9267.00	9.86	0.00	1.6396	1.9445	0.0000	0.0000
9272.00	9.86	0.00	1.6474	1.9451	0.0000	0.0000
9277.00	9.86	0.00	1.6551	1.9457	0.0000	0.0000

TB6

TABLE 2-2 (EP0)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST 0PP

TIME FROM MOTION (SEC)	X POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Y POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Z POSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)
709.00	19213381.3	3C3797.0	5717580.3	-11542.661	248.592	22812.782
712.00	19178627.0	3C4562.3	5785939.4	-11623.914	247.488	22771.560
717.00	19120169.0	3C5794.9	5899624.0	-11758.959	245.548	22702.226
721.00	19072517.3	3C6774.0	5990320.9	-11866.700	243.990	22646.181
726.00	19013246.8	3C7989.0	IC103374.9	-12001.003	242.035	22575.405
731.00	18952506.5	3C9194.3	IC216073.1	-12134.884	240.071	22503.834
736.00	18891897.5	310389.7	IC328411.5	-12268.338	238.099	22431.470
741.00	18830222.5	311575.3	IC440385.9	-12401.361	236.118	22358.316
746.00	18767883.8	312750.9	IC551992.9	-12533.947	234.130	22284.374
751.00	18704882.8	313916.5	IC663228.1	-12666.092	232.132	22209.647
756.00	18641222.3	315072.2	IC774087.5	-12797.791	230.127	22134.137
761.00	18576904.5	316217.8	IC884567.6	-12929.048	228.114	22057.860
766.00	18511531.8	317353.3	IC994664.4	-13059.854	226.092	21980.814
771.00	18446305.8	318478.7	11104374.0	-13190.202	224.063	21902.993
776.00	18380029.5	319593.9	11213692.5	-13320.086	222.026	21824.400
781.00	18313104.8	320698.9	11322616.3	-13449.501	219.981	21745.039
786.00	18245534.3	321793.7	11431141.3	-13578.443	217.928	21664.910
791.00	18177320.0	322878.1	11539263.6	-13706.907	215.867	21584.019
796.00	18108464.8	323952.3	11646979.8	-13834.889	213.798	21502.365
927.00	16084612.4	348295.5	14311937.5	-17001.368	157.081	19101.848
1247.00	9625834.1	374511.3	19257904.8	-22879.361	4.426	11436.559
1567.00	1791557.6	350739.0	214555767.5	-25494.112	-151.842	2133.267
1887.00	-6300115.3	279194.4	20589972.8	-24467.737	-290.713	-7479.750
2207.00	-13492873.4	188602.8	16783154.0	-19943.706	-392.987	-16025.167
2527.00	-18758776.0	33173.0	IC579643.5	-12571.352	-444.017	-22277.773
2847.00	-21346825.8	-1C9227.6	2866742.8	-3410.242	-435.875	-25345.800
3167.00	-20890178.5	-239446.0	-5254167.6	6228.379	-368.558	-24798.545
3487.00	-17456549.0	-339620.4	-12626024.3	14972.554	-250.062	-20721.621
3807.00	-11536520.4	-355609.9	-18200275.0	21582.208	-95.405	-13698.808
4127.00	-3973173.5	-358966.1	-21184517.5	25119.975	75.225	-4728.447
4447.00	4157641.7	-348197.3	-21154375.8	25082.298	239.010	4914.531
4767.00	11696737.3	-249135.1	-18114284.5	21473.621	373.518	13857.102
5087.00	17572594.8	-114311.4	-12497474.9	14808.880	459.807	20825.152
5407.00	2094825.0	38609.8	-5103996.1	6039.278	485.137	24828.190
5727.00	21345927.8	189143.6	3014126.1	-3587.009	444.836	25300.372
6047.00	18707114.8	316715.6	IC702864.3	-12704.131	343.046	22176.830
6367.00	13408398.1	4C3446.7	16869247.0	-20019.328	192.309	15899.192
6687.00	620052.9	4C6634.2	2C635348.8	-24491.404	12.111	7354.525
7007.00	-1892505.6	410613.9	21463088.8	-25477.862	-173.511	-2244.643
7327.00	-9716402.3	327731.3	19232533.3	-22831.721	-339.104	-11528.697
7647.00	-16155734.0	158231.3	14260953.3	-16929.046	-461.348	-19168.654
7967.00	-20292644.5	39019.3	7257874.5	-8616.260	-522.458	-24072.631
8287.00	-21539427.5	-128561.7	-777771.1	916.266	-173.511	-2244.643
8383.00	-21312226.0	-177031.3	-3219779.8	3811.848	-11528.697	-19168.654
8423.00	-21135841.3	-156692.3	-42266759.4	5005.694	-495.893	-25274.646
8463.00	-20911921.3	-215975.1	-5224221.8	6188.163	-486.975	-25064.914
8503.00	-20640574.0	-234837.2	-6209926.3	7356.606	-476.987	-24798.874
8543.00	-20323613.3	-253237.1	-7181659.4	8508.407	-465.948	-24477.137
8583.00	-19960556.5	-271134.0	-8137239.3	9640.986	-453.878	-24100.438
					-440.801	-23669.633

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	X POSI CØRD				Y POSI CØRD				Z POSI CØRD				X VEL. CØMP.				Y VEL. CØMP.				Z VEL. CØMP.							
	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT/S)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT/S)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT/S)				
8623.00	-19552623.5	-288488.1	-9074521.0	-9074521.0	-288488.1	-305260.7	-9991400.8	-9991400.8	10751.808	11838.389	12898.298	13929.165	14928.683	15894.619	16824.812	17717.181	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374
8663.00	-19100734.0	-305260.7	-9991400.8	-9991400.8	-305260.7	-321414.4	-10885821.0	-10885821.0	11838.389	12898.298	13929.165	14928.683	15894.619	16824.812	17717.181	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374	
8703.00	-18605907.0	-321414.4	-10885821.0	-10885821.0	-321414.4	-336912.8	-11755774.9	-11755774.9	12898.298	13929.165	14928.683	15894.619	16824.812	17717.181	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374		
8743.00	-18069256.5	-336912.8	-11755774.9	-11755774.9	-336912.8	-351721.0	-12599310.3	-12599310.3	13929.165	14928.683	15894.619	16824.812	17717.181	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374			
8783.00	-17491590.5	-351721.0	-12599310.3	-12599310.3	-351721.0	-365805.4	-13414535.0	-13414535.0	14928.683	15894.619	16824.812	17717.181	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374				
8823.00	-16875408.3	-365805.4	-13414535.0	-13414535.0	-365805.4	-379133.9	-14199620.0	-14199620.0	15894.619	16824.812	17717.181	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374					
8863.00	-16220896.6	-379133.9	-14199620.0	-14199620.0	-379133.9	-391676.0	-14952804.0	-14952804.0	16824.812	17717.181	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374						
8903.00	-15529526.9	-391676.0	-14952804.0	-14952804.0	-391676.0	-403402.7	-15672397.6	-15672397.6	17717.181	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374							
8943.00	-14804052.1	-403402.7	-15672397.6	-15672397.6	-403402.7	-414287.0	-16356786.8	-16356786.8	18569.732	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374								
8983.00	-14044903.5	-414287.0	-16356786.8	-16356786.8	-414287.0	-424303.1	-17004436.0	-17004436.0	19380.555	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374									
9023.00	-13254187.5	-424303.1	-17004436.0	-17004436.0	-424303.1	-433427.5	-17613892.5	-17613892.5	20147.836	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374										
9063.00	-12433679.8	-433427.5	-17613892.5	-17613892.5	-433427.5	-441638.4	-18183788.8	-18183788.8	20869.857	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374											
9103.00	-11585223.5	-441638.4	-18183788.8	-18183788.8	-441638.4	-448915.8	-18712846.3	-18712846.3	21545.001	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374												
9143.00	-10710724.5	-448915.8	-18712846.3	-18712846.3	-448915.8	-455241.7	-19643789.5	-19643789.5	22171.754	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374													
9183.00	-9812146.8	-455241.7	-19643789.5	-19643789.5	-455241.7	-460600.3	-20043586.0	-20043586.0	22748.710	23274.575	23748.168	23792.610	23847.410	23901.374														
9223.00	-8891508.4	-460600.3	-20043586.0	-20043586.0	-460600.3	-464977.6	-20464977.6	-20464977.6	23274.575	23748.168	23792.610	23847.410	23901.374															
9263.00	-7950877.3	-464977.6	-20464977.6	-20464977.6	-464977.6	-465360.9	-20811103.5	-20811103.5	23748.168	23792.610	23847.410	23901.374																
9267.00	-7855795.4	-465360.9	-20811103.5	-20811103.5	-465360.9	-465826.0	-2127366.0	-2127366.0	23792.610	23847.410	23901.374																	
9272.00	-7736694.7	-465826.0	-2127366.0	-2127366.0	-465826.0	-466275.6			23847.410	23901.374																		
9277.00	-7617322.1	-466275.6			-466275.6				23901.374																			

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TABLE 2.2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	TOTAL ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
9287.00	0.00	134474.33	205654.8	7788.89	89.96	0.000
9297.00	0.00	134474.12	205657.7	7788.88	89.96	0.000
9307.00	0.00	134473.90	205659.6	7788.86	89.96	0.000
9317.00	0.00	134473.69	205660.1	7788.84	89.96	0.000
9327.00	47.81	134473.47	205659.3	7788.82	89.96	0.000
9337.00	139.45	134473.26	205657.5	7788.81	89.96	0.000
9347.00	46.85	134473.05	205654.5	7788.79	89.96	0.000
9357.00	24.28	134472.83	205650.3	7788.78	89.96	0.000
9367.00	30.40	134472.62	205645.2	7788.76	89.96	0.000
9377.00	36.51	134472.41	205638.9	7788.75	89.97	0.000
9387.00	42.63	134472.19	205631.6	7788.74	89.97	0.000
9397.00	48.74	134471.98	205623.3	7788.73	89.97	0.000
9407.00	54.86	134471.77	205614.0	7788.72	89.97	0.000
9417.00	60.97	134471.55	205603.7	7788.71	89.97	0.000
9427.00	67.09	134471.34	205592.4	7788.70	89.97	0.000
9437.00	73.21	134471.13	205580.4	7788.69	89.97	0.000
9447.00	158.75	134470.91	205567.2	7788.68	89.97	0.000
9457.00	83.46	134470.70	205553.3	7788.67	89.97	0.000
9467.00	85.63	134470.48	205538.3	7788.67	89.97	0.000
9477.00	111.11	134470.27	205522.7	7788.66	89.97	0.000
9487.00	89.95	134470.06	205506.3	7788.65	89.97	0.000
9497.00	92.12	134469.84	205489.1	7788.65	89.97	0.000
9507.00	94.28	134469.63	205471.2	7788.64	89.97	0.000
9517.00	96.44	134469.42	205452.5	7788.64	89.97	0.000
9527.00	98.61	134469.20	205433.4	7788.63	89.97	0.000
9537.00	100.77	134468.99	205413.6	7788.63	89.97	0.000
9547.00	102.93	134468.78	205393.1	7788.62	89.97	0.000
9557.00	105.10	134468.56	205372.1	7788.62	89.97	0.000
9567.00	125.75	134468.35	205350.5	7788.61	89.97	0.000
9577.00	109.43	134468.13	205328.4	7788.61	89.97	0.000
9587.00	111.59	134467.92	205305.5	7788.61	89.97	0.000
9597.00	113.75	134467.71	205282.4	7788.60	89.97	0.000
9607.00	115.92	134467.49	205258.8	7788.60	89.97	0.000
9617.00	118.08	134467.28	205234.7	7788.60	89.97	0.000
9627.00	120.24	134467.07	205210.4	7788.59	89.97	0.000
9637.00	136.27	134466.85	205185.5	7788.59	89.97	0.000
9647.00	124.57	134466.64	205160.5	7788.59	89.97	0.000
9657.00	126.73	134466.43	205134.9	7788.59	89.97	0.000
9667.00	128.90	134466.21	205109.3	7788.59	89.98	0.000
9677.00	131.06	134466.00	205083.1	7788.58	89.98	0.000
9687.00	133.23	134465.79	205057.0	7788.58	89.98	0.000
9697.00	135.39	134465.57	205030.4	7788.58	89.98	0.000
9707.00	137.55	134465.36	205003.5	7788.58	89.98	0.000
9717.00	139.72	134465.14	204976.7	7788.58	89.98	0.000
9727.00	141.88	134464.93	204949.6	7788.58	89.98	0.000
9737.00	144.04	134464.72	204922.3	7788.58	89.98	0.000
9747.00	146.21	134464.50	204895.1	7788.58	89.98	0.000
9757.00	148.37	134464.29	204867.7	7788.58	89.98	0.000
9767.00	150.54	134464.08	204840.2	7788.58	89.98	0.000

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST 0PP

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
9777.00	613.29	134462.95	204812.5	7788.59	89.98	0.000
9787.00	613.29	134460.26	204784.9	7788.62	89.98	0.000
9797.00	613.29	134457.58	204757.2	7788.66	89.98	0.000
9807.00	613.29	134454.89	204729.7	7788.69	89.98	0.000
9817.00	613.29	134452.21	204702.3	7788.72	89.98	0.000
9827.00	613.29	134449.52	204674.8	7788.76	89.98	0.000
9837.00	613.29	134446.84	204647.5	7788.79	89.98	0.000
9847.00	613.29	134444.15	204620.2	7788.82	89.98	0.000
9855.00	1718.00	134297.43	204598.6	7788.93	89.98	0.000
9857.00	715911.01	134239.06	204593.2	7792.57	89.98	0.000
9857.50	710921.39	134224.47	204591.8	7795.23	89.98	0.000
9867.00	789731.19	132479.45	204570.0	7850.79	89.98	0.000
9877.00	794240.59	130612.36	204542.0	7910.80	89.99	0.000
9887.00	795064.54	128739.34	204497.1	7971.90	90.00	0.000
9897.00	795274.30	126864.75	204449.5	8033.96	90.00	0.000
9907.00	795280.16	124989.23	204415.9	8096.95	89.98	0.000
9917.00	794877.20	123113.90	204413.7	8160.87	89.95	0.000
9927.00	794264.41	121239.44	204461.9	8225.68	89.91	0.000
9937.00	793205.34	115365.99	204579.3	8291.39	89.86	0.000
9947.00	792597.68	117494.00	204786.5	8357.97	89.79	0.000
9957.00	791952.89	115622.93	205104.1	8425.46	89.71	0.000
9967.00	791205.53	113752.77	205553.7	8493.87	89.62	0.000
9977.00	902036.64	111783.66	206157.3	8566.71	89.51	0.000
9987.00	901369.05	109638.43	206937.3	8647.19	89.39	0.000
9997.00	901008.22	107490.96	207921.8	8729.07	89.25	0.000
10007.00	901357.75	105343.58	209139.7	8812.36	89.10	0.000
10017.00	900939.25	103196.60	210618.0	8897.08	88.93	0.000
10027.00	900713.45	101050.29	212385.9	8983.29	88.75	0.000
10037.00	900589.05	98904.51	214474.4	9071.05	88.55	0.000
10047.00	900463.61	96759.05	216915.2	9160.41	88.33	0.000
10057.00	900321.11	94613.95	219741.2	9251.44	88.10	0.000
10067.00	900177.57	92469.13	222985.9	9344.19	87.85	0.000
10077.00	900024.74	90324.70	226683.7	9438.73	87.59	0.000
10087.00	899850.29	88180.60	230870.1	9535.15	87.31	0.000
10097.00	899663.09	86036.99	235581.5	9633.54	87.01	0.000
10107.00	899450.29	83893.78	240854.8	9733.98	86.70	0.000
10117.00	899165.89	81751.21	246727.8	9836.59	86.38	0.000
10127.00	898860.13	75609.39	253238.5	9941.47	86.04	0.000
10137.00	897291.87	77469.81	260425.9	10048.60	85.68	0.000
10147.00	896977.43	75331.14	268328.9	10158.28	85.31	0.000
10157.00	896533.64	73193.56	276986.7	10270.67	84.93	0.000
10167.00	896024.16	71057.17	286437.6	10385.94	84.54	0.000
10177.00	897165.59	68921.29	296719.3	10504.29	84.14	0.000
10187.00	896379.07	66783.70	307842.2	10625.87	83.75	0.000
10197.00	895895.27	64647.63	319833.8	10751.18	83.34	0.000
10203.80	895547.87	63195.55	328523.8	10838.66	83.04	0.000
10207.00	0.00	63150.07	332760.8	10838.70	82.90	0.000
10213.80	0.00	63150.07	342055.5	10831.07	82.59	0.000

S-IVB GCS2
TLI

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	*** RADIAL DISTANCE OF THE C G (M)	MAGNITUDE OF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LØNGITUDE ØBLATE EARTH (DEG)	ARC RANGE ALØNG SPHER. EARTH TØ SUB VEHICLE PT. (KM)
9287.00	6574040.88	7386.820	-23.6993	139.73	8625.15
9297.00	6574090.38	7386.855	-23.4299	140.37	8586.89
9307.00	6574139.56	7386.890	-23.1576	141.00	8548.63
9317.00	6574188.69	7386.925	-22.8824	141.64	8510.36
9327.00	6574237.44	7386.961	-22.6044	142.27	8472.08
9337.00	6574285.94	7386.998	-22.3236	142.89	8433.80
9347.00	6574334.25	7387.035	-22.0400	143.52	8395.51
9357.00	6574382.25	7387.073	-21.7537	144.14	8357.21
9367.00	6574430.00	7387.112	-21.4648	144.76	8318.91
9377.00	6574477.44	7387.152	-21.1732	145.37	8280.60
9387.00	6574524.56	7387.192	-20.8791	145.99	8242.29
9397.00	6574571.31	7387.232	-20.5825	146.60	8203.97
9407.00	6574617.81	7387.273	-20.2835	147.21	8165.65
9417.00	6574663.94	7387.315	-19.9820	147.81	8127.32
9427.00	6574709.63	7387.357	-19.6782	148.42	8088.98
9437.00	6574755.06	7387.400	-19.3720	149.02	8050.65
9447.00	6574800.00	7387.442	-19.0636	149.62	8012.30
9457.00	6574844.50	7387.486	-18.7530	150.22	7973.96
9467.00	6574888.50	7387.529	-18.4402	150.81	7935.60
9477.00	6574932.06	7387.572	-18.1252	151.40	7897.25
9487.00	6574975.38	7387.615	-17.8082	151.99	7858.89
9497.00	6575018.13	7387.658	-17.4892	152.58	7820.52
9507.00	6575060.50	7387.701	-17.1681	153.16	7782.16
9517.00	6575102.44	7387.744	-16.8452	153.75	7743.79
9527.00	6575143.81	7387.786	-16.5203	154.33	7705.41
9537.00	6575184.81	7387.828	-16.1935	154.91	7667.03
9547.00	6575225.31	7387.870	-15.8650	155.48	7628.65
9557.00	6575265.38	7387.912	-15.5347	156.06	7590.27
9567.00	6575304.88	7387.953	-15.2026	156.63	7551.88
9577.00	6575343.88	7387.994	-14.8689	157.20	7513.49
9587.00	6575382.25	7388.034	-14.5336	157.77	7475.09
9597.00	6575420.13	7388.074	-14.1966	158.34	7436.69
9607.00	6575457.56	7388.113	-13.8581	158.90	7398.29
9617.00	6575494.44	7388.152	-13.5181	159.47	7359.89
9627.00	6575530.69	7388.190	-13.1767	160.03	7321.48
9637.00	6575566.50	7388.228	-12.8338	160.59	7283.08
9647.00	6575601.75	7388.264	-12.4895	161.15	7244.66
9657.00	6575636.38	7388.301	-12.1439	161.71	7206.25
9667.00	6575670.38	7388.336	-11.7970	162.26	7167.83
9677.00	6575703.88	7388.371	-11.4488	162.82	7129.41
9687.00	6575736.75	7388.405	-11.0995	163.37	7090.99
9697.00	6575769.06	7388.437	-10.7489	163.92	7052.56
9707.00	6575800.69	7388.469	-10.3972	164.47	7014.14
9717.00	6575831.75	7388.501	-10.0445	165.02	6975.71
9727.00	6575862.25	7388.532	-9.6906	165.56	6937.28
9737.00	6575892.13	7388.561	-9.3358	166.11	6898.84
9747.00	6575921.38	7388.590	-8.9800	166.66	6860.40
9757.00	6575949.94	7388.618	-8.6232	167.20	6821.97
9767.00	6575977.94	7388.645	-8.2656	167.74	6783.52

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2-2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	**** RADIAL DISTANCE ØF THE C G (M)	MAGNITUDE ØF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (KM)
9777.00	6576005.19	7388.683	-7.9071	168.28	6745.08
9787.00	6576031.88	7388.742	-7.5478	168.82	6706.64
9797.00	6576058.00	7388.800	-7.1877	169.36	6668.19
9807.00	6576083.50	7388.856	-6.8269	169.90	6629.74
9817.00	6576108.38	7388.911	-6.4653	170.44	6591.29
9827.00	6576132.69	7388.965	-6.1031	170.98	6552.83
9837.00	6576156.31	7389.017	-5.7403	171.51	6514.38
9847.00	6576179.44	7389.069	-5.3769	172.05	6475.92
9855.00	6576197.44	7389.187	-5.0857	172.48	6445.15
9857.00	6576201.81	7392.834	-5.0129	172.58	6437.46
9857.50	6576202.94	7395.501	-4.9947	172.61	6435.53
9867.00	6576227.56	7451.058	-4.6469	173.12	6398.82
9877.00	6576248.25	7511.096	-4.2775	173.66	6359.88
9887.00	6576251.38	7572.214	-3.9048	174.20	6320.62
9897.00	6576251.50	7634.267	-3.5286	174.75	6281.04
9907.00	6576265.00	7697.238	-3.1492	175.30	6241.12
9917.00	6576309.44	7761.114	-2.7664	175.86	6200.88
9927.00	6576403.44	7825.875	-2.3803	176.42	6160.29
9937.00	6576566.31	7891.515	-1.9910	176.99	6119.37
9947.00	6576818.31	7958.011	-1.5984	177.56	6078.10
9957.00	6577180.13	8025.399	-1.2026	178.13	6036.48
9967.00	6577673.19	8093.682	-0.8036	178.71	5994.50
9977.00	6578319.50	8166.376	-0.4014	179.30	5952.16
9987.00	6579141.38	8246.691	0.0042	179.89	5909.42
9997.00	6580167.25	8328.408	0.4134	-179.52	5866.26
10007.00	6581425.69	8411.487	0.8262	-178.92	5822.68
10017.00	6582943.81	8495.964	1.2423	-178.31	5778.67
10027.00	6584750.69	8581.899	1.6619	-177.69	5734.22
10037.00	6586877.38	8669.359	2.0847	-177.07	5689.33
10047.00	6589355.50	8758.399	2.5109	-176.44	5644.00
10057.00	6592217.88	8849.072	2.9402	-175.81	5598.22
10067.00	6595498.06	8941.441	3.3726	-175.16	5551.99
10077.00	6599230.50	9035.578	3.8080	-174.51	5505.29
10087.00	6603450.75	9131.562	4.2464	-173.85	5458.14
10097.00	6608194.94	9229.477	4.6876	-173.19	5410.52
10107.00	6613500.00	9329.422	5.1315	-172.52	5362.44
10117.00	6619403.94	9431.497	5.5780	-171.83	5313.88
10127.00	6625944.63	9535.817	6.0270	-171.15	5264.85
10137.00	6633161.00	9642.352	6.4783	-170.45	5215.34
10147.00	6641092.00	9751.400	6.9318	-169.74	5165.35
10157.00	6649776.88	9863.125	7.3873	-169.02	5114.88
10167.00	6659254.06	9977.687	7.8447	-168.30	5063.91
10177.00	6669560.94	10095.306	8.3038	-167.57	5012.46
10187.00	6680708.38	10216.061	8.7644	-166.82	4960.52
10197.00	6692723.50	10340.541	9.2263	-166.07	4908.07
10203.80	6701429.13	10427.456	9.5412	-165.55	4872.11
10207.00	6705673.38	10427.354	9.6893	-165.31	4855.15
10213.80	6714982.94	10419.432	10.0028	-164.78	4819.17

S-IVB GCS2

TLI

**** MEASURED FROM THE CENTER ØF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM MOTION (SEC)	X COORDINATE		Y COORDINATE		Z COORDINATE		X COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)		Y COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)		Z COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)	
	EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	X COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)	Y COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)	Z COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)	X COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)	Y COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)	Z COMPONENT OF VEL. IN THE XXE SYSTEM (M/S)
9287.00	-11664156.375	-1116065.391	-3738365.188	4340.873	-2608.320	-5377.602						
9297.00	-11620415.875	-1142099.531	-3791895.438	4407.111	-2598.447	-5328.327						
9307.00	-11576016.000	-1168033.141	-3844925.594	4472.796	-2588.200	-5278.395						
9317.00	-11530962.000	-1193862.297	-3897461.094	4537.920	-2577.579	-5227.813						
9327.00	-11485259.375	-1219583.406	-3949483.625	4602.475	-2566.585	-5176.587						
9337.00	-11438914.375	-1245192.781	-4000990.719	4666.450	-2555.220	-5124.727						
9347.00	-11391932.375	-1270686.625	-4051976.094	4729.841	-2543.484	-5072.236						
9357.00	-11344319.500	-1296061.234	-4102433.344	4792.641	-2531.378	-5019.120						
9367.00	-11296081.625	-1321312.953	-4152356.469	4854.840	-2518.903	-4965.389						
9377.00	-11247224.625	-1346438.078	-4201739.188	4916.430	-2506.060	-4911.048						
9387.00	-11197755.000	-1371432.906	-4250575.375	4977.404	-2492.850	-4856.105						
9397.00	-11147678.750	-1396293.859	-4298859.250	5037.753	-2479.274	-4800.567						
9407.00	-11097002.000	-1421017.188	-4346584.750	5097.471	-2465.334	-4744.441						
9417.00	-11045731.375	-1445599.297	-4393746.125	5156.550	-2451.031	-4687.735						
9427.00	-10993873.250	-1470036.578	-4440337.563	5214.982	-2436.366	-4630.457						
9437.00	-10941434.000	-1494325.406	-4486353.375	5272.760	-2421.340	-4572.612						
9447.00	-10888420.125	-1518462.203	-4531787.938	5329.883	-2405.954	-4514.205						
9457.00	-10834838.500	-1542443.281	-4576635.625	5386.338	-2390.209	-4455.245						
9467.00	-10780695.625	-1566265.156	-4620891.063	5442.110	-2374.110	-4395.750						
9477.00	-10725998.625	-1589524.313	-4664548.813	5497.200	-2357.657	-4335.720						
9487.00	-10670754.000	-1613417.063	-4707603.750	5551.601	-2340.839	-4275.166						
9497.00	-10614968.875	-1636739.969	-4750050.438	5605.305	-2323.682	-4214.089						
9507.00	-10558650.250	-1659889.531	-4791883.813	5658.306	-2306.174	-4152.501						
9517.00	-10501805.125	-1682662.281	-4833098.750	5710.596	-2288.320	-4090.409						
9527.00	-1044446.625	-1705454.750	-4873690.313	5762.170	-2270.119	-4027.821						
9537.00	-10386564.125	-1728263.500	-4913653.563	5813.021	-2251.574	-3964.746						
9547.00	-10328182.750	-1750685.094	-4952983.625	5863.142	-2232.687	-3901.192						
9557.00	-10269303.750	-1772516.125	-4951675.813	5912.528	-2213.460	-3837.166						
9567.00	-10209934.625	-1794553.172	-5029725.375	5961.177	-2193.893	-3772.671						
9577.00	-10150082.750	-1816792.875	-5067127.813	6009.074	-2173.991	-3707.727						
9587.00	-10089755.625	-1838431.906	-5103878.438	6056.216	-2153.757	-3642.339						
9597.00	-10028961.000	-1859666.906	-5139973.063	6102.597	-2133.191	-3576.513						
9607.00	-9967706.250	-1881694.625	-5175407.313	6148.212	-2112.295	-3510.259						
9617.00	-9905995.250	-1902111.703	-5210176.813	6193.055	-2091.073	-3443.585						
9627.00	-9843847.750	-1922514.984	-5244277.625	6237.121	-2069.525	-3376.499						
9637.00	-9781259.500	-1943501.172	-5277705.563	6280.405	-2047.655	-3309.007						
9647.00	-9718242.250	-1963867.031	-5310456.438	6322.903	-2025.464	-3241.117						
9657.00	-9654804.000	-1984009.406	-5342526.625	6364.605	-2002.957	-3172.848						
9667.00	-9590952.750	-2003925.094	-5373912.125	6405.506	-1980.135	-3104.202						
9677.00	-9526696.625	-2023611.063	-5404609.438	6445.604	-1957.001	-3035.188						
9687.00	-9462043.375	-2043664.094	-5434614.688	6484.892	-1933.558	-2965.815						
9697.00	-9397001.500	-2062281.063	-5463924.563	6523.369	-1909.790	-2896.094						
9707.00	-9331578.750	-2081258.938	-5492535.375	6561.026	-1885.735	-2826.027						
9717.00	-9265783.625	-2099594.750	-5520444.000	6597.857	-1861.379	-2755.631						
9727.00	-9199624.375	-2118485.500	-5547647.000	6633.859	-1836.725	-2684.912						
9737.00	-9133109.250	-2136728.281	-5574141.188	6669.030	-1811.775	-2613.877						
9747.00	-9066246.625	-2154720.031	-5599923.500	6703.363	-1786.532	-2542.535						
9757.00	-8999044.875	-2172457.938	-5624990.875	6736.854	-1761.000	-2470.900						
9767.00	-8931512.375	-2189539.094	-5649340.563	6769.498	-1735.182	-2398.980						

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	X CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Y CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Z CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	X CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)	Y CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)	Z CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)
9777.00	-8863657.625	-2207160.625	-5672969.500	6801.305	-1709.082	-2326.785
9787.00	-8795489.000	-2224119.813	-5695875.250	6832.277	-1682.707	-2254.329
9797.00	-8727015.000	-2240813.844	-5718055.250	6862.390	-1656.054	-2181.616
9807.00	-8658244.125	-2257239.969	-5739506.750	6891.642	-1629.128	-2108.653
9817.00	-8589185.000	-2273395.500	-5760227.500	6920.027	-1601.930	-2035.449
9827.00	-8519846.375	-2289277.688	-5780215.063	6947.544	-1574.465	-1962.015
9837.00	-8450237.000	-2304883.875	-5799467.000	6974.187	-1546.736	-1888.360
9847.00	-8380365.625	-2320211.531	-5817981.500	6999.954	-1518.746	-1814.493
9855.00	-8324285.438	-2332271.344	-5832260.563	7020.010	-1496.192	-1755.267
9857.00	-8310238.688	-2335258.406	-5835756.688	7028.391	-1491.219	-1741.259
9857.50	-8306723.563	-2336003.813	-5836626.563	7032.156	-1490.296	-1738.168
9867.00	-8239560.438	-2350079.156	-5852869.313	7107.848	-1473.074	-1681.315
9877.00	-8168074.188	-2364716.500	-5869358.938	7189.524	-1454.149	-1616.402
9887.00	-8095768.938	-2379159.531	-5885196.500	7271.555	-1434.299	-1551.034
9897.00	-8022642.625	-2393399.875	-5900378.625	7353.729	-1413.655	-1485.352
9907.00	-7948694.063	-2407430.406	-5914902.750	7435.995	-1392.347	-1419.443
9917.00	-7873922.500	-2421244.813	-5928766.813	7518.328	-1370.438	-1353.337
9927.00	-7798327.313	-2434837.281	-5941968.875	7600.711	-1347.962	-1287.051
9937.00	-7721908.063	-2448202.250	-5954507.375	7683.142	-1324.936	-1220.612
9947.00	-7644664.375	-2461334.188	-5966380.688	7765.602	-1301.367	-1154.036
9957.00	-7566595.813	-2474227.813	-5977587.688	7848.141	-1277.262	-1087.348
9967.00	-7479701.313	-2486877.656	-5988127.313	7930.768	-1252.620	-1020.561
9977.00	-7407973.938	-2499278.906	-5997999.375	8016.990	-1227.713	-954.099
9987.00	-7327339.563	-2511432.063	-6007211.625	8109.970	-1203.010	-888.298
9997.00	-7245773.000	-2523341.938	-6015765.000	8203.445	-1178.786	-822.399
10007.00	-7163268.188	-2535002.594	-6023660.250	8297.641	-1153.119	-756.694
10017.00	-7079817.813	-2546401.000	-6030899.188	8392.552	-1126.440	-691.091
10027.00	-6995414.688	-2557529.469	-6037482.313	8488.199	-1099.161	-625.545
10037.00	-6910051.063	-2568382.406	-6043410.313	8584.658	-1071.336	-560.067
10047.00	-6823718.563	-2578554.188	-6048683.938	8682.003	-1042.916	-494.676
10057.00	-6736407.875	-2589238.594	-6053304.125	8780.307	-1013.856	-429.387
10067.00	-6648108.938	-2599229.094	-6057272.000	8879.652	-984.131	-364.200
10077.00	-6558811.000	-2608518.906	-6060588.438	8980.125	-953.716	-299.114
10087.00	-6468502.313	-2618301.031	-6063254.563	9081.820	-922.583	-234.117
10097.00	-6377170.250	-2627368.094	-6065271.063	9184.840	-890.697	-169.192
10107.00	-6284800.813	-2636112.344	-6066638.500	9289.297	-858.020	-104.306
10117.00	-6191379.188	-2644525.688	-6067357.188	9395.306	-824.500	-39.413
10127.00	-6096889.125	-2652599.219	-6067426.438	9502.997	-790.059	25.566
10137.00	-6001313.938	-2660323.625	-6066845.313	9612.349	-754.645	90.679
10147.00	-5904635.625	-2667688.531	-6065612.000	9723.668	-718.155	156.037
10157.00	-5806833.563	-2674683.063	-6063723.188	9837.117	-680.553	221.822
10167.00	-5707885.625	-2681295.313	-6061173.500	9952.859	-641.674	288.246
10177.00	-5607768.125	-2687511.625	-6057955.125	10071.106	-601.279	355.652
10187.00	-5506455.375	-2693300.469	-6054036.688	10191.909	-556.378	428.181
10197.00	-5403919.563	-2698641.188	-6049397.313	10315.783	-511.867	499.412
10203.80	-5333465.813	-2702020.563	-6045838.563	10401.946	-481.874	546.954
10207.00	-5300190.375	-2703544.625	-6044054.375	10401.164	-471.389	568.588
10213.80	-5229480.375	-2706677.313	-6040031.688	10391.578	-449.786	614.286

S-IVB REIGNITION

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TLI

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	COMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN PITCH (DEG)	COMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN YAW (DEG)	COMMANDED ROLL ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN ROLL (DEG)
9287.00	19.745	20.2575	0.152	0.136000	0.000	-0.14135924
9297.00	19.065	19.7773	0.163	0.138121	0.000	-0.23855080
9307.00	18.386	19.2968	0.175	0.146727	0.000	-0.14720224
9317.00	17.706	18.8163	0.186	0.155311	0.000	-0.05578159
9327.00	17.027	18.3352	0.198	0.164164	0.000	0.03571325
9337.00	16.347	17.7838	0.209	0.174812	0.000	0.10563084
9347.00	15.668	17.0119	0.220	0.188930	0.000	0.10638135
9357.00	14.989	16.1829	0.232	0.208759	0.000	0.09278508
9367.00	14.309	15.3377	0.243	0.236346	0.000	0.07959298
9377.00	13.630	14.4720	0.254	0.273652	0.000	0.06696830
9387.00	12.950	13.5817	0.266	0.322637	0.000	0.05512107
9397.00	12.271	12.6627	0.277	0.385258	0.000	0.04431576
9407.00	11.592	11.7110	0.288	0.463464	0.000	0.03488035
9417.00	10.912	10.7224	0.299	0.559201	0.000	0.02721666
9427.00	10.233	9.6927	0.310	0.674401	0.000	0.02181215
9437.00	9.554	8.6179	0.321	0.810985	0.000	0.01925295
9447.00	8.874	7.6102	0.332	0.970917	0.000	0.05667327
9457.00	8.195	7.0974	0.343	1.150777	0.000	0.09535978
9467.00	7.516	6.6993	0.354	1.348721	0.000	-0.01591685
9477.00	6.836	6.2438	0.365	1.568232	0.000	0.01376365
9487.00	6.157	5.7232	0.375	1.370739	0.000	0.00292878
9497.00	5.478	5.1412	0.386	1.056331	0.000	-0.00401011
9507.00	4.798	4.4974	0.397	0.771501	0.000	-0.01300745
9517.00	4.119	3.7904	0.407	0.516971	0.000	-0.02432448
9527.00	3.440	3.0186	0.418	0.293472	0.000	-0.03813426
9537.00	2.760	2.1806	0.428	0.101748	0.000	-0.05451351
9547.00	2.081	1.2749	0.439	-0.057451	0.000	-0.07343421
9557.00	1.402	0.3050	0.449	-0.183370	0.000	-0.09310445
9567.00	0.722	-0.4816	0.460	-0.275642	0.000	-0.03014459
9577.00	0.043	-0.9666	0.470	-0.331063	0.000	0.08446702
9587.00	-0.636	-1.4840	0.480	-0.345321	0.000	0.04192117
9597.00	-1.315	-2.0763	0.490	-0.320508	0.000	-0.01481247
9607.00	-1.995	-2.7452	0.500	-0.253037	0.000	-0.01481247
9617.00	-2.674	-3.4921	0.510	-0.148157	0.000	0.04307003
9627.00	-3.353	-4.3183	0.520	-0.005265	0.000	0.10246068
9637.00	-4.032	-5.1841	0.530	0.176325	0.000	0.17783980
9647.00	-4.712	-5.8350	0.540	0.404173	0.000	0.17554989
9657.00	-5.391	-6.3807	0.549	0.672872	0.000	0.22505154
9667.00	-6.070	-7.0028	0.559	0.977884	0.000	0.15983131
9677.00	-6.749	-7.7117	0.568	1.323654	0.000	0.08747068
9687.00	-7.429	-8.5040	0.578	1.711321	0.000	0.02352126
9697.00	-8.108	-9.2626	0.587	1.611725	0.000	0.05002781
9707.00	-8.787	-9.8555	0.596	1.249207	0.000	0.12237551
9717.00	-9.466	-10.4653	0.606	0.929330	0.000	0.17292056
9727.00	-10.146	-11.1676	0.615	0.654078	0.000	0.22204968
9737.00	-10.825	-11.9328	0.624	0.427494	0.000	0.18548168
9747.00	-11.504	-12.6162	0.633	0.251747	0.000	0.08334143
9757.00	-12.183	-13.2345	0.642	0.123233	0.000	0.03294859
9767.00	-12.863	-13.9106	0.650	0.042511	0.000	-0.00467485

TABLE 2-2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	COMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN PITCH (DEG)	COMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN YAW (DEG)	COMMANDED ROLL ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN ROLL (DEG)
9777.00	-13.542	-14.6306	0.659	0.007021	0.000	-0.02593487
9787.00	-14.221	-15.2599	0.667	-0.026205	0.000	-0.01171641
9797.00	-14.900	-15.8585	0.676	-0.070653	0.000	0.00996029
9807.00	-15.579	-16.4503	0.684	-0.126308	0.000	0.03112578
9817.00	-16.259	-17.0353	0.693	-0.193161	0.000	0.05167758
9827.00	-16.938	-17.6134	0.701	-0.270117	0.000	0.04045432
9837.00	-17.617	-18.1846	0.709	-0.357164	0.000	-0.00262197
9847.00	-18.296	-18.7492	0.717	-0.453704	0.000	0.00217981
9855.00	-18.840	-19.1733	0.723	-0.425913	0.000	0.01951280
9857.00	-18.840	-19.1796	0.723	-0.233637	0.000	0.00466751
9857.50	-18.840	-19.1031	0.723	-0.092381	0.000	0.03317127
9867.00	-22.840	-21.1144	-0.405	-0.658771	0.000	-0.13131257
9877.00	-24.722	-24.3841	0.002	-0.345842	0.000	0.11975682
9887.00	-25.421	-25.1936	0.551	0.228988	0.000	-0.18843592
9897.00	-25.913	-25.6970	0.988	0.674196	0.000	-0.13024713
9907.00	-26.346	-26.1325	1.355	1.045740	0.000	-0.02276250
9917.00	-26.746	-26.5346	1.680	1.373911	0.000	0.08836313
9927.00	-27.121	-26.9119	1.986	1.678960	0.000	0.18277871
9937.00	-27.479	-27.2698	2.280	1.972655	0.000	0.18306238
9947.00	-27.824	-27.6144	2.570	2.261791	0.000	0.18359346
9957.00	-28.166	-27.9552	2.853	2.543274	0.000	0.13137064
9967.00	-28.502	-28.2907	3.134	2.823289	0.000	0.08162546
9977.00	-28.811	-28.6060	3.419	3.110629	0.000	0.03826684
9987.00	-28.925	-28.7720	2.782	2.575894	0.000	-0.09507006
9997.00	-29.276	-29.0436	3.479	3.074042	0.000	-0.10299948
10007.00	-29.703	-29.4823	4.226	3.881978	0.000	-0.09171860
10017.00	-30.034	-29.8200	4.566	4.249320	0.000	-0.12215049
10027.00	-30.342	-30.1274	4.803	4.487564	0.000	-0.16433610
10037.00	-30.657	-30.4391	5.058	4.736781	0.000	-0.20058377
10047.00	-30.985	-30.7644	5.335	5.010564	0.000	-0.17824911
10057.00	-31.328	-31.1038	5.618	5.291795	0.000	-0.15506645
10067.00	-31.685	-31.4574	5.901	5.573375	0.000	-0.13180078
10077.00	-32.062	-31.8308	6.183	5.853961	0.000	-0.10907272
10087.00	-32.462	-32.2255	6.464	6.133571	0.000	-0.08675644
10097.00	-32.890	-32.6490	6.744	6.412143	0.000	-0.06525089
10107.00	-33.353	-33.1054	7.023	6.689630	0.000	-0.04451402
10117.00	-33.856	-33.6177	7.303	6.975540	0.000	-0.02453366
10127.00	-34.391	-34.1506	7.574	7.247706	0.000	-0.00491681
10137.00	-34.995	-34.7468	7.884	7.554488	0.000	0.01812969
10147.00	-35.669	-35.4122	8.141	7.812505	0.000	0.03597861
10157.00	-36.438	-36.1712	8.400	8.068648	0.000	0.05383410
10167.00	-37.334	-37.0530	8.674	8.341222	0.000	0.07489940
10177.00	-40.168	-38.9836	9.563	8.924915	0.000	0.10240751
10187.00	-40.765	-40.6220	9.466	9.166730	0.000	-0.10018786
10197.00	-40.250	-40.0916	9.320	9.019212	0.000	-0.21027236
10203.80	-40.145	-39.9469	9.285	8.972311	0.000	-0.18159041
10207.00	-40.145	-39.9447	9.285	8.971084	0.000	-0.16424425
10213.80	-40.145	-39.9389	9.285	8.968140	0.000	-0.12742721

S-IVB REIGNITION

S-IVB 90 THRUST

S-IVB GCS2

TLI

TABLE 2-2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAW ALONG VELOCITY VECTØR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTOR (KG)
9287.00	9.86	0.00	1.6768	0.0000	0.0000	0.0000
9297.00	9.86	0.00	1.7227	0.0000	0.0000	0.0000
9307.00	9.86	0.00	1.7953	0.0000	0.0000	0.0000
9317.00	9.86	0.00	1.8861	0.0000	0.0000	0.0000
9327.00	9.86	0.00	1.9924	0.0000	0.0000	0.0000
9337.00	9.86	0.00	2.0729	0.0000	0.0000	0.0000
9347.00	9.86	0.00	2.0362	0.0000	0.0000	0.0000
9357.00	9.86	0.00	1.9808	0.0000	0.0000	0.0000
9367.00	9.86	0.00	1.9361	0.0000	0.0000	0.0000
9377.00	9.86	0.00	1.9082	0.0000	0.0000	0.0000
9387.00	9.86	0.00	1.9054	0.0000	0.0000	0.0000
9397.00	9.86	0.00	1.9386	0.0000	0.0000	0.0000
9407.00	9.86	0.00	2.0197	0.0000	0.0000	0.0000
9417.00	9.86	0.00	2.1603	0.0000	0.0000	0.0000
9427.00	9.86	0.00	2.3695	0.0000	0.0000	0.0000
9437.00	9.87	0.00	2.6535	0.0000	0.0000	0.0000
9447.00	9.87	0.00	2.9536	0.0000	0.0000	0.0000
9457.00	9.87	0.00	3.0299	0.0000	0.0000	0.0000
9467.00	9.87	0.00	3.0980	0.0000	0.0000	0.0000
9477.00	9.87	0.00	3.2373	0.0000	0.0000	0.0000
9487.00	9.87	0.00	3.0055	0.0000	0.0000	0.0000
9497.00	9.87	0.00	2.6762	0.0000	0.0000	0.0000
9507.00	9.87	0.00	2.3898	0.0000	0.0000	0.0000
9517.00	9.87	0.00	2.1498	0.0000	0.0000	0.0000
9527.00	9.87	0.00	1.9653	0.0000	0.0000	0.0000
9537.00	9.87	0.00	1.8529	0.0000	0.0000	0.0000
9547.00	9.87	0.00	1.8346	0.0000	0.0000	0.0000
9557.00	9.87	0.00	1.9270	0.0000	0.0000	0.0000
9567.00	9.87	0.00	1.9418	0.0000	0.0000	0.0000
9577.00	9.87	0.00	1.7583	0.0000	0.0000	0.0000
9587.00	9.87	0.00	1.6315	0.0000	0.0000	0.0000
9597.00	9.87	0.00	1.5895	0.0000	0.0000	0.0000
9607.00	9.87	0.00	1.6323	0.0000	0.0000	0.0000
9617.00	9.87	0.00	1.7537	0.0000	0.0000	0.0000
9627.00	9.87	0.00	1.9543	0.0000	0.0000	0.0000
9637.00	9.87	0.00	2.2101	0.0000	0.0000	0.0000
9647.00	9.88	0.00	2.3685	0.0000	0.0000	0.0000
9657.00	9.88	0.00	2.5169	0.0000	0.0000	0.0000
9667.00	9.88	0.00	2.7551	0.0000	0.0000	0.0000
9677.00	9.88	0.00	3.0770	0.0000	0.0000	0.0000
9687.00	9.88	0.00	3.4730	0.0000	0.0000	0.0000
9697.00	9.88	0.00	3.4089	0.0000	0.0000	0.0000
9707.00	9.88	0.00	3.0396	0.0000	0.0000	0.0000
9717.00	9.88	0.00	2.7181	0.0000	0.0000	0.0000
9727.00	9.88	0.00	2.4852	0.0000	0.0000	0.0000
9737.00	9.88	0.00	2.3398	0.0000	0.0000	0.0000
9747.00	9.88	0.00	2.1982	0.0000	0.0000	0.0000
9757.00	9.88	0.00	2.0554	0.0000	0.0000	0.0000
9767.00	9.88	0.00	1.9875	0.0000	0.0000	0.0000

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTOR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTOR (KG)
9777.00	9.88	0.00	1.9839	0.0000	0.0000	0.0000
9787.00	9.88	0.00	1.9203	0.0000	0.0000	0.0000
9797.00	9.88	0.00	1.8275	0.0000	0.0000	0.0000
9807.00	9.88	0.00	1.7219	0.0000	0.0000	0.0000
9817.00	9.88	0.00	1.6036	0.0000	0.0000	0.0000
9827.00	9.88	0.00	1.4733	0.0000	0.0000	0.0000
9837.00	9.88	0.00	1.3310	0.0000	0.0000	0.0000
9847.00	9.88	0.00	1.1773	0.0000	0.0000	0.0000
9855.00	9.88	0.00	1.1246	0.0000	0.0000	0.0000
9857.00	9.89	0.00	1.2316	0.0000	0.0000	0.0000
9857.50	9.89	0.00	1.3266	0.0000	0.0000	0.0000
9867.00	9.96	0.00	1.8761	0.0000	0.0000	0.0000
9877.00	10.04	0.00	4.4229	0.0000	0.0000	0.0000
9887.00	10.13	0.00	4.6834	0.0000	0.0000	0.0000
9897.00	10.21	0.00	4.6669	0.0000	0.0000	0.0000
9907.00	10.29	0.00	4.6050	0.0000	0.0000	0.0000
9917.00	10.38	0.00	4.5345	0.0000	0.0000	0.0000
9927.00	10.46	0.00	4.4703	0.0000	0.0000	0.0000
9937.00	10.55	0.00	4.4239	0.0000	0.0000	0.0000
9947.00	10.63	0.00	4.4049	0.0000	0.0000	0.0000
9957.00	10.72	0.00	4.4163	0.0000	0.0000	0.0000
9967.00	10.80	0.00	4.4601	0.0000	0.0000	0.0000
9977.00	10.88	0.00	4.5342	0.0000	0.0000	0.0000
9987.00	10.97	0.00	3.8191	0.0000	0.0000	0.0000
9997.00	11.06	0.00	4.1432	0.0000	0.0000	0.0000
10007.00	11.15	0.00	4.8428	0.0000	0.0000	0.0000
10017.00	11.23	0.00	5.1075	0.0000	0.0000	0.0000
10027.00	11.31	0.00	5.2561	0.0000	0.0000	0.0000
10037.00	11.38	0.00	5.4295	0.0000	0.0000	0.0000
10047.00	11.45	0.00	5.6374	0.0000	0.0000	0.0000
10057.00	11.51	0.00	5.8584	0.0000	0.0000	0.0000
10067.00	11.56	0.00	6.0818	0.0000	0.0000	0.0000
10077.00	11.61	0.00	6.3031	0.0000	0.0000	0.0000
10087.00	11.65	0.00	6.5196	0.0000	0.0000	0.0000
10097.00	11.69	0.00	6.7285	0.0000	0.0000	0.0000
10107.00	11.74	0.00	6.9280	0.0000	0.0000	0.0000
10117.00	11.78	0.00	7.1237	0.0000	0.0000	0.0000
10127.00	11.81	0.00	7.2989	0.0000	0.0000	0.0000
10137.00	11.84	0.00	7.4975	0.0000	0.0000	0.0000
10147.00	11.86	0.00	7.6392	0.0000	0.0000	0.0000
10157.00	11.87	0.00	7.7751	0.0000	0.0000	0.0000
10167.00	11.88	0.00	7.9337	0.0000	0.0000	0.0000
10177.00	11.88	0.00	8.5363	0.0000	0.0000	0.0000
10187.00	11.88	0.00	8.9196	0.0000	0.0000	0.0000
10197.00	11.89	0.00	8.4193	0.0000	0.0000	0.0000
10203.80	11.90	0.00	8.2198	0.0000	0.0000	0.0000
10207.00	11.86	0.00	8.1942	0.0000	0.0000	0.0000
10213.80	11.76	0.00	8.1534	0.0000	0.0000	0.0000

S-IVB REIGNITION

S-IVB 90 THRUST

S-IVB GCS2

TLI

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM MOTION (SEC)	X PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Y PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Z PØSI CØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)
9287.00	-2248746.9	-142380.6	-6175830.3	7317.261	-25.030	-2669.068
9297.00	-2175417.9	-142621.4	-6202086.8	7348.359	-23.124	-2582.182
9307.00	-2101783.2	-142843.1	-6227472.7	7378.425	-21.214	-2494.934
9317.00	-2027852.9	-143045.6	-6251984.3	7407.456	-19.300	-2407.335
9327.00	-1953637.5	-143225.0	-6275618.3	7435.446	-17.382	-2319.399
9337.00	-1879147.5	-143393.3	-6298371.2	7462.392	-15.460	-2231.141
9347.00	-1804393.2	-143538.2	-6320240.0	7488.291	-13.535	-2142.569
9357.00	-1729385.1	-143664.0	-6341221.6	7513.140	-11.608	-2053.694
9367.00	-1654133.9	-143770.4	-6361312.9	7536.934	-9.677	-1964.531
9377.00	-1578650.0	-143857.5	-6380511.3	7559.671	-7.744	-1875.092
9387.00	-1502544.0	-143925.3	-6398813.9	7581.346	-5.809	-1785.389
9397.00	-1427026.6	-143973.7	-6416218.2	7601.957	-3.872	-1695.437
9407.00	-1350908.4	-144002.7	-6432721.8	7621.502	-1.933	-1605.246
9417.00	-1274600.1	-144012.3	-6448322.4	7639.976	0.007	-1514.830
9427.00	-1198112.5	-144002.5	-6463017.7	7657.379	1.949	-1424.201
9437.00	-1121456.1	-143973.3	-6476805.8	7673.706	3.891	-1333.373
9447.00	-1044641.9	-143924.7	-6489684.5	7688.958	5.834	-1242.349
9457.00	-967680.6	-143856.7	-6501652.1	7703.130	7.778	-1151.149
9467.00	-890582.9	-143769.2	-6512707.0	7716.220	9.721	-1059.798
9477.00	-813359.8	-143662.2	-6522847.6	7728.226	11.665	-968.297
9487.00	-736022.0	-143535.8	-6532072.5	7739.147	13.620	-876.660
9497.00	-658580.5	-143389.9	-6540380.4	7748.981	15.563	-784.900
9507.00	-581046.0	-143224.5	-6547770.1	7757.727	17.504	-693.030
9517.00	-503429.6	-143039.8	-6554240.7	7765.384	19.444	-601.063
9527.00	-425742.0	-142835.6	-6559191.1	7771.951	21.383	-509.011
9537.00	-347994.2	-142612.1	-6564420.6	7777.426	23.320	-416.887
9547.00	-270197.1	-142369.2	-6568128.6	7781.809	25.255	-324.705
9557.00	-192361.6	-142107.0	-6570914.6	7785.100	27.188	-232.476
9567.00	-114498.8	-141825.5	-6572778.0	7787.297	29.118	-140.206
9577.00	-36619.4	-141524.7	-6573718.7	7788.400	31.045	-47.922
9587.00	41265.6	-141204.6	-6573736.4	7788.410	32.970	44.365
9597.00	119145.2	-140865.3	-6572831.4	7787.326	34.891	136.646
9607.00	197008.5	-140506.8	-6571003.6	7785.149	36.808	228.908
9617.00	274844.5	-140129.1	-6568253.4	7781.878	38.722	321.137
9627.00	352642.4	-129732.4	-6564581.0	7777.514	40.632	413.322
9637.00	430391.2	-129316.5	-6559987.1	7772.058	42.537	505.452
9647.00	508079.9	-138881.6	-6554472.2	7765.510	44.438	597.515
9657.00	585697.7	-138427.8	-6548037.1	7757.871	46.334	689.487
9667.00	663233.7	-137955.0	-6540682.8	7749.144	48.225	781.362
9677.00	740677.0	-137463.3	-6532410.3	7739.328	50.111	873.128
9687.00	818016.6	-136952.8	-6523220.6	7728.424	51.991	964.771
9697.00	895241.8	-136423.4	-6513115.3	7716.435	53.882	1056.283
9707.00	972341.7	-135875.2	-6502095.4	7703.362	55.751	1147.646
9717.00	1049305.5	-135308.4	-6490162.9	7689.208	57.612	1238.844
9727.00	1126122.3	-134723.0	-6477319.2	7673.973	59.467	1329.868
9737.00	1202781.3	-134119.1	-6463566.1	7657.661	61.315	1420.707
9747.00	1279271.9	-133496.7	-6448905.7	7640.272	63.156	1511.349
9757.00	1355583.2	-132856.0	-6433339.9	7621.811	64.989	1601.775
9767.00	1431704.6	-132197.0	-6416870.9	7602.280	66.814	1691.976

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	X PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Y PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	Z PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (M)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (M/S)
9777.00	1507625.3	-131519.7	-6399501.1	7581.693	68.632	1781.944
9787.00	1583334.9	-130824.3	-6381232.8	7560.062	76.441	1871.666
9797.00	1658823.0	-130110.9	-6362068.6	7537.369	72.242	1961.124
9807.00	1734678.8	-129379.5	-6342011.3	7513.617	74.035	2050.307
9817.00	1809091.8	-128630.3	-6321063.5	7488.809	75.818	2139.202
9827.00	1883851.5	-127863.2	-6299228.3	7462.949	77.592	2227.796
9837.00	1958347.3	-127078.4	-6276508.6	7436.041	79.357	2316.078
9847.00	2032568.8	-126276.1	-6252907.8	7408.088	81.111	2404.035
9855.00	2091741.8	-125621.6	-6233394.7	7385.051	82.498	2474.184
9857.00	2106507.7	-125456.3	-6228428.2	7382.598	82.782	2492.892
9857.50	2110199.3	-125414.9	-6227180.4	7383.636	82.840	2498.142
9867.00	2180456.6	-124618.7	-6202970.9	7407.693	84.652	2598.824
9877.00	2254653.6	-123768.8	-6176427.8	7431.596	85.476	2710.127
9887.00	2329087.5	-122907.4	-6148766.4	7455.094	86.917	2822.273
9897.00	2403754.0	-122028.7	-6119980.4	7478.126	88.887	2934.999
9907.00	2478648.6	-121128.2	-6090064.9	7500.730	91.294	3048.171
9917.00	2553767.2	-120201.6	-6059015.7	7522.917	94.090	3161.738
9927.00	2629105.6	-119245.2	-6026829.0	7544.691	97.245	3275.665
9937.00	2704659.7	-118255.5	-5993501.3	7566.067	100.754	3389.929
9947.00	2780425.6	-117228.9	-5959029.4	7587.042	104.614	3504.501
9957.00	2856399.4	-116162.0	-5923410.2	7607.666	108.832	3619.391
9967.00	2932577.8	-115051.1	-5886640.5	7627.955	113.413	3734.596
9977.00	3008962.9	-113892.2	-5848714.3	7651.025	118.540	3851.784
9987.00	3085616.5	-112678.1	-5809593.5	7679.721	124.155	3972.453
9997.00	3162558.5	-111412.1	-5769264.6	7708.684	129.227	4093.426
10007.00	3237990.3	-110088.7	-5727722.4	7737.704	135.657	4215.117
10017.00	3317312.9	-108695.9	-5684960.0	7766.846	143.011	4337.483
10027.00	3395128.0	-107226.8	-5640970.8	7796.220	150.893	4460.464
10037.00	3473238.4	-105676.4	-5595748.6	7825.904	159.263	4584.085
10047.00	3551647.3	-104039.7	-5549286.8	7855.953	168.177	4708.387
10057.00	3630358.8	-102310.9	-5501578.4	7886.418	177.689	4833.420
10067.00	3709377.3	-100483.8	-5452615.8	7917.355	187.829	4959.239
10077.00	3788707.7	-98552.1	-5402390.8	7948.823	198.624	5085.906
10087.00	3868355.7	-96509.1	-5350894.6	7980.883	210.102	5213.496
10097.00	3948327.5	-94347.7	-5298117.5	8013.593	222.297	5342.095
10107.00	4028629.9	-92060.7	-5244048.9	8047.011	235.239	5471.808
10117.00	4109270.3	-89640.3	-5188677.3	8081.189	248.966	5602.755
10127.00	4190256.3	-87078.5	-5131989.1	8116.161	263.529	5735.098
10137.00	4271595.8	-84366.9	-5073970.6	8151.865	278.955	5868.877
10147.00	4353296.6	-81496.3	-5014605.8	8188.456	295.319	6004.390
10157.00	4435367.9	-78457.5	-4953876.3	8225.949	312.596	6141.887
10167.00	4517818.4	-75241.1	-4891760.5	8264.308	330.865	6281.672
10177.00	4600656.6	-71836.6	-4828233.9	8303.446	350.230	6424.197
10187.00	4683874.4	-68226.5	-4763247.9	8340.330	371.852	6573.414
10197.00	4767474.2	-64398.9	-4696763.9	8380.184	393.725	6723.494
10203.80	4824569.0	-61669.9	-4650686.4	8409.215	408.768	6826.019
10207.00	4851441.9	-60360.4	-4628814.6	8391.402	409.576	6847.923
10213.80	4908366.2	-57572.8	-4582098.1	8347.547	410.131	6889.289

S-1VB REIGNITION

S-1VB 90 THRUST

S-1VB GCS2

TLI

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED PITCH CHANL (DEG)	GIMBAL DEFLECTION ANG REQUIRED YAW CHANNEL (DEG)	GIMBAL DEFLECTION ANG REQUIRED ROLL CHANNEL (DEG)	EULERIAN ATTITUDE RATE IN PITCH (DEG/S)	EULERIAN ATTITUDE RATE IN YAW (DEG/S)	EULERIAN ATTITUDE RATE IN ROLL (DEG/S)
9287.00	0.3689	-0.012528	0.000000	-0.04800695	-0.00011310	-0.01582052
9297.00	0.5303	-0.018671	0.000000	-0.00066167	0.00066167	0.00901542
9307.00	0.6914	-0.021163	0.000000	-0.04805047	0.00073610	0.00901542
9317.00	0.8526	-0.023910	0.000000	-0.04804927	0.00081053	0.00901542
9327.00	1.0130	-0.026586	0.000000	-0.04823921	0.00097680	0.00901542
9337.00	1.0990	-0.027579	0.000000	-0.06643097	0.00132937	0.00321908
9347.00	1.0090	-0.024807	0.000000	-0.08195375	0.00181295	-0.00156181
9357.00	0.8863	-0.017047	0.000000	-0.08363760	0.00247351	-0.00164653
9367.00	0.7501	-0.002808	0.000000	-0.08547555	0.00333072	-0.00164653
9377.00	0.5970	0.019498	0.000000	-0.08772477	0.00438448	-0.00164653
9387.00	0.4236	0.051456	0.000000	-0.09038531	0.00563467	-0.00164653
9397.00	0.2266	0.094650	0.000000	-0.09345723	0.00708117	-0.00164653
9407.00	0.0026	0.150660	0.000000	-0.09694057	0.00872387	-0.00164653
9417.00	-0.2516	0.221066	0.000000	-0.10083541	0.01056264	-0.00164653
9427.00	-0.5395	0.307442	0.000000	-0.10514179	0.01259737	-0.00164653
9437.00	-0.8644	0.411361	0.000000	-0.10985977	0.01482795	-0.00164653
9447.00	-1.0977	0.534814	0.000000	-0.07634367	0.01731685	-0.01108534
9457.00	-0.9242	0.673131	0.000000	-0.03694561	0.01859889	-0.01472232
9467.00	-0.7026	0.825777	0.000000	-0.04263450	0.02060864	0.00345104
9477.00	-0.5267	0.997115	0.000000	-0.04846843	0.02282689	0.00404543
9487.00	-0.4048	0.774323	0.000000	-0.05514409	-0.03289310	-0.00192121
9497.00	-0.3319	0.513779	0.000000	-0.06125144	-0.02997812	-0.00192121
9507.00	-0.3092	0.277303	0.000000	-0.06750516	-0.02699544	-0.00192121
9517.00	-0.3379	0.065470	0.000000	-0.07390529	-0.02394511	-0.00192121
9527.00	-0.4191	-0.121143	0.000000	-0.08045184	-0.02082718	-0.00192121
9537.00	-0.5540	-0.281960	0.000000	-0.08714484	-0.01764169	-0.00192121
9547.00	-0.7439	-0.416400	0.000000	-0.09398431	-0.01438870	-0.00192121
9557.00	-0.9811	-0.523875	0.000000	-0.09603511	-0.01106534	-0.00029475
9567.00	-1.0301	-0.603102	0.000000	-0.05668792	-0.00756334	0.01502658
9577.00	-0.8650	-0.651272	0.000000	-0.04804588	-0.00327640	0.00035343
9587.00	-0.7408	-0.667580	0.000000	-0.05544990	0.00067825	-0.01194114
9597.00	-0.6771	-0.652293	0.000000	-0.06305847	0.00481813	0.00605277
9607.00	-0.6765	-0.601807	0.000000	-0.07076871	0.00858903	0.00605277
9617.00	-0.7392	-0.520869	0.000000	-0.0782039	0.01243777	0.00605277
9627.00	-0.8661	-0.408885	0.000000	-0.08661344	0.01636449	0.00605277
9637.00	-1.0097	-0.265106	0.000000	-0.07883922	0.02038470	0.01129381
9647.00	-0.9616	-0.084005	0.000000	-0.05322633	0.02502083	0.00346583
9657.00	-0.8575	0.129714	0.000000	-0.05777730	0.02886468	-0.00245200
9667.00	-0.8192	0.372190	0.000000	-0.06642886	0.03269099	-0.00866023
9677.00	-0.8518	0.648057	0.000000	-0.07521559	0.03677513	-0.00866023
9687.00	-0.9486	0.958026	0.000000	-0.08044223	0.04091894	-0.00743748
9697.00	-0.9981	0.793344	0.000000	-0.06535827	-0.03811127	0.01027064
9707.00	-0.9197	0.496619	0.000000	-0.05675911	-0.03407811	0.00378353
9717.00	-0.8722	0.234745	0.000000	-0.06558840	-0.02956867	0.00395118
9727.00	-0.9002	0.009248	0.000000	-0.07480242	-0.02498272	0.00403904
9737.00	-0.9696	-0.176599	0.000000	-0.07419301	-0.01968009	-0.01413365
9747.00	-0.9610	-0.322571	0.000000	-0.06194192	-0.01512860	-0.00689304
9757.00	-0.9129	-0.429688	0.000000	-0.06324418	-0.01043682	-0.00406413
9767.00	-0.9184	-0.497812	0.000000	-0.07161459	-0.00566436	-0.00351446

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED		GIMBAL DEFLECTION ANG REQUIRED		GIMBAL DEFLECTION ANG REQUIRED		EULERIAN ATTITUDE RATE IN		EULERIAN ATTITUDE RATE IN		EULERIAN ATTITUDE RATE IN	
	YAW CHANNEL (DEG)	PITCH CHANNEL (DEG)	YAW CHANNEL (DEG)	PITCH CHANNEL (DEG)	YAW CHANNEL (DEG)	PITCH CHANNEL (DEG)	YAW (DEG/S)	PITCH (DEG/S)	YAW (DEG/S)	PITCH (DEG/S)	YAW (DEG/S)	PITCH (DEG/S)
9777.00	-0.9470	-0.530952	0.000000	-0.06717636	-0.00279059	-0.00014262						
9787.00	-0.8998	-0.565765	0.000000	-0.06020056	-0.00389653	0.00221725						
9797.00	-0.8340	-0.609544	0.000000	-0.05952110	-0.00499491	0.00221725						
9807.00	-0.7627	-0.662353	0.000000	-0.05884206	-0.00609357	0.00221725						
9817.00	-0.6858	-0.724187	0.000000	-0.05816343	-0.00719252	0.00221725						
9827.00	-0.6032	-0.794128	0.000000	-0.05746963	-0.00810538	-0.00399594						
9837.00	-0.5148	-0.872516	0.000000	-0.05678250	-0.00926590	-0.00399594						
9847.00	-0.4213	-0.957973	0.000000	-0.05611755	-0.01009875	0.00412891						
9855.00	-0.3183	-0.996663	0.000000	-0.04937341	0.03513705	-0.02713175						
9857.00	-0.1643	-0.540998	0.000000	0.11457224	0.24127607	0.04149767						
9857.50	-0.0341	-0.355937	0.000000	0.18498392	0.31405346	0.06867917						
9867.00	0.2657	-0.485904	0.000000	-1.16692996	-0.29046723	0.02909369						
9877.00	0.1805	-0.230206	0.000000	-0.09537979	0.05365619	-0.04277696						
9887.00	0.1311	-0.213507	0.000000	-0.05501631	0.04865311	-0.00246207						
9897.00	0.1305	-0.215584	0.000000	-0.04609909	0.03979106	0.00990024						
9907.00	0.1321	-0.216785	0.000000	-0.04171686	0.03452502	0.01015610						
9917.00	0.1335	-0.217730	0.000000	-0.03869681	0.03154414	0.01041571						
9927.00	0.1337	-0.219506	0.000000	-0.03622544	0.03025122	-0.00113076						
9937.00	0.1351	-0.221396	0.000000	-0.03518125	0.02875467	-0.00109759						
9947.00	0.1363	-0.221671	0.000000	-0.03417534	0.02877066	-0.00662937						
9957.00	0.1374	-0.223500	0.000000	-0.03369050	0.02841087	-0.00655409						
9967.00	0.1385	-0.224807	0.000000	-0.03324084	0.02762868	-0.00654891						
9977.00	0.1403	-0.222449	0.000000	-0.02660061	0.02782284	-0.00496093						
9987.00	0.1423	-0.267781	0.000000	0.01872313	-0.10400310	-0.01338574						
9997.00	0.1408	-0.228116	0.000000	-0.04853983	0.10283912	0.00051830						
10007.00	0.1414	-0.226835	0.000000	-0.03792192	0.05306485	-0.00409573						
10017.00	0.1434	-0.230727	0.000000	-0.03092407	0.02611113	-0.00628323						
10027.00	0.1439	-0.232876	0.000000	-0.03070245	0.02345324	-0.00659552						
10037.00	0.1452	-0.234673	0.000000	-0.03205681	0.02609758	-0.00059390						
10047.00	0.1463	-0.235634	0.000000	-0.03343375	0.02796426	-0.00064026						
10057.00	0.1472	-0.237128	0.000000	-0.03480899	0.02789021	-0.00094757						
10067.00	0.1489	-0.238483	0.000000	-0.03608743	0.02783170	-0.00122720						
10077.00	0.1495	-0.239590	0.000000	-0.03803154	0.02774209	-0.00164711						
10087.00	0.1509	-0.240529	0.000000	-0.04064539	0.02811677	-0.00213748						
10097.00	0.1521	-0.242006	0.000000	-0.04363328	0.02772889	-0.00278907						
10107.00	0.1532	-0.243270	0.000000	-0.04718999	0.02772191	-0.00348981						
10117.00	0.1500	-0.242853	0.000000	-0.05880695	0.03141585	-0.00527226						
10127.00	0.1546	-0.245713	0.000000	-0.05518231	0.02710642	-0.00507244						
10137.00	0.1558	-0.246543	0.000000	-0.06213277	0.02874091	-0.00613087						
10147.00	0.1569	-0.248459	0.000000	-0.06989422	0.02484790	-0.00790764						
10157.00	0.1576	-0.249734	0.000000	-0.07998277	0.02669795	-0.00938947						
10167.00	0.1589	-0.250504	0.000000	-0.09436377	0.02786546	-0.01154498						
10177.00	0.2530	-0.285303	0.000000	-0.09016677	0.33254286	-0.13798033						
10187.00	0.1610	-0.253069	0.000000	0.06598764	-0.01586048	-0.01738698						
10197.00	0.1605	-0.253861	0.000000	0.04776391	-0.01518248	0.01008585						
10203.80	0.1575	-0.252915	0.000000	-0.00170559	0.00044478	0.00531462						
10207.00	0.1610	-0.254543	0.000000	0.00083619	-0.00043041	0.00554495						
10213.80	0.1655	-0.256974	0.000000	0.00083591	-0.00043094	0.005554495						

S-IVB GCS2

TLI

TABLE 2.2 (ILI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABOVE SPHERICAL EARTH (FT)	MAGNITUDE ØF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (FT/S**2)
9287.00	0.00	296465.14	674720.4	25554.11	89.96	0.000
9297.00	0.00	296464.67	674730.0	25554.05	89.96	0.000
9307.00	0.00	296464.20	674736.2	25553.99	89.96	0.000
9317.00	0.00	296463.73	674737.7	25553.94	89.96	0.000
9327.00	10.75	296463.26	674735.3	25553.88	89.96	0.000
9337.00	31.35	296462.79	674729.4	25553.83	89.96	0.000
9347.00	10.53	296462.32	674719.6	25553.78	89.96	0.000
9357.00	5.46	296461.85	674705.7	25553.73	89.96	0.000
9367.00	6.83	296461.37	674689.1	25553.69	89.96	0.000
9377.00	8.21	296460.90	674668.4	25553.65	89.97	0.000
9387.00	9.58	296460.43	674644.5	25553.61	89.97	0.000
9397.00	10.96	296459.96	674617.0	25553.57	89.97	0.000
9407.00	12.33	296459.49	674586.5	25553.53	89.97	0.000
9417.00	13.71	296459.02	674552.7	25553.50	89.97	0.000
9427.00	15.08	296458.55	674515.8	25553.47	89.97	0.000
9437.00	16.46	296458.08	674476.3	25553.44	89.97	0.000
9447.00	35.69	296457.61	674433.1	25553.41	89.97	0.000
9457.00	18.76	296457.14	674387.4	25553.39	89.97	0.000
9467.00	19.25	296456.67	674338.4	25553.36	89.97	0.000
9477.00	24.98	296456.19	674287.0	25553.34	89.97	0.000
9487.00	20.22	296455.72	674233.3	25553.32	89.97	0.000
9497.00	20.71	296455.25	674176.7	25553.30	89.97	0.000
9507.00	21.20	296454.78	674118.1	25553.28	89.97	0.000
9517.00	21.68	296454.31	674056.9	25553.27	89.97	0.000
9527.00	22.17	296453.84	673994.1	25553.25	89.97	0.000
9537.00	22.65	296453.37	673929.0	25553.23	89.97	0.000
9547.00	23.14	296452.90	673861.8	25553.22	89.97	0.000
9557.00	23.63	296452.43	673792.9	25553.21	89.97	0.000
9567.00	28.27	296451.96	673722.0	25553.19	89.97	0.000
9577.00	24.60	296451.49	673649.4	25553.18	89.97	0.000
9587.00	25.09	296451.01	673574.6	25553.17	89.97	0.000
9597.00	25.57	296450.54	673498.8	25553.16	89.97	0.000
9607.00	26.06	296450.07	673421.3	25553.15	89.97	0.000
9617.00	26.55	296449.60	673342.2	25553.14	89.97	0.000
9627.00	27.03	296449.13	673262.5	25553.13	89.97	0.000
9637.00	30.63	296448.66	673180.7	25553.12	89.97	0.000
9647.00	28.00	296448.19	673098.7	25553.12	89.97	0.000
9657.00	28.49	296447.72	673014.9	25553.11	89.97	0.000
9667.00	28.98	296447.25	672930.6	25553.11	89.98	0.000
9677.00	29.46	296446.78	672845.0	25553.10	89.98	0.000
9687.00	29.95	296446.30	672759.1	25553.09	89.98	0.000
9697.00	30.44	296445.83	672671.8	25553.09	89.98	0.000
9707.00	30.92	296445.36	672583.6	25553.08	89.98	0.000
9717.00	31.41	296444.89	672495.7	25553.08	89.98	0.000
9727.00	31.90	296444.42	672406.7	25553.08	89.98	0.000
9737.00	32.38	296443.95	672317.3	25553.08	89.98	0.000
9747.00	32.87	296443.48	672228.0	25553.07	89.98	0.000
9757.00	33.36	296443.01	672138.0	25553.07	89.98	0.000
9767.00	33.84	296442.54	672047.8	25553.07	89.98	0.000

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	TOTAL ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABOVE SPHERICAL EARTH (FT)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (FT/S**2)
9777.00	137.87	296440.05	671956.9	25553.11	89.98	0.000
9787.00	137.87	296434.13	671866.3	25553.22	89.98	0.000
9797.00	137.87	296428.21	671775.7	25553.33	89.98	0.000
9807.00	137.87	296422.29	671685.3	25553.44	89.98	0.000
9817.00	137.87	296416.37	671595.3	25553.55	89.98	0.000
9827.00	137.87	296410.46	671505.3	25553.66	89.98	0.000
9837.00	137.87	296404.54	671415.5	25553.77	89.98	0.000
9847.00	137.87	296398.61	671326.2	25553.87	89.98	0.000
9855.00	386.22	296075.14	671255.2	25554.22	89.98	0.000
9857.00	160943.19	295946.47	671237.4	25566.17	89.98	0.000
9857.50	159821.48	295914.30	671233.1	25574.91	89.98	0.000
9867.00	177538.63	292067.18	671161.3	25757.20	89.98	0.000
9877.00	178552.38	287950.96	671069.5	25954.06	89.99	0.000
9887.00	178737.62	283821.65	670922.3	26154.53	90.00	0.000
9897.00	178784.77	275688.88	670766.0	26358.13	90.00	0.000
9907.00	178786.09	275554.07	670655.9	26564.80	89.98	0.000
9917.00	178695.50	271419.70	670648.8	26774.49	89.95	0.000
9927.00	178557.74	267287.20	670806.6	26987.14	89.91	0.000
9937.00	178319.65	263156.96	671192.0	27202.72	89.86	0.000
9947.00	178183.04	259029.92	671871.7	27421.16	89.79	0.000
9957.00	178038.09	254904.91	672913.8	27642.59	89.71	0.000
9967.00	177870.08	250781.92	674388.7	27867.02	89.62	0.000
9977.00	202785.90	246440.78	676368.9	28105.99	89.51	0.000
9987.00	202635.82	241711.35	678928.0	28370.05	89.39	0.000
9997.00	202554.70	236977.00	682158.0	28638.68	89.25	0.000
10007.00	202633.28	23242.83	686153.8	28911.94	89.10	0.000
10017.00	202539.20	227509.54	691003.9	29189.91	88.93	0.000
10027.00	202488.44	222777.75	696804.2	29472.75	88.75	0.000
10037.00	202460.47	218047.12	703656.1	29760.67	88.55	0.000
10047.00	202432.27	213317.20	711664.2	30053.85	88.33	0.000
10057.00	202400.23	208588.05	720935.8	30352.49	88.10	0.000
10067.00	202367.96	203859.52	731580.9	30656.78	87.85	0.000
10077.00	202333.61	195131.88	743712.8	30966.97	87.59	0.000
10087.00	202294.39	194404.95	757447.8	31283.32	87.31	0.000
10097.00	202252.31	185679.08	772905.3	31606.10	87.01	0.000
10107.00	202204.46	184954.13	790205.9	31935.64	86.70	0.000
10117.00	202140.53	180230.57	809474.3	32272.27	86.38	0.000
10127.00	202071.79	175508.66	830835.0	32616.38	86.04	0.000
10137.00	201719.23	170791.70	854415.6	32967.86	85.68	0.000
10147.00	201648.54	166076.72	880344.2	33327.70	85.31	0.000
10157.00	201548.78	161364.18	908749.1	33696.43	84.93	0.000
10167.00	201434.24	156654.24	939756.0	34074.59	84.54	0.000
10177.00	201690.84	151945.43	973488.5	34462.91	84.14	0.000
10187.00	201514.03	147232.86	1009980.9	34861.78	83.75	0.000
10197.00	201405.26	142523.63	1049323.4	35272.90	83.34	0.000
10203.80	201327.17	135222.33	1077833.8	35559.90	83.04	0.000
10207.00	0.00	135222.08	1091734.9	35560.04	82.90	0.000
10213.80	0.00	135222.08	1122229.3	35535.00	82.59	0.000

S-IVB REIGNITION

S-IVB 90 THRUST

S-IVB GC52

TLI

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	*** RADIUS DISTANCE ØF THE C G (FT)	MAGNITUDE ØF THE EARTH FIXED VELOCITY (FT/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØBLATE EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (NM)
9287.00	21568375.50	24234.975	-23.6993	139.73	4657.21
9297.00	21568537.75	24235.088	-23.4299	140.37	4636.55
9307.00	21568699.25	24235.203	-23.1576	141.00	4615.89
9317.00	21568860.25	24235.318	-22.8824	141.64	4595.23
9327.00	21569020.25	24235.436	-22.6044	142.27	4574.56
9337.00	21569179.50	24235.557	-22.3236	142.89	4553.89
9347.00	21569338.00	24235.679	-22.0400	143.52	4533.21
9357.00	21569495.50	24235.805	-21.7537	144.14	4512.53
9367.00	21569652.00	24235.932	-21.4648	144.76	4491.85
9377.00	21569807.75	24236.061	-21.1732	145.37	4471.17
9387.00	21569962.25	24236.193	-20.8791	145.99	4450.48
9397.00	21570115.75	24236.326	-20.5825	146.60	4429.79
9407.00	21570268.25	24236.461	-20.2835	147.21	4409.10
9417.00	21570419.50	24236.598	-19.9820	147.81	4388.40
9427.00	21570569.50	24236.736	-19.6782	148.42	4367.70
9437.00	21570718.50	24236.875	-19.3720	149.02	4347.00
9447.00	21570866.00	24237.016	-19.0636	149.62	4326.30
9457.00	21571012.00	24237.157	-18.7530	150.22	4305.59
9467.00	21571156.25	24237.300	-18.4402	150.81	4284.88
9477.00	21571299.25	24237.441	-18.1252	151.40	4264.17
9487.00	21571441.25	24237.581	-17.8082	151.99	4243.46
9497.00	21571581.50	24237.723	-17.4892	152.58	4222.75
9507.00	21571720.75	24237.864	-17.1681	153.16	4202.03
9517.00	21571858.25	24238.004	-16.8452	153.75	4181.31
9527.00	21571994.00	24238.144	-16.5203	154.33	4160.59
9537.00	21572128.50	24238.282	-16.1935	154.91	4139.87
9547.00	21572261.25	24238.419	-15.8650	155.48	4119.14
9557.00	21572392.75	24238.556	-15.5347	156.06	4098.42
9567.00	21572522.50	24238.690	-15.2026	156.63	4077.69
9577.00	21572650.25	24238.824	-14.8689	157.20	4056.96
9587.00	21572776.25	24238.957	-14.5336	157.77	4036.23
9597.00	21572900.50	24239.087	-14.1966	158.34	4015.49
9607.00	21573023.25	24239.217	-13.8581	158.90	3994.76
9617.00	21573144.25	24239.344	-13.5181	159.47	3974.02
9627.00	21573263.25	24239.469	-13.1767	160.03	3953.29
9637.00	21573380.75	24239.592	-12.8338	160.59	3932.55
9647.00	21573496.50	24239.713	-12.4895	161.15	3911.81
9657.00	21573610.00	24239.832	-12.1439	161.71	3891.06
9667.00	21573721.50	24239.948	-11.7970	162.26	3870.32
9677.00	21573831.50	24240.063	-11.4488	162.82	3849.57
9687.00	21573939.25	24240.174	-11.0995	163.37	3828.83
9697.00	21574045.25	24240.279	-10.7489	163.92	3808.08
9707.00	21574149.00	24240.385	-10.3972	164.47	3787.33
9717.00	21574251.00	24240.489	-10.0445	165.02	3766.58
9727.00	21574351.00	24240.589	-9.6906	165.56	3745.83
9737.00	21574449.00	24240.686	-9.3358	166.11	3725.08
9747.00	21574545.00	24240.780	-8.9800	166.66	3704.32
9757.00	21574638.75	24240.872	-8.6232	167.20	3683.57
9767.00	21574730.50	24240.959	-8.2656	167.74	3662.81

**** MEASURED FROM THE CENTER ØF THE EARTH IN THE SPACE FIXED. EARTH CENTERED PLUMBLINE SYSTEM

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	*** RADIUS OF THE C G (FT)	MAGNITUDE OF THE EARTH FIXED VELOCITY (FT/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (NM)
9777.00	21574820.00	24241.087	-7.9071	168.28	3642.05
9787.00	21574907.50	24241.280	-7.5478	168.82	3621.29
9797.00	21574993.25	24241.469	-7.1877	169.36	3600.53
9807.00	21575077.00	24241.654	-6.8269	169.90	3579.77
9817.00	21575158.50	24241.835	-6.4653	170.44	3559.01
9827.00	21575238.25	24242.011	-6.1031	170.98	3538.25
9837.00	21575315.75	24242.183	-5.7403	171.51	3517.48
9847.00	21575391.75	24242.351	-5.3769	172.05	3496.72
9855.00	21575450.75	24242.740	-5.0857	172.48	3480.10
9857.00	21575465.00	24254.705	-5.0129	172.58	3475.95
9857.50	21575468.75	24263.453	-4.9947	172.61	3474.91
9867.00	21575549.50	24445.728	-4.6469	173.12	3455.09
9877.00	21575617.50	24642.703	-4.2775	173.66	3434.06
9887.00	21575627.75	24843.222	-3.9048	174.20	3412.86
9897.00	21575628.00	25046.806	-3.5286	174.75	3391.49
9907.00	21575672.50	25253.406	-3.1492	175.30	3369.94
9917.00	21575818.25	25462.971	-2.7664	175.86	3348.21
9927.00	21576126.50	25675.441	-2.3803	176.42	3326.29
9937.00	21576661.00	25890.798	-1.9910	176.99	3304.19
9947.00	21577487.75	26108.959	-1.5984	177.56	3281.91
9957.00	21578674.75	26330.050	-1.2026	178.13	3259.44
9967.00	21580292.50	26554.075	-0.8036	178.71	3236.77
9977.00	21582413.00	26792.573	-0.4014	179.30	3213.91
9987.00	21585109.25	27056.073	0.0042	179.89	3190.83
9997.00	21588475.00	27324.173	0.4134	-179.52	3167.53
10007.00	21592603.75	27596.743	0.8262	-178.92	3144.00
10017.00	21597584.50	27873.896	1.2423	-178.31	3120.23
10027.00	21603512.50	28155.836	1.6619	-177.69	3096.23
10037.00	21610490.00	28442.780	2.0847	-177.07	3071.99
10047.00	21618620.25	28734.904	2.5109	-176.44	3047.52
10057.00	21628011.25	29032.387	2.9402	-175.81	3022.80
10067.00	21638773.00	29335.437	3.3726	-175.16	2997.83
10077.00	21651018.50	29644.286	3.8080	-174.51	2972.62
10087.00	21664864.50	29959.193	4.2464	-173.85	2947.16
10097.00	21680429.50	30280.436	4.6876	-173.19	2921.45
10107.00	21697834.50	30608.339	5.1315	-172.52	2895.49
10117.00	21717204.25	30943.230	5.5780	-171.83	2869.27
10127.00	21738663.25	31285.488	6.0270	-171.15	2842.79
10137.00	21762339.00	31635.013	6.4783	-170.45	2816.06
10147.00	21788359.50	31992.783	6.9318	-169.74	2789.07
10157.00	21816853.00	32359.333	7.3873	-169.02	2761.81
10167.00	21847946.25	32735.193	7.8447	-168.30	2734.30
10177.00	21881761.50	33121.083	8.3038	-167.57	2706.51
10187.00	21918334.50	33517.259	8.7644	-166.82	2678.47
10197.00	21957754.00	33925.660	9.2263	-166.07	2650.15
10203.80	21986315.75	34210.813	9.5412	-165.55	2630.73
10207.00	22000240.50	34210.478	9.6893	-165.31	2621.57
10213.80	22030783.75	34184.487	10.0028	-164.78	2602.14

S-IVB REIGNITION

S-IVB 90 THRUST

S-IVB GCS2

TLI

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	X CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Y CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Z CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	COMPONENT ØF VEL. IN THE XXE SYSTEM (FT/S)	COMPONENT ØF VEL. IN THE XXE SYSTEM (FT/S)	COMPONENT ØF VEL. IN THE XXE SYSTEM (FT/S)	Z COMPONENT ØF VEL. IN THE XXE SYSTEM (FT/S)
9287.00	-38268229.500	-3661631.844	-12264977.625	14241.710	-8557.481	-17643.051	-17481.388
9297.00	-38124723.500	-3747645.688	-12440601.750	14459.026	-8525.089	-17481.388	-17317.570
9307.00	-37979055.000	-3832129.719	-12614598.375	14674.528	-8491.469	-17317.570	-17151.618
9317.00	-37831240.000	-3916871.031	-12786945.750	14888.190	-8456.623	-16983.554	-16813.408
9327.00	-37681297.000	-4001257.875	-12957623.375	15099.984	-8420.555	-16641.195	-16466.931
9337.00	-37529246.500	-4085278.125	-13126609.875	15309.877	-8383.268	-16290.645	-16112.361
9347.00	-37375106.000	-4168519.344	-13253884.750	15517.851	-8344.765	-15932.102	-15749.891
9357.00	-37218895.500	-4252169.375	-13459426.875	15723.887	-8305.046	-15565.752	-15379.709
9367.00	-37060635.000	-4335016.188	-13623216.625	15927.952	-8264.117	-15191.787	-15002.009
9377.00	-36900343.000	-4417447.750	-13785233.500	16130.019	-8221.981	-14810.382	-14616.947
9387.00	-36738041.000	-4499451.750	-13945457.125	16330.065	-8178.641	-14421.752	-14224.802
9397.00	-36573749.000	-4581016.563	-14103868.875	16528.062	-8134.102	-13825.753	-13623.690
9407.00	-36407486.500	-4662129.813	-14260448.625	16723.987	-8088.367	-13419.576	-13214.636
9417.00	-36239276.000	-4742779.813	-14415177.500	16917.815	-8041.440	-13007.698	-12799.187
9427.00	-36069137.500	-4822554.625	-14568036.500	17109.522	-7993.327	-12589.128	-12377.530
9437.00	-35897093.000	-4902642.375	-14719007.000	17299.083	-7944.030	-12164.457	-11949.930
9447.00	-35723163.000	-4981831.313	-14868070.625	17486.492	-7893.549	-11733.968	-11516.599
9457.00	-35547370.000	-5060509.438	-15015208.625	17671.711	-7841.893	-11297.851	-11077.752
9467.00	-35369736.000	-5138665.188	-15160403.625	17854.693	-7789.075	-10856.322	-10633.587
9477.00	-35190284.000	-5216287.063	-15303637.750	18035.434	-7735.094	-10409.605	-10184.389
9487.00	-35009035.000	-5293363.000	-15444894.125	18213.915	-7679.918	-9957.966	-9730.363
9497.00	-34826013.000	-5369881.750	-15584154.875	18390.109	-7623.627	-9501.620	-9271.741
9507.00	-34641240.500	-5445831.750	-15721403.500	18563.995	-7566.189	-9040.783	-8808.765
9517.00	-34454741.000	-5521201.688	-15856623.125	18735.551	-7507.610	-8575.711	-8341.650
9527.00	-34266537.000	-5595580.125	-15989797.500	18904.757	-7447.897	-8106.627	-7870.669
9537.00	-34076553.500	-5670155.813	-16120910.500	19071.590	-7387.055	-7692.662	-7400.000
9547.00	-33885113.500	-5743717.438	-16249946.250	19236.030	-7325.090	-7244.434	-6957.966
9557.00	-33691941.000	-5816653.938	-16376889.000	19398.058	-7262.009	-6860.475	-6504.115
9567.00	-33497160.750	-5889553.938	-16501723.625	19557.668	-7197.811	-6456.506	-6026.001
9577.00	-33300796.250	-5960606.500	-16624435.000	19714.811	-7132.518	-6026.001	-5544.145
9587.00	-33102872.500	-6031600.688	-16745007.875	19869.474	-7066.132	-5544.145	-5062.000
9597.00	-32903415.250	-6101525.500	-16863428.500	20021.642	-6998.658	-5062.000	-4579.966
9607.00	-32702448.250	-6171570.250	-16979682.750	20171.298	-6930.103	-4579.966	-4097.932
9617.00	-32499997.250	-6240523.938	-17093755.750	20318.422	-6860.475	-4097.932	-3615.900
9627.00	-32296088.250	-6308776.125	-17205635.000	20462.995	-6789.782	-3615.900	-3133.868
9637.00	-32090746.250	-6376316.125	-17315306.750	20605.005	-6718.028	-3133.868	-2651.836
9647.00	-31883996.750	-6443133.250	-17422757.250	20744.434	-6645.222	-2651.836	-2169.804
9657.00	-31675866.000	-6509217.188	-17527974.250	20881.248	-6571.380	-2169.804	-1687.772
9667.00	-31466380.250	-6574557.375	-17630945.250	21015.440	-6496.506	-1687.772	-1205.740
9677.00	-31255566.250	-6639143.875	-17731658.000	21146.994	-6420.608	-1205.740	-723.708
9687.00	-31043449.250	-6702566.125	-17830100.500	21275.893	-6343.693	-723.708	-241.676
9697.00	-30830057.250	-6766013.938	-17926261.500	21402.130	-6265.715	-241.676	20.356
9707.00	-30615415.750	-6828277.313	-18020129.000	21525.675	-6186.793	20.356	72.324
9717.00	-30399552.500	-6889746.500	-18111692.750	21646.512	-6106.886	72.324	124.292
9727.00	-30182494.500	-6950411.688	-18200941.500	21764.631	-6026.001	124.292	176.260
9737.00	-29964269.000	-7010263.375	-18287864.750	21880.018	-5944.145	176.260	228.228
9747.00	-29744903.500	-7069291.375	-18372452.250	21992.662	-5861.326	228.228	280.196
9757.00	-29524425.250	-7127486.625	-18454694.250	22102.540	-5777.559	280.196	332.164
9767.00	-29302862.000	-7184839.500	-18534581.750	22209.640	-5692.854	332.164	384.132

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	X CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Y CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	Z CØRDINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (FT)	X COMPONENT ØF VEL. IN THE XXE SYSTEM (FT/S)	Y COMPONENT ØF VEL. IN THE XXE SYSTEM (FT/S)	Z COMPONENT ØF VEL. IN THE XXE SYSTEM (FT/S)
9777.00	-29080241.500	-7241340.563	-18612104.500	22313.993	-5607.225	-7633.809
9787.00	-28856591.000	-7296581.000	-18687254.500	22415.607	-5520.691	-7396.093
9797.00	-28631938.750	-7351751.438	-18760023.750	22514.404	-5433.248	-7157.532
9807.00	-28406312.500	-7405642.875	-18830402.500	22610.373	-5344.907	-6918.151
9817.00	-28179740.750	-7458646.625	-18858384.000	22703.502	-5255.676	-6677.983
9827.00	-27952251.750	-7510753.500	-18963960.000	22793.778	-5165.568	-6437.058
9837.00	-27723874.500	-7561554.938	-19027122.500	22881.191	-5074.593	-6195.406
9847.00	-27494637.750	-7612242.500	-19087865.750	22965.729	-4982.763	-5953.059
9855.00	-27310647.500	-7651608.813	-19134713.000	23031.528	-4908.765	-5758.750
9857.00	-27264562.500	-7661608.875	-19146183.250	23059.026	-4892.450	-5712.791
9857.50	-27253030.000	-7664054.438	-19149037.250	23071.378	-4889.424	-5702.652
9867.00	-27032678.500	-7710233.438	-19202327.000	23319.710	-4832.920	-5516.124
9877.00	-26798143.500	-7758256.188	-19256426.750	23587.678	-4770.830	-5303.156
9887.00	-26560921.500	-7805641.438	-19308387.250	23856.807	-4705.705	-5088.693
9897.00	-26321005.750	-7852361.750	-19358197.500	24126.409	-4637.976	-4873.201
9907.00	-26078392.500	-7898293.688	-19405848.750	24396.310	-4568.069	-4656.967
9917.00	-25833079.000	-7943716.563	-19451334.500	24666.432	-4496.188	-4440.082
9927.00	-25585063.250	-7988211.250	-19494648.500	24936.716	-4422.448	-4222.609
9937.00	-25334344.000	-8032159.563	-19535785.250	25207.158	-4346.903	-4004.632
9947.00	-25080919.750	-8075243.375	-19574735.750	25477.698	-4269.577	-3786.209
9957.00	-24824789.250	-8117545.250	-19611508.000	25748.492	-4190.492	-3567.415
9967.00	-24565949.000	-8159047.375	-19646086.750	26019.581	-4109.646	-3348.298
9977.00	-24304376.250	-8199733.875	-19678475.500	26302.459	-4027.931	-3130.246
9987.00	-24039827.750	-8239006.438	-19708699.500	26607.512	-3946.884	-2914.363
9997.00	-23773221.000	-8278680.875	-19736761.750	26914.189	-3867.410	-2698.160
10007.00	-23501536.000	-8316537.625	-19762664.750	27223.232	-3783.200	-2482.593
10017.00	-23227748.500	-8354333.938	-19786414.500	27534.618	-3695.670	-2267.359
10027.00	-22950835.500	-8390644.625	-19808012.750	27848.423	-3606.170	-2052.314
10037.00	-22670771.000	-8426451.375	-19827461.500	28164.889	-3514.881	-1837.490
10047.00	-22387528.000	-8461135.750	-19844763.500	28484.261	-3421.642	-1622.954
10057.00	-22101075.500	-8494877.250	-19859921.500	28806.782	-3326.300	-1408.749
10067.00	-21811380.750	-8527654.375	-19872939.500	29132.716	-3228.777	-1194.882
10077.00	-21518408.750	-8559445.125	-19883820.250	29462.351	-3128.988	-981.344
10087.00	-21222120.250	-8590226.375	-19895267.250	29795.997	-3026.848	-768.101
10097.00	-20922474.500	-8619574.000	-19899183.000	30133.989	-2922.235	-555.093
10107.00	-20619425.000	-8648662.500	-19903665.500	30476.695	-2815.026	-342.211
10117.00	-20312923.750	-8676265.250	-19906027.250	30824.496	-2705.052	-129.306
10127.00	-20002917.000	-8702753.250	-19906254.500	31177.812	-2592.058	83.879
10137.00	-19689350.000	-8728095.750	-19904348.000	31536.577	-2475.869	297.504
10147.00	-19372164.000	-8752258.875	-1990301.750	31901.798	-2356.151	511.931
10157.00	-19051291.000	-8775206.875	-19854104.750	32274.007	-2232.784	727.761
10167.00	-18726658.750	-8796900.500	-19885739.750	32653.738	-2105.228	945.691
10177.00	-18398189.250	-8817295.250	-19875180.750	33041.687	-1972.701	1166.837
10187.00	-18065798.250	-8836287.500	-19862325.000	33438.020	-1825.387	1404.792
10197.00	-17729394.750	-8853609.625	-19847104.000	33844.432	-1679.352	1638.491
10203.80	-17498247.250	-8864896.750	-19835428.250	34127.118	-1580.951	1794.469
10207.00	-17389076.000	-8869897.000	-19829574.500	34124.553	-1546.553	1865.447
10213.80	-17157087.750	-8880174.875	-19816376.750	34093.105	-1475.676	2015.374

S-IVB REIGNITION

S-IVB 90 THRUST

S-IVB GCS2

TLI

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAW ALONG VELOCITY VECTØR (LB)	AERODYNAMIC LOAD (LB/FT**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTOR (FT)
9287.00	9.86	0.00	1.6768	0.0000	0.0000	0.0000
9297.00	9.86	0.00	1.7227	0.0000	0.0000	0.0000
9307.00	9.86	0.00	1.7953	0.0000	0.0000	0.0000
9317.00	9.86	0.00	1.8861	0.0000	0.0000	0.0000
9327.00	9.86	0.00	1.9924	0.0000	0.0000	0.0000
9337.00	9.86	0.00	2.0729	0.0000	0.0000	0.0000
9347.00	9.86	0.00	2.0362	0.0000	0.0000	0.0000
9357.00	9.86	0.00	1.9808	0.0000	0.0000	0.0000
9367.00	9.86	0.00	1.9361	0.0000	0.0000	0.0000
9377.00	9.86	0.00	1.9082	0.0000	0.0000	0.0000
9387.00	9.86	0.00	1.9054	0.0000	0.0000	0.0000
9397.00	9.86	0.00	1.9386	0.0000	0.0000	0.0000
9407.00	9.86	0.00	2.0197	0.0000	0.0000	0.0000
9417.00	9.86	0.00	2.1603	0.0000	0.0000	0.0000
9427.00	9.86	0.00	2.3695	0.0000	0.0000	0.0000
9437.00	9.87	0.00	2.6535	0.0000	0.0000	0.0000
9447.00	9.87	0.00	2.9536	0.0000	0.0000	0.0000
9457.00	9.87	0.00	3.0299	0.0000	0.0000	0.0000
9467.00	9.87	0.00	3.0980	0.0000	0.0000	0.0000
9477.00	9.87	0.00	3.2373	0.0000	0.0000	0.0000
9487.00	9.87	0.00	3.0055	0.0000	0.0000	0.0000
9497.00	9.87	0.00	2.6762	0.0000	0.0000	0.0000
9507.00	9.87	0.00	2.3898	0.0000	0.0000	0.0000
9517.00	9.87	0.00	2.1498	0.0000	0.0000	0.0000
9527.00	9.87	0.00	1.9653	0.0000	0.0000	0.0000
9537.00	9.87	0.00	1.8529	0.0000	0.0000	0.0000
9547.00	9.87	0.00	1.8346	0.0000	0.0000	0.0000
9557.00	9.87	0.00	1.9270	0.0000	0.0000	0.0000
9567.00	9.87	0.00	1.9418	0.0000	0.0000	0.0000
9577.00	9.87	0.00	1.7583	0.0000	0.0000	0.0000
9587.00	9.87	0.00	1.6315	0.0000	0.0000	0.0000
9597.00	9.87	0.00	1.5895	0.0000	0.0000	0.0000
9607.00	9.87	0.00	1.6323	0.0000	0.0000	0.0000
9617.00	9.87	0.00	1.7537	0.0000	0.0000	0.0000
9627.00	9.87	0.00	1.9543	0.0000	0.0000	0.0000
9637.00	9.87	0.00	2.2101	0.0000	0.0000	0.0000
9647.00	9.88	0.00	2.3685	0.0000	0.0000	0.0000
9657.00	9.88	0.00	2.5169	0.0000	0.0000	0.0000
9667.00	9.88	0.00	2.7551	0.0000	0.0000	0.0000
9677.00	9.88	0.00	3.0770	0.0000	0.0000	0.0000
9687.00	9.88	0.00	3.4730	0.0000	0.0000	0.0000
9697.00	9.88	0.00	3.4089	0.0000	0.0000	0.0000
9707.00	9.88	0.00	3.0396	0.0000	0.0000	0.0000
9717.00	9.88	0.00	2.7181	0.0000	0.0000	0.0000
9727.00	9.88	0.00	2.4852	0.0000	0.0000	0.0000
9737.00	9.88	0.00	2.3398	0.0000	0.0000	0.0000
9747.00	9.88	0.00	2.1982	0.0000	0.0000	0.0000
9757.00	9.88	0.00	2.0554	0.0000	0.0000	0.0000
9767.00	9.88	0.00	1.9875	0.0000	0.0000	0.0000

TABLE 2.2 (TLI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTOR (LB)	AERODYNAMIC LOAD (LB/FT**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTOR (FT)
9777.00	9.88	0.00	1.9839	0.0000	0.0000	0.0000
9787.00	9.88	0.00	1.9203	0.0000	0.0000	0.0000
9797.00	9.88	0.00	1.8275	0.0000	0.0000	0.0000
9807.00	9.88	0.00	1.7219	0.0000	0.0000	0.0000
9817.00	9.88	0.00	1.6036	0.0000	0.0000	0.0000
9827.00	9.88	0.00	1.4733	0.0000	0.0000	0.0000
9837.00	9.88	0.00	1.3310	0.0000	0.0000	0.0000
9847.00	9.88	0.00	1.1773	0.0000	0.0000	0.0000
9855.00	9.88	0.00	1.1246	0.0000	0.0000	0.0000
9857.00	9.89	0.00	1.2316	0.0000	0.0000	0.0000
9857.50	9.89	0.00	1.3266	0.0000	0.0000	0.0000
9867.00	9.96	0.00	1.8761	0.0000	0.0000	0.0000
9877.00	10.04	0.00	4.4229	0.0000	0.0000	0.0000
9887.00	10.13	0.00	4.6834	0.0000	0.0000	0.0000
9897.00	10.21	0.00	4.6669	0.0000	0.0000	0.0000
9907.00	10.29	0.00	4.6050	0.0000	0.0000	0.0000
9917.00	10.38	0.00	4.5345	0.0000	0.0000	0.0000
9927.00	10.46	0.00	4.4703	0.0000	0.0000	0.0000
9937.00	10.55	0.00	4.4239	0.0000	0.0000	0.0000
9947.00	10.63	0.00	4.4049	0.0000	0.0000	0.0000
9957.00	10.72	0.00	4.4163	0.0000	0.0000	0.0000
9967.00	10.80	0.00	4.4601	0.0000	0.0000	0.0000
9977.00	10.88	0.00	4.5342	0.0000	0.0000	0.0000
9987.00	10.97	0.00	3.8191	0.0000	0.0000	0.0000
9997.00	11.06	0.00	4.1432	0.0000	0.0000	0.0000
10007.00	11.15	0.00	4.8428	0.0000	0.0000	0.0000
10017.00	11.23	0.00	5.1075	0.0000	0.0000	0.0000
10027.00	11.31	0.00	5.2561	0.0000	0.0000	0.0000
10037.00	11.38	0.00	5.4295	0.0000	0.0000	0.0000
10047.00	11.45	0.00	5.6374	0.0000	0.0000	0.0000
10057.00	11.51	0.00	5.8584	0.0000	0.0000	0.0000
10067.00	11.56	0.00	6.0818	0.0000	0.0000	0.0000
10077.00	11.61	0.00	6.3031	0.0000	0.0000	0.0000
10087.00	11.65	0.00	6.5196	0.0000	0.0000	0.0000
10097.00	11.69	0.00	6.7285	0.0000	0.0000	0.0000
10107.00	11.74	0.00	6.9280	0.0000	0.0000	0.0000
10117.00	11.78	0.00	7.1237	0.0000	0.0000	0.0000
10127.00	11.81	0.00	7.2989	0.0000	0.0000	0.0000
10137.00	11.84	0.00	7.4975	0.0000	0.0000	0.0000
10147.00	11.86	0.00	7.6392	0.0000	0.0000	0.0000
10157.00	11.87	0.00	7.7751	0.0000	0.0000	0.0000
10167.00	11.88	0.00	7.9337	0.0000	0.0000	0.0000
10177.00	11.88	0.00	8.5363	0.0000	0.0000	0.0000
10187.00	11.88	0.00	8.9196	0.0000	0.0000	0.0000
10197.00	11.89	0.00	8.4193	0.0000	0.0000	0.0000
10203.80	11.90	0.00	8.2198	0.0000	0.0000	0.0000
10207.00	11.86	0.00	8.1942	0.0000	0.0000	0.0000
10213.80	11.76	0.00	8.1534	0.0000	0.0000	0.0000

S-IVB GCS2
TLI

TABLE 2.2 (ILI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	X PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Y PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Z PØSI CØØRD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	X VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Y VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Z VEL. CØMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)
9287.00	-7377778.5	-467128.0	-20261910.0	24006.761	-82.119	-8756.785
9297.00	-7137197.9	-467917.9	-20348053.8	24108.789	-75.867	-8471.726
9307.00	-6895614.0	-468645.2	-20431340.8	24207.431	-69.599	-8185.479
9317.00	-6653060.7	-469309.8	-20511759.5	24302.676	-63.319	-7898.082
9327.00	-6409571.9	-469911.6	-20589298.5	24394.509	-57.026	-7609.577
9337.00	-6165181.9	-470450.3	-20663947.3	24482.914	-50.722	-7320.016
9347.00	-5919525.0	-470926.0	-20735695.5	24567.884	-44.407	-7029.427
9357.00	-5673835.7	-471338.4	-20804532.5	24649.408	-38.083	-6737.841
9367.00	-5426948.4	-471687.6	-20870449.0	24727.474	-31.749	-6445.310
9377.00	-5179297.8	-471973.4	-20933435.8	24802.069	-25.408	-6151.875
9387.00	-4930518.6	-472195.7	-20993483.8	24873.182	-19.059	-5857.577
9397.00	-4681845.8	-472354.6	-21050584.5	24940.805	-12.703	-5562.456
9407.00	-4432114.1	-472449.8	-21104730.3	25004.927	-6.342	-5266.555
9417.00	-4181758.9	-472481.4	-21155913.3	25065.539	0.023	-4969.915
9427.00	-3930815.1	-472449.3	-21204126.0	25122.634	6.393	-4672.576
9437.00	-3679318.0	-472353.5	-21249362.5	25176.202	12.766	-4374.582
9447.00	-3427302.9	-472194.0	-21291615.8	25226.240	19.140	-4075.948
9457.00	-3174805.0	-471970.7	-21330879.5	25272.737	25.518	-3776.737
9467.00	-2921860.0	-471683.6	-21367148.8	25315.681	31.895	-3477.026
9477.00	-2668503.3	-471332.8	-21400418.3	25355.072	38.271	-3176.827
9487.00	-2414770.4	-470917.9	-21430684.0	25390.901	44.686	-2876.182
9497.00	-2160697.1	-470439.2	-21457940.8	25423.166	51.059	-2575.132
9507.00	-1906319.0	-469896.7	-21482185.3	25451.861	57.429	-2273.721
9517.00	-1651671.8	-469290.6	-21503414.3	25476.982	63.794	-1971.990
9527.00	-1396791.3	-468620.9	-21521624.0	25498.527	70.155	-1669.983
9537.00	-1147113.2	-467887.6	-21536813.0	25516.490	76.510	-1367.740
9547.00	-886473.4	-467090.7	-21548978.3	25530.870	82.858	-1065.305
9557.00	-631107.8	-466230.4	-21558118.5	25541.666	89.199	-762.717
9567.00	-375652.1	-465306.8	-21564232.3	25548.874	95.532	-459.993
9577.00	-120142.2	-463269.7	-21567318.3	25552.494	101.855	-157.225
9587.00	390896.3	-462156.5	-21564407.3	25548.970	108.169	145.554
9597.00	646353.3	-460980.3	-21558410.8	25541.826	114.471	448.313
9607.00	901720.9	-459741.3	-21549387.5	25531.095	120.762	751.009
9617.00	1156963.2	-458439.5	-21537339.0	25516.778	127.041	1053.600
9627.00	1412044.5	-457075.2	-21522267.0	25498.877	133.307	1356.043
9637.00	1666928.9	-455648.4	-21504173.8	25477.394	139.558	1658.307
9647.00	1921580.5	-454159.3	-21483061.3	25452.334	145.794	1960.350
9657.00	2175963.6	-452608.2	-21458932.8	25423.699	152.015	2262.098
9667.00	2430042.6	-450995.0	-21431792.0	25391.495	158.219	2563.525
9677.00	2683781.6	-449320.1	-21401642.3	25355.723	164.405	2864.592
9687.00	2937145.1	-447583.3	-21368488.3	25316.387	170.574	3165.260
9697.00	3190097.5	-445784.8	-21332334.0	25273.497	176.780	3465.497
9707.00	3442603.2	-443925.1	-21293185.3	25227.059	182.909	3765.243
9717.00	3694626.9	-442004.5	-21251047.0	25177.078	189.016	4064.449
9727.00	3946133.0	-440023.2	-21205925.5	25123.559	195.102	4363.083
9737.00	4197086.3	-437981.3	-21157827.0	25066.510	201.164	4661.111
9747.00	4447451.4	-435879.2	-21106758.0	25005.942	207.203	4958.493
9757.00	4697193.4	-433717.0	-21052726.0	24941.863	213.218	5255.168
9767.00					219.207	5551.101

TABLE 2-2 (TLI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST 0PP

TIME FROM MOTION (SEC)	X POSI CORD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Y POSI CORD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	Z POSI CORD SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT)	X VEL. COMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Y VEL. COMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)	Z VEL. COMP. SPACE FIXED EARTH CENTER PLUMBLIN SYS (FT/S)
9777.00	4946277.1	-421495.1	-20995738.3	24874.321	225.171	5846.272
9787.00	5194668.4	-429213.7	-20935803.0	24803.352	231.107	6140.635
9797.00	5442332.6	-426873.1	-20872928.5	24728.900	237.016	6434.133
9807.00	5689234.8	-424473.5	-20807123.3	24650.973	242.896	6726.727
9817.00	5935340.4	-422015.3	-20738397.3	24569.583	248.747	7018.378
9827.00	6180614.9	-419498.7	-20666759.3	24484.742	254.567	7309.043
9837.00	6425023.9	-416924.0	-20592219.5	24396.460	260.356	7598.681
9847.00	6668532.8	-414291.6	-20514789.0	24304.751	266.113	7887.253
9855.00	6862669.8	-412144.5	-20450769.8	24229.168	270.663	8117.401
9857.00	6911114.5	-411602.1	-20434475.5	24221.122	271.594	8178.779
9857.50	6923225.9	-411466.3	-20430381.8	24224.529	271.785	8196.003
9867.00	7153729.1	-408854.1	-20350954.3	24303.454	277.729	8526.327
9877.00	7397157.5	-406065.6	-20263870.5	24381.875	280.433	8891.491
9887.00	7641363.3	-403239.4	-20173117.8	24458.970	285.159	9259.427
9887.00	7886332.0	-400356.8	-20078675.8	24534.535	291.624	9629.261
9907.00	8132049.3	-397402.1	-19980527.8	24608.695	299.523	10000.560
9917.00	8378501.3	-394362.1	-19878660.3	24681.487	308.693	10373.155
9927.00	8625674.5	-391224.3	-19773060.8	24752.924	319.045	10746.933
9937.00	8873555.4	-387977.3	-19663717.8	24823.055	330.558	11121.813
9947.00	9122131.0	-384609.4	-19550621.3	24891.869	343.223	11497.706
9957.00	9371389.0	-381108.9	-19433760.3	24959.533	357.061	11874.643
9967.00	9621318.1	-377464.2	-19313124.8	25026.100	372.090	12252.610
9977.00	9871925.5	-373662.1	-19188695.0	25101.788	388.911	12637.086
9987.00	10123413.6	-369678.9	-19060346.0	25195.935	407.334	13032.982
9997.00	10375847.9	-36525.2	-18928033.3	25290.958	423.974	13429.874
10007.00	10629233.3	-361183.5	-18791740.3	25386.168	445.070	13829.124
10017.00	10883572.5	-356613.9	-18651443.5	25481.779	469.196	14230.587
10027.00	11138871.4	-351793.9	-18507121.8	25578.150	495.056	14634.069
10037.00	11395139.0	-346707.5	-18358755.0	25675.539	522.515	15039.648
10047.00	11652386.3	-341337.7	-18206321.5	25774.125	551.761	15447.464
10057.00	11910626.0	-335665.7	-18049797.8	25874.074	582.971	15857.678
10067.00	12169872.9	-329671.4	-17898159.3	25975.574	616.237	16270.469
10077.00	12430143.3	-323333.8	-17724379.0	26078.816	651.654	16686.044
10087.00	12691455.8	-316630.9	-17555428.3	26184.000	689.311	17104.645
10097.00	12953830.4	-309539.7	-17382275.0	26291.315	729.320	17526.557
10107.00	13217289.8	-302036.3	-17204884.8	26400.956	771.780	17952.125
10117.00	13481857.8	-294095.5	-17023219.3	26513.087	816.816	18381.744
10127.00	13747560.0	-285690.7	-16837234.5	26627.824	864.595	18815.938
10137.00	14014421.6	-276794.4	-16646885.1	26744.965	915.207	19254.845
10147.00	14282469.0	-267376.3	-16452118.8	26865.012	968.895	19699.443
10157.00	14551731.8	-257406.5	-16252874.8	26988.020	1025.576	20150.547
10167.00	14822238.9	-246853.9	-16049082.9	27113.870	1085.515	20609.159
10177.00	15094017.8	-235684.5	-15840662.3	27242.278	1149.048	21076.760
10187.00	15367042.0	-223840.3	-15627453.8	27363.286	1219.988	21566.320
10197.00	15641319.4	-211282.4	-15409330.4	27494.041	1291.750	22058.706
10203.80	15828638.4	-202329.2	-15258157.5	27589.287	1341.102	22395.075
10207.00	15916804.1	-198032.9	-15186399.4	27530.845	1343.754	22466.938
10213.80	16103563.5	-188887.3	-15033130.3	27386.965	1345.574	22602.655

S-IVB GCS2

TLI

TABLE 2.2 (POST TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST OPP

TIME FROM FIRST MOTION (SEC)	TOTAL VEHICLE ENGINE THRUST (NEWTONS)	TOTAL VEHICLE WEIGHT (KG)	ALTITUDE ABOVE SPHERICAL EARTH (M)	MAGNITUDE OF SPACE FIXED VELOCITY IN XPP SYSTEM (M/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LONGITUDINAL ACCELERATION (M/S**2)
10213.50	196.90	63145.88	337458.3	10831.07	82.50	0.003
10233.50	180.40	63136.79	366705.0	10807.35	81.70	0.003
10283.50	139.09	63114.07	455084.6	10736.55	79.49	0.002
11713.50	0.00	62832.55	701469.5	7622.41	44.97	-0.000

TABLE 2.2 (PØST TLI)
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST NOTION (SEC)	*** RADIAL DISTANCE ØF THE C G (M)	MAGNITUDE ØF THE EARTH FIXED VELOCITY (M/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN ØBLATE EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (KM)
10213.50	6714982.94	0.000	9.9401	-164.53	4801.69
10233.50	6744110.94	0.000	10.8328	-163.28	4715.54
10283.50	6832158.44	0.000	13.0236	-159.49	4457.10
11713.50	13386881.75	0.000	31.1807	-90.88	559.96

**** MEASURED FROM THE CENTER ØF THE EARTH IN THE SPACE FIXED, EARTH CENTERED PLUMBLINE SYSTEM

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST OPP

TIME FROM FIRST NOTION (SEC)	X CØØRCINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Y CØØRCINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	Z CØØRCINATE EARTH FIXED, PT. CENTERED PLUMBIN SYS, (M)	X CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)	Y CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)	Z CØMPØNENT ØF VEL. IN THE XXE SYSTEM (M/S)
10213.50	1177008.531	-268917.313	-6039054.250	10390.226	-435.033	647.035
10233.50	1348200.797	-2697860.438	-6032158.313	10360.823	-387.579	745.124
10283.50	1863919.859	-2713335.219	-5986819.000	10263.526	-232.174	1065.398
11713.50	13212018.000	-1284886.266	-1732120.391	6046.391	1537.232	3300.961

TABLE 2.2 (PØST TLI)									
AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969				AZIMUTH=72.06 DEG 1ST ØPP					
TIME FROM FIRST MOTIØN (SEC)	CØMMANDED PITCH ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN PITCH (DEG)	CØMMANDED YAW ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN YAW (DEG)	CØMMANDED RØLL ATTITUDE (DEG)	EULERIAN ANGLE ØF ATTITUDE IN RØLL (DEG)			
10213.50	-39.939	-39.9389	8.968	8.968140	-0.127	-0.12742721			
10233.50	-48.758	-48.7582	0.957	0.957448	0.000	0.00000000			
10283.50	-53.244	-53.2441	0.971	0.971239	0.000	0.00000000			
11713.50	-123.467	-123.4673	0.459	0.458655	0.000	0.00000000			

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG IST ØPP

TIME FROM FIRST MOTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (NT/M**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTOR (KG)	AERODYNAMIC LOAD (KG/M**2)	MAGNITUDE ØF THE AERODYNAMIC FORCE VECTOR (KG)
10213.50	11.76	0.00	8.1534	0.0950	0.0000	0.0000
10233.50	11.45	0.00	8.6301	0.0576	0.0000	0.0000
10283.50	10.73	0.00	10.9394	0.0149	0.0000	0.0000
11713.50	6.78	0.00	49.8326	0.0004	0.0000	0.0000

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969									
TABLE 2.2 (PØST TLI)									
AZIMUTH=72.06 DEG 1ST ØPP									
TIME	X PØSI CØRD	Y PØSI CØRD	Z PØSI CØRD	X VEL. CØMP.	Y VEL. CØMP.	Z VEL. CØMP.			
FROM	SPACE FIXED	SPACE FIXED	SPACE FIXED	SPACE FIXED	SPACE FIXED	SPACE FIXED	SPACE FIXED	SPACE FIXED	SPACE FIXED
FIRST	EARTH CENTER	EARTH CENTER	EARTH CENTER	EARTH CENTER	EARTH CENTER	EARTH CENTER	EARTH CENTER	EARTH CENTER	EARTH CENTER
MØTION	PLUMBLIN SYS	PLUMBLIN SYS	PLUMBLIN SYS	PLUMBLIN SYS	PLUMBLIN SYS	PLUMBLIN SYS	PLUMBLIN SYS	PLUMBLIN SYS	PLUMBLIN SYS
(SEC)	(M)	(M)	(M)	(M/S)	(M/S)	(M/S)	(M/S)	(M/S)	(M/S)
10213.50	4931064.4	-57035.3	-4557158.5	8323.354	410.664	6918.315			
10233.50	5071539.9	-49478.9	-4445229.8	8218.805	411.594	7005.765			
10283.50	5474114.0	-28825.4	-4087986.7	7882.125	414.315	7278.325			
11713.50	11162068.9	488471.3	7374158.8	1524.686	292.526	7462.636			

AS-506 TRAJ DATA (METRIC UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	GIMBAL DEFLECTION ANG REQUIRED PITCH CHANL (DEG)	GIMBAL DEFLECTION ANG REQUIRED YAW CHANNEL (DEG)	GIMBAL DEFLECTION ANG REQUIRED ROLL CHANNEL (DEG)	EULERIAN ATTITUDE RATE IN PITCH (DEG/S)	EULERIAN ATTITUDE RATE IN YAW (DEG/S)	EULERIAN ATTITUDE RATE IN ROLL (DEG/S)
10213.50	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
10233.50	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
10283.50	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000
11713.50	0.0000	0.000000	0.000000	0.00000000	0.00000000	0.00000000

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MØTION (SEC)	TOTAL VEHICLE ENGINE THRUST (LB)	TOTAL VEHICLE WEIGHT (LB)	ALTITUDE ABØVE SPHERICAL EARTH (FT)	MAGNITUDE ØF SPACE FIXED VELOCITY IN XPP SYSTEM (FT/SEC)	SPACE FIXED PATH ANGLE (DEG)	VEHICLE LØNGITUDINAL ACCELERATION (FT/S**2)
10213.50	44.26	139212.84	1107146.5	35535.00	82.50	0.010
10233.50	40.55	135192.80	1203100.4	35457.20	81.70	0.009
10283.50	31.27	135142.70	1493059.6	35224.91	79.49	0.007
11713.50	0.00	136522.07	23013351.3	25007.92	44.97	-0.000

TABLE 2.2 (POST TLI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	*** RADIAL DISTANCE OF THE C G (FT)	MAGNITUDE OF THE EARTH FIXED VELOCITY (FT/SEC)	SUB VEHICLE LATITUDE ØN SPHERICAL EARTH (DEG)	LONGITUDE ØN ØBLATE EARTH (DEG)	ARC RANGE ALONG SPHER. EARTH TO SUB VEHICLE PT. (NM)
10213.50	22030783.75	0.000	9.9401	-164.53	2592.70
10233.50	22126348.00	0.000	10.8328	-163.28	2546.19
10283.50	22415217.75	0.000	13.0236	-159.49	2406.64
11713.50	43920215.50	0.000	31.1807	-90.88	302.35

**** MEASURED FROM THE CENTER OF THE EARTH IN THE SPACE FIXED, EARTH-CENTERED PLUMBLINE SYSTEM

TABLE 2-2 (PØST TLI)
AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MOTION (SEC)	X COORDINATE			Y COORDINATE			Z COORDINATE			X			Y			Z		
	EARTH FIXED,			EARTH FIXED,			PT. CENTERED			PT. CENTERED			VEL. IN THE			VEL. IN THE		
	PT. CENTERED			PT. CENTERED			PLUMBIN SYS,			PLUMBIN SYS,			XXE SYSTEM			XXE SYSTEM		
	(FT)			(FT)			(FT)			(FT/S)			(FT/S)			(FT/S)		
10213.50	3861576.531	-8822235.250	-19813170.000	34088.669	-1427.275	2122.818												
10233.50	4423230.938	-8851248.125	-19790545.500	33992.200	-1271.583	2444.632												
10283.50	6115222.625	-8902018.375	-19641794.500	33672.984	-761.725	3495.399												
11713.50	43346515.500	-4215506.063	-5682809.625	19837.241	5043.411	10829.923												

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST ØPP

TIME FROM FIRST MØTION (SEC)	MACH NUMBER	DYNAMIC PRESSURE (LB/FT**2)	TOTAL ANGLE ØF ATTACK (DEG)	DRAG ALONG VELOCITY VECTØR (LB)	AERØDYNAMIC LOAD (LB/FT**2)	MAGNITUDE ØF THE AERØDYNAMIC FORCE VECTØR (FT)
10213.50	11.76	0.00	8.1534	0.2094	0.0000	0.0000
10233.50	11.45	0.00	8.6301	0.1270	0.0000	0.0000
10283.50	10.73	0.00	10.9394	0.0330	0.0000	0.0000
11713.50	6.78	0.00	49.8326	0.0008	0.0000	0.0000

AS-506 TRAJ DATA (ENGLISH UNITS) JULY 16 1969 AZIMUTH=72.06 DEG 1ST OPP

TIME FROM FIRST MOTION (SEC)	X POSI CØRD				Y POSI CØRD				Z POSI CØRD				X VEL. CØMP.				Y VEL. CØMP.				Z VEL. CØMP.			
	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT/S)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT/S)	SPACE FIXED	EARTH CENTER	PLUMBLIN SYS	(FT/S)
10213.50	16178032.6				-187123.8				-14951307.4				27307.593				1347.323				22697.882			
10233.50	16638910.5				-182332.3				-14584087.0				26964.583				1350.374				22984.792			
10283.50	17959691.5				-54571.4				-13412029.6				25859.990				1359.300				23879.019			
11713.50	36620960.5				1602596.1				24193434.3				5002.251				959.731				24483.713			

TABLE 2.3
TRANSLUNAR MIDCOURSE CORRECTION (TLMC)

The values presented below pertain to the preflight and real time midcourse velocities required to return the S/C probe to the nominal free return trajectory. The data presented are for a launch azimuth of 72.06 degrees.

<u>Trajectory Prediction</u>	<u>Midcourse Velocity Requirements</u>
PRE-FLIGHT*	9.7 m/sec (31.8 fps)
REAL TIME**	7.7 m/sec (25.3 fps)

Note: *Midcourse applied at TB7 + 2 hrs.

**The midcourse velocity required at TB7 + 1 hr. and 50 minutes based on real time performance predictions.

TABLE 2.4
TRAJECTORY SUMMARY

A. LAUNCH

Launch Complex and Pad	39, A
Latitude; deg North of Equator	28.608422
Longitude; deg East of GM	279.395867

B. INTO-ORBIT TRAJECTORY SUMMARY

S-IC Stage

Tilt and Roll Initiation; sec	13.50
Time of Maximum Dynamic Pressure; sec	81.00
Maximum Dynamic Pressure; new/m ²	34725.243
Time of Center-Engine Cutoff (CECO); sec	135.0
Time of Outboard-Engine Cutoff (OECO); sec	160.819
Pitch Attitude from Horizontal at OECO; deg	21.201758
Inertial Flight-Path Angle at OECO; deg	70.365148
Inertial Velocity at OECO; m/sec	2755.6109
Altitude at OECO; m	66731.000
Range at OECO; m	49.786266
Latitude at OECO; deg N	28.861279
Longitude at OECO; deg E	280.294490

S-II Stage

Time of 90% Thrust; sec	165.219
Time of S-II CECO; sec	462.000
Time of Engine Cutoff (ECO); sec	551.384
Pitch Attitude Angle from Horizontal at ECO; deg	-6.097084
Inertial Flight-Path Angle at ECO; deg	89.33912
Inertial Velocity at ECO; m/sec	6938.8816
Altitude at ECO; m	187986.0
Range at ECO; m	88.594922
Latitude at ECO; deg N	31.914594
Longitude at ECO; deg E	296.0333762

TABLE 2.4 (Continued)
TRAJECTORY SUMMARY

B. INTO-ORBIT TRAJECTORY SUMMARY ,(Continued)

S-IVB First Burn

Time of 90% Thrust; sec	557.884
Time of Guidance Cutoff Signal (GCS1); sec	699.22793
Pitch Attitude from Local Horizontal at GCS1; deg	-22.17356
Inertial Flight-Path Angle at GCS1; deg	90.002205
Inertial Velocity at GCS1; m/sec	7791.4071
Altitude at GCS1; m	191339.56
Range at GCS1; m	1423.1607
Latitude at GCS1; deg N	32.666871
Longitude at GCS1; deg E	296.033762

Parking Orbit Insertion Conditions

Geocentric Radius; m	6563308.7
Inertial Velocity; m/sec	7793.1014
Inertial Path Angle; deg	90.001449
Inclination Angle; deg	32.531447
Descending Node; deg	123.10042
Eccentricity	.0010449
Orbital Period; sec	5284.3769

C. S-IVB SECOND BURN - FIRST OPPORTUNITY

Time of 90% Thrust; sec	9857.50
Time of Guidance Cutoff Signal (GCS2); sec	10203.801
Pitch Attitude from Horizontal at GCS2; deg	50.053105
Inertial Flight-Path Angle at GCS2; deg	83.041294
Inertial Velocity at GCS2; m/sec	10838.658
Altitude at GCS2; m	323847.31
Range at GCS2; m	4872.1147
Latitude at GCS2; deg N	9.5412146
Longitude at GCS2; deg E	194.45002

TABLE 2.4 (Continued)
TRAJECTORY SUMMARY

C. S-IVB SECOND BURN - FIRST OPPORTUNITY (Continued)

First-Opportunity Injection Conditions

Geocentric Radius; m	6714982.9
Inertial Velocity; m/sec	10831.069
Inertial Path Angle; deg	82.588522
Inclination Angle; deg	31.379103
Descending Node; deg	121.86592
Eccentricity	0.97667179
Twice Specific Energy; m^2/sec^2	1408484.0
Radius of Apogee; km	559401.776
Radius of Perigee; km	6601.924
Orbital Period; sec	1498291.321

TABLE 2.5

S-IC SPENT STAGE TRAJECTORY

	Altitude Above Oblate		Range n. mi.	Earth-Fixed Positions			Earth-Fixed Velocities			Longitude Degrees	Latitude Degrees
	Time sec	Earth m		X _e m	Y _e m	Z _e m	$\dot{X}_e$ m/s	$\dot{Y}_e$ m/s	$\dot{Z}_e$ m/s		
Sep.	161.59	67445	50.7	66748	35	94818	893.5	4.4	2223.6	79.69	28.87
	211.59	102836	109.0	99591	406	205064	425.7	10.6	2195.9	78.63	29.15
	261.59	117002	166.8	109391	1115	314377	-33.4	17.9	2175.7	77.57	29.43
Apex	270.59	117307	177.2	108719	1282	333940	-115.9	19.3	2171.6	77.38	29.47
	311.59	110037	224.5	96266	2206	422553	-491.7	25.9	2150.4	76.51	29.69
	361.59	81900	282.5	60183	3717	529329	-952.1	34.6	2119.3	75.44	29.93
	411.59	33724	339.9	2527	5630	631834	-1206.2	37.1	1754.9	74.37	30.17
	461.59	12231	356.3	-21931	6267	659985	-175.4	1.4	43.5	74.06	30.23
	511.59	4958	356.8	-29256	6287	660110	-123.1	0	-11.1	74.05	30.23
Impact	556.77	0	356.8	-34190	6285	659621	-97.7	0	-10.0	74.05	30.23

TABLE 2.5 (Continued)

S-II SPENT STAGE TRAJECTORY

Time sec	Altitude Above Earth m	Range n. mi.	Earth-Fixed Positions			Earth-Fixed Velocities			Longitude degrees	Latitude degrees
			X _e m	Y _e m	Z _e m	$\dot{X}_e$ m/s	$\dot{Y}_e$ m/s	$\dot{Z}_e$ m/s		
Sep.	552.3	188055	-29332	26980	1675798	-1588.9	177.4	6340.3	63.91	31.92
Apex	594	189775	-102728	34794	1938451	-1927.2	197.0	6242.6	61.12	32.22
	600	189745	-114436	35985	1975861	-1975.5	199.8	6227.5	60.72	30.26
	650	186972	-223210	46554	2283885	-2374.4	222.9	6090.2	57.36	32.51
	700	179007	-351772	58267	2584558	-2766.9	245.5	5933.5	53.98	32.65
	750	166402	-499790	71094	2876907	-3152.6	267.4	5757.2	50.58	32.70
	800	148993	-666915	84994	3159954	-3531.1	288.4	5561.4	47.16	32.64
	850	126805	-852774	99918	3432715	-3901.9	308.4	5345.8	43.73	32.47
	900	99868	-1056965	115808	3694194	-4264.2	327.0	5109.9	40.29	32.20
	950	68315	-1278273	132535	3942516	-4529.5	337.5	4758.7	36.85	31.82
	1000	37720	-1469425	146511	4130489	-2247.4	157.5	1980.9	34.10	31.44
	1050	23481	-1516952	149445	4164405	-324.1	10.2	55.0	33.52	31.35
	1100	14495	-1526568	149496	4161814	-114.9	-2.6	-83.7	33.47	31.34
	1150	8628	-1531113	149367	4158101	-76.3	-2.3	-65.1	33.47	31.34
	1200	4135	-1534527	149263	4155183	-61.5	-1.9	-52.6	33.47	31.34
Impact	1256.3	0	-1537668	149166	4152496	-50.9	-1.6	-43.5	33.47	31.34

TABLE 2.6
AS-506 SLINGSHOT TRAJECTORY PARAMETERS

Day	G.M.T. (Hr:Min:Sec)	Inertial Velocity (km/sec)	Inertial Heading Angle (deg)	Inertial Flight Path Angle (deg)
July 16	16:17:14.0	10.671940	61.98126	12.224094
July 16	16:26:14.0	9.3678589	76.30392	30.536367
July 16	16:36:14.0	8.0478669	89.77580	41.901312
July 16	16:46:14.0	7.1249554	97.88207	48.472239
July 16	16:56:14.0	6.4680100	102.83207	52.737066
July 16	17:06:14.0	5.9767351	106.07833	55.754757
July 16	17:16:14.0	5.5931458	108.34975	58.023583
July 16	17:26:14.0	5.2832451	110.02301	59.805514
July 16	17:36:14.0	5.0260922	111.30583	61.251434
July 16	17:46:14.0	4.8081271	112.32075	62.454528
July 16	17:56:14.0	4.6201795	113.14420	63.475636
July 16	18:06:14.0	4.4558143	113.82617	64.356299
July 16	18:16:14.0	4.3103725	114.40066	65.125935
July 16	18:26:14.0	4.1803902	114.89158	65.806021
July 16	18:36:14.0	4.0632327	115.31622	66.412640
July 16	19:04:14.0	3.7618881	116.24152	67.988881
July 17	00:45:34.0	2.4032334	119.58713	73.981953
July 17	09:17:34.0	1.7590114	120.55581	76.076807
July 17	17:49:34.0	1.4320021	120.92301	76.732912
July 18	02:21:34.0	1.2163213	121.14196	76.928745
July 18	10:53:34.0	1.0567654	121.32366	76.959290
July 18	19:25:34.0	0.9318212	121.53790	77.052667
July 19	01:06:54.0	0.8634191	121.75732	77.349728
July 19	15:41:34.0	0.8001595	126.93781	84.576372
July 19	18:10:54.0	0.9590657	261.89170	88.767522
July 19	19:36:14.0	1.5960845	127.79794	87.193780
July 19	19:57:34.0	1.9518132	119.54631	75.522003
July 19	20:50:54.0	2.0453685	118.48207	49.764098
July 19	22:16:14.0	1.8767662	118.43348	41.363696
July 20	00:02:54.0	1.7864215	118.44118	39.146266
July 20	01:06:54.0	1.7545924	118.44487	38.745221
July 20	02:10:54.0	1.7305034	118.44561	38.614271
July 20	08:13:34.0	1.6510978	118.39139	39.559903
July 20	16:45:34.0	1.5885312	118.18044	41.911566

TABLE 2.6 (Continued)

AS-506 SLINGSHOT TRAJECTORY PARAMETERS

Day	G.M.T. (Hr:Min:Sec)	Geocentric Radius Vector (km)	Selenocentric Radius Vector (km)	Declination (deg)	Longitude (deg)	C ₃ (Twice Specific Energy Relative to The Earth) (km ² /sec ²)
July 16	16:17:14.0	6913.15	405996.54	14.735671	205.30539	-1.42699441584
July 16	16:26:14.0	8935.40	400994.01	28.497210	240.42183	-1.46201508270
July 16	16:36:14.0	12035.83	395844.08	31.364166	264.52283	-1.46788571317
July 16	16:46:14.0	15262.33	391315.55	30.457030	277.71427	-1.46860933045
July 16	16:56:14.0	18409.52	387283.20	28.868426	285.18495	-1.46886446325
July 16	17:06:14.0	21435.86	383627.43	27.300166	289.56504	-1.46893799137
July 16	17:16:14.0	24340.55	380263.43	25.894577	292.12847	-1.46890445545
July 16	17:26:14.0	27132.89	37132.00	24.661573	293.53867	-1.46886916902
July 16	17:36:14.0	29823.92	374190.82	23.581032	294.17003	-1.46883251250
July 16	17:46:14.0	32424.05	371408.69	22.628921	294.24902	-1.46879474290
July 16	17:56:14.0	34942.48	368761.94	21.783794	293.92062	-1.46875604115
July 16	18:06:14.0	37387.16	366232.16	21.027920	293.28211	-1.46871654185
July 16	18:16:14.0	39764.90	363804.75	20.346975	292.40132	-1.46867634547
July 16	18:26:14.0	42081.60	361467.86	19.729441	291.32707	-1.46863552943
July 16	18:36:14.0	44342.32	359211.75	19.166017	290.09550	-1.46859415749
July 16	19:04:14.0	50387.58	353280.42	17.830402	214.52153	-1.66968342893
July 17	00:45:34.0	107099.48	301672.54	10.928024	142.33536	-1.66807642711
July 17	09:17:34.0	167504.79	249488.18	7.512786	19.95458	-1.66518419260
July 17	17:49:34.0	214751.03	207718.08	5.757329	254.57543	-1.66160551904
July 18	02:21:34.0	254178.72	170553.24	4.594014	128.16653	-1.65696355083
July 18	10:53:34.0	288086.44	135682.06	3.731703	1.24206	-1.65049403834
July 18	19:25:34.0	317775.18	101895.92	3.054626	234.00191	-1.64042119375
July 19	01:06:54.0	335680.50	79576.06	2.679466	149.04466	-1.62940369324
July 19	15:41:34.0	377188.77	20939.41	1.996061	290.85048	-1.47329228641
July 19	18:10:54.0	384830.99	9983.95	1.959194	253.45017	-1.1517682416
July 19	19:36:14.0	390877.31	3890.95	1.961242	232.03420	0.50795464148
July 19	19:57:34.0	393125.87	3260.95	1.936832	226.72689	1.78170926357
July 19	20:50:54.0	398818.44	5811.41	1.709178	213.77357	2.18543007600
July 19	22:16:14.0	403801.59	12224.88	1.226519	193.27314	1.55772875908
July 20	00:02:54.0	413313.36	20021.06	0.630322	167.63457	1.26248339957
July 20	01:06:54.0	417582.21	24559.00	0.283013	152.23196	1.1694939447
July 20	02:10:54.0	421761.45	29023.50	-0.056661	136.81518	1.10445890384
July 20	08:13:34.0	444821.72	53578.90	-1.853074	49.21953	0.93393093211
July 20	16:45:34.0	477228.88	87559.42	-4.074338	285.00033	0.85294069641

TABLE 2.6 (Continued)

AS-506 SLINGSHOT TRAJECTORY PARAMETERS

Day	G.M.T.	X-Component S.F.E. Apollo IV Coordinate System (km)	Y-Component S.F.E. Apollo IV Coordinate System (km)	Z-Component S.F.E. Apollo IV Coordinate System (km)	X Component of Velocity in S.F.E. Apollo IV Coordinate System (km/sec)	Y Component of Velocity in S.F.E. Apollo IV Coordinate System (km/sec)	Z Component of Velocity in S.F.E. Apollo IV Coordinate System (km/sec)
July 16	16:17:14.0	6110.72	2712.699	1758.431	-2.8775923	8.7965721	5.3131983
July 16	16:26:14.0	3769.86	6888.723	4263.226	-5.3062319	6.6331930	3.9498651
July 16	16:36:14.0	373.35	10270.336	6264.361	-5.8197138	4.7917724	2.8174476
July 16	16:46:14.0	-3097.24	12786.502	7736.353	-5.7073324	3.6862969	2.1453581
July 16	16:56:14.0	-6452.76	14774.095	8888.116	-5.4715995	2.9876283	1.7236095
July 16	17:06:14.0	-9662.21	16415.769	9831.607	-5.2288125	2.5124128	1.4382852
July 16	17:16:14.0	-12731.39	17815.135	10629.926	-5.0056953	2.1692546	1.2331376
July 16	17:26:14.0	-15673.86	19035.483	11321.407	-4.8064618	1.9097909	1.0785652
July 16	17:36:14.0	-18503.60	20118.051	11930.930	-4.6294727	1.7065023	0.9578284
July 16	17:46:14.0	-21233.09	21091.145	12475.521	-4.4718200	1.5426989	0.8608089
July 16	17:56:14.0	-23873.07	21975.029	12967.339	-4.3306328	1.4077056	0.7810527
July 16	18:06:14.0	-26432.64	22784.712	13415.368	-4.2034132	1.2943865	0.7142561
July 16	18:16:14.0	-28919.54	23531.629	13826.437	-4.0880770	1.1977933	0.6574407
July 16	18:26:14.0	-31340.37	24224.695	14205.864	-3.9829079	1.1143870	0.6084804
July 16	18:36:14.0	-33700.78	24871.006	14557.871	-3.8864955	1.0415685	0.5658166
July 16	19:04:14.0	-39993.85	26483.110	15428.701	-3.6256117	0.8840887	0.4744777
July 17	00:45:34.0	-98733.15	36191.603	20303.458	-2.3862353	0.2604977	0.1164163
July 17	09:17:34.0	-161020.29	40628.486	21900.823	-1.7576842	0.0675945	0.0099028
July 17	17:49:34.0	-209605.77	41464.812	21542.822	-1.4317182	-0.0041558	-0.0282091
July 18	02:21:34.0	-250068.63	40719.037	20358.390	-1.2147331	-0.0407474	-0.0469099
July 18	10:53:34.0	-284800.59	39126.246	18749.957	-1.0534858	-0.0608804	-0.0566964
July 18	19:25:34.0	-315148.74	37088.957	16933.615	-0.9272025	-0.0700497	-0.0606570
July 19	01:06:54.0	-333413.45	35645.658	15692.552	-0.8584878	-0.0699131	-0.0600289
July 19	15:41:34.0	-375507.50	33058.721	13137.784	-0.7999049	0.0097374	-0.0176767
July 19	18:10:54.0	-383144.84	33493.685	13156.498	-0.9529632	0.1038063	0.0298728
July 19	19:36:14.0	-389149.78	34184.427	13377.173	-1.5941607	0.0780539	0.0066948
July 19	19:57:34.0	-391418.35	34104.050	13286.736	-1.9265725	-0.2582596	-0.1766214
July 19	20:50:54.0	-397377.47	31714.309	11895.303	-1.6670569	-1.0320308	-0.5832533
July 19	22:16:14.0	-404875.83	25981.593	8686.248	-1.3310560	-1.1557858	-0.6439724
July 20	00:02:54.0	-412872.46	18536.402	4546.850	-1.1884588	-1.1660925	-0.6473760
July 20	01:06:54.0	-417340.37	14059.272	2062.647	-1.1412206	-1.1655009	-0.6463882
July 20	02:10:54.0	-421652.28	9586.374	-417.093	-1.1060695	-1.1640693	-0.6451318
July 20	08:13:34.0	-444313.68	-15646.629	-14384.029	-0.9913872	-1.1554272	-0.6389549
July 20	16:45:34.0	-473284.88	-50981.471	-33907.451	-0.9013055	-1.1451369	-0.6322510